



Additional Problems for Self Practice (APSP)

This Section is not meant for classroom discussion. It is being given to promote self-study and self testing amongst the Resonance students.

PART - I : PRACTICE TEST-1 (IIT-JEE (MAIN Pattern))

Max. Time : 1 Hr.

Max. Marks : 120

Important Instructions

- The test is of **1 hour** duration.
- The Test Booklet consists of **30** questions. The maximum marks are **120**.
- Each question is allotted **4 (four)** marks for correct response.
- Candidates will be awarded marks as stated above in Instructions No. 3 for correct response of each question.
 $\frac{1}{4}$ (**one fourth**) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 4 above.

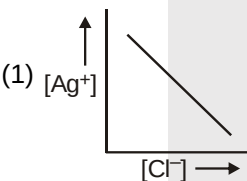
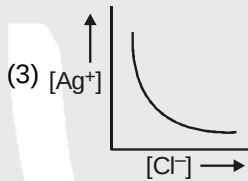
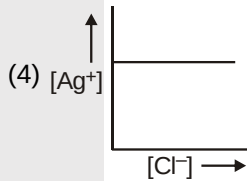
Note : Take water as solvent and temperature as 25°C , if not specified.
 Take $\log 2 = 0.3$, $\log 3 = 0.48$, $\log 5 = 0.7$, $\log 7 = 0.845$, if not specified.

- Which of the following is not a Lewis acid ?
 (1) SiCl_4 (2) SO_3 (3) Zn^{2+} (4) CO
- If the degree of ionization of water at a particular temperature was found to be 1.8×10^{-9} , the K_a of water is:
 (1) 10^{-14} (2) 1.8×10^{-12} (3) 1.8×10^{-16} (4) 10^{-7}
- An aqueous solution whose $\text{pH} = 0$ is :
 (1) Alkaline (2) Acidic (3) Neutral (4) Not possible
- The pH , when 10^{-2} M H_2SO_4 solution is prepared in 10^{-3} M Na_2SO_4 solution, is :
 (1) 2 (2) 1.7 (3) 2.3 (4) 7
- A 100 mL solution of a strong acid of $\text{pH} = 1$ is mixed with a 100 mL solution of another strong acid of $\text{pH} = 2$. The resulting pH will be nearly : ($\log 11 = 1.04$)
 (1) 1.74 (2) 0.96 (3) 1.26 (4) 1.5
- V_1 mL of 0.1 M HNO_3 is mixed with V_2 mL of 0.1 M $\text{Sr}(\text{OH})_2$. The final solution :
 (1) is neutral, if $2V_1 = V_2$ (2) is acidic, if $V_1 > V_2$
 (3) is neutral, if $V_1 = V_2$ (4) is neutral, if $V_1 = 2V_2$
- The dissociation constants of two bases B_1OH and B_2OH are 2.56×10^{-4} and 1.6×10^{-5} respectively. The relative strength of the bases will be respectively :
 (1) 1 : 4 (2) 4 : 1 (3) 1 : 16 (4) 16 : 1
- What will be the effect of adding 10^{-3} M HCl solution to a solution of HA having concentration 0.1 M ? (K_a of $\text{HA} = 10^{-5}$)
 (1) Both degree of dissociation of HA and pH will decrease.
 (2) Degree of dissociation of HA will decrease but pH of solution will remain constant.
 (3) Effect on degree of dissociation of HA and pH of solution can be predicted only by knowing the volumes of both solutions.
 (4) Neither pH nor degree of dissociation of HA will change.
- The pH of the solution of 0.2 M acetic acid and 0.1 M benzoic acid is :
 ($K_a(\text{C}_6\text{H}_5\text{COOH}) = 6.5 \times 10^{-5}$, $K_a(\text{CH}_3\text{COOH}) = 1.75 \times 10^{-5}$)
 (1) 2.65 (2) 2.5 (3) 5.3 (4) 2.4



10. H_3A is a weak triprotic acid ($K_{a_1} = 10^{-5}$, $K_{a_2} = 10^{-9}$, $K_{a_3} = 10^{-13}$). What is the value of pX of 0.1 M H_3A (aq) solution ? Where $pX = -\log X$ and $X = \frac{[A^{3-}]}{[HA^{2-}]}$
- (1) 11 (2) 8 (3) 9 (4) 10
11. When $FeCl_3$ gets soluble in water, then its solution represents which of the following characteristics ?
- (1) Amphoteric (2) Acidic (3) Basic (4) Neutral
12. The salt NaA of weak acid HA is dissolved to form its 0.01 M solution. If the degree of hydrolysis is 0.01, the pH of solution and K_a of HA at $25^\circ C$ respectively are :
- (1) 4, 10^{-8} (2) 10, 10^{-6} (3) 10, 10^{-8} (4) 4, 10^{-6}
13. 10 mL of $\frac{M}{5}$ CH_3COOH solution is mixed with 10 mL of $\frac{M}{5}$ NH_4OH solution. The change in pH of solution upon diluting each solution to double volume will be : (pK_a of $CH_3COOH = 4.76$, pK_b of $NH_4OH = 4.74$)
- (1) - 0.3 (2) + 0.3 (3) + 1 (4) No change
14. H_2A is a diprotic acid for which $K_{a_1} = 10^{-7}$ and $K_{a_2} = 10^{-11}$. The solution which will have a pH closest to 9 is :
- (1) 0.1 M H_2A (2) 0.1 M Na_2A (3) 0.1 M $NaHA$ (4) 0.1 M $NaHA + 0.1$ M Na_2A
15. S_1 : The pH of solution made by dissolving 1 mole each of HCl, NaOH & CH_3COONH_4 in the same beaker is 7, if $pK_a(CH_3COOH) = pK_b(NH_3)$.
 S_2 : Methyl orange can be used as an indicator in the titration of CH_3COOH with NaOH.
 S_3 : Water act as an acid when ammonia is dissolved in water.
- (1) T F T (2) T F F (3) F T T (4) T T T
16. Which of following cannot act as buffer ?
- (1) NH_4CN (2) $NaHCO_3 + H_2CO_3$
 (3) $NaH_2PO_4 + Na_2HPO_4$ (4) None of these
17. 500 mL of 0.01 M $CH_3COOH + 500$ mL of 0.02 M CH_3COONa gives a pH equal to 5.3. The pK_b of CH_3COO^- is :
- (1) 10 (2) 9 (3) 4 (4) 5
18. HCl gas is passed through an aqueous solution of 0.1 M 1-aminopropane ($PrNH_2$) till the pH reaches 9.7. Calculate the ratio of $[PrNH_2]/[PrNH_3^+]$ in this solution. [$K_b(PrNH_2) = 5 \times 10^{-4}$]
- (1) 0.1 (2) 0.25 (3) 10 (4) 4
19. What fraction of indicator is in acidic form at a solution pH of 5, if pK_a of the indicator is 4.7 ?
- (1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) $\frac{1}{2}$ (4) $\frac{1}{11}$
20. **Statement-1** : In a titration of weak monoacidic base with strong acid, the pOH at the half equivalent point is pK_b .
Statement-2 : At half equivalence point, it will form buffer at its maximum capacity where $[base] = [conjugate\ acid]$.
- (1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (2) Statement-1 is true, statement-2 is true and statement-2 is not correct explanation for statement-1.
 (3) Statement-1 is true, statement-2 is false
 (4) Statement-1 is false, statement-2 is true
21. 8 g weak acid HX (molecular mass = 80 u ; $K_a = 10^{-4}$) is dissolved in 100 mL water. If it is titrated with 0.25 M NaOH, find pH at equivalence point :
- (1) 9.35 (2) 8.65 (3) 9 (4) 8.7



22. 10 mL of $\frac{M}{5}$ CH_3COONa solution is titrated with $\frac{M}{5}$ HCl solution. The pH value at equivalence point is :
($\text{pK}_a(\text{CH}_3\text{COOH}) = 4.76$)
(1) 0.7 (2) 1 (3) 1.88 (4) 2.88
23. The minimum volume of water required to dissolve 10 mg CaCO_3 ($K_{\text{sp}} = 2.5 \times 10^{-9}$) is : (neglect hydrolysis of Ca^{2+} and CO_3^{2-}).
(1) 2 L (2) 0.2 L (3) 20 L (4) 0.02 L
24. Which of the following sparingly soluble salts is most soluble in water, assuming all to have same K_{sp} values ?
(1) AB (ionising into A^+ & B^-) (2) CD_2 (ionising into C^{2+} & D^-)
(3) E_2F_2 (ionising into E_2^{2+} & F^-) (4) GH_3 (ionising into G^{3+} & H^-)
25. K_{sp} of Ag_2CrO_4 at a certain temperature is 8.64×10^{-13} . How many times is its solubility in water greater than in 0.6 M Na_2CrO_4 solution ?
(1) 100 (2) 10 (3) 1000 (4) ≈ 129
26. In a saturated solution of AgCl , NaCl is added gradually. The concentration of Ag^+ is plotted against the concentration of Cl^- . The graph appears as :
(1)  (2)  (3)  (4) 
27. SrCO_3 ($K_{\text{sp}} = 10^{-10}$) and ZnCO_3 ($K_{\text{sp}} = 1.5 \times 10^{-11}$) are dissolved together in a solution. The ratio of $[\text{Sr}^{2+}]/[\text{Zn}^{2+}]$ in the solution is :
(1) $\frac{10}{3}$ (2) $\frac{3}{10}$ (3) $\frac{20}{3}$ (4) $\frac{3}{20}$
28. The solubility of $\text{Fe}(\text{OH})_3$ would be maximum in :
(1) 0.1 M NaOH (2) 0.1 M HCl (3) 0.1 M KOH (4) 0.1 M H_2SO_4 .
29. **Statement-1** : Solubility of AgCN in $\text{KCN}(\text{aq})$ is greater than in pure water.
Statement-2 : When AgCN dissolve in $\text{KCN}(\text{aq})$, complex ion $[\text{Ag}(\text{CN})_2]^-$ formation takes place and solubility equilibrium of AgCN shifted in backward direction.
(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(2) Statement-1 is true, statement-2 is true and statement-2 is not correct explanation for statement-1.
(3) Statement-1 is true, statement-2 is false. (4) Statement-1 is false, statement-2 is true.
30. What is the effect of pH on the solubility of $\text{M}(\text{CN})_2$? (neglect hydrolysis of M^{2+} ion)
(1) As pH decreases, solubility decreases.
(2) As pH decreases below 7 or increases above 7, solubility increases in both cases.
(3) As pH decreases, solubility increases. (4) No dependence on pH of solution.

Practice Test-1 (IIT-JEE (Main Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23	24	25	26	27	28	29	30
Ans.										


PART-II : NATIONAL STANDARD EXAMINATION IN CHEMISTRY (NSEC) STAGE-I

1. Which of the following is not a strong electrolyte ? [NSEC-2000]
 (A) fused NaCl (B) acetic acid
 (C) dilute perchloric acid (D) aqueous NaOH solution
2. If the solubility product of a salt MX is 3×10^{-10} at 0°C , its solubility is : [NSEC-2000]
 (A) $1.73 \times 10^{-5} \text{ mol dm}^{-3}$ (B) $6.9 \times 10^{-10} \text{ mol dm}^{-3}$
 (C) $9 \times 10^{-20} \text{ mol dm}^{-3}$ (D) $1.73 \times 10^{-10} \text{ mol dm}^{-3}$
3. What will be the pH of a solution formed by mixing 40 cm^3 of 0.1 M HCl with 10 cm^3 of 0.45 M NaOH ? [NSEC-2000]
 (A) 10 (B) 8 (C) 5 (D) 12
4. The concentration of H_3O^+ ions in pure water is $10^{-6} \text{ mol dm}^{-3}$. The corresponding concentration of OH^- ions will be : [NSEC-2000]
 (A) $10^{-14} \text{ mol dm}^{-3}$ (B) $10^{-8} \text{ mol dm}^{-3}$ (C) $10^{-6} \text{ mol dm}^{-3}$ (D) $10^{-7} \text{ mol dm}^{-3}$
5. In a reaction, $\text{HCO}_3^- + \text{H}_2\text{O} \rightarrow \text{CO}_3^{2-} + \text{H}_3\text{O}^+$
 Which two substances are Bronsted bases ? [NSEC-2000]
 (A) CO_3^{2-} & H_3O^+ (B) HCO_3^- & H_3O^+ (C) HCO_3^- & CO_3^{2-} (D) CO_3^{2-} & H_2O
6. Which of the reagents listed below could be added to water to make 0.1M solution of NH_4^+ ? [NSEC-2000]
 (A) NH_3 (B) NH_4Cl (C) both (D) none of these
7. The solubility of AgCl is $1.56 \times 10^{-5} \text{ mol m}^{-3}$ at 30°C . Its solubility product K_{sp} will be : [NSEC-2001]
 (A) 1.25×10^{-10} (B) 2.43×10^{-10} (C) 1.25×10^{-5} (D) 5.90×10^{-20}
8. The Ostwald's dilution law can be written as : [NSEC-2001]
 (A) $K_a = \left(\frac{\alpha^2}{1-\alpha} \right) V$ (B) $K_a = \left(\frac{\alpha^2}{1-\alpha} \right) \frac{1}{V}$ (C) $K_a = \frac{\alpha}{V}$ (D) $K_a = \alpha^2 \sqrt{V}$
9. Which of the alkali hydroxides is most basic ? [NSEC-2001]
 (A) KOH (B) NaOH (C) LiOH (D) CsOH
10. The pH of 0.1M CH_3COOH (dissociation constant of acetic acid is 1.80×10^{-5} at 25°C) will be : [NSEC-2001]
 (A) 5 (B) 2.873 (C) 1 (D) 0.18
11. Which of the following does not act as a buffer solution ? [NSEC-2001]
 (A) Sodium acetate and acetic acid (B) Boric acid and borax
 (C) Na_3PO_4 and Na_2HPO_4 (D) Sodium acetate and sodium citrate
12. The pH of a soft drink is 3.82. Its hydrogen ion concentration in mol.dm^{-3} will be : [NSEC-2001]
 (A) 1.5×10^{-4} (B) 1.96×10^{-3} (C) 1.96×10^{-2} (D) 1.96×10^{-1}
13. The colour change of an acid-base indicator is due to formation of : [NSEC-2002]
 (A) benzoic structure (B) quinoid structure (C) ionic structure (D) covalent bond.
14. The pK_a of aspirin is 3.5. The pH of gastric juice in human stomach is about 2 to 3, while in small intestine, the pH is 8. Aspirin will be : [NSEC-2002]
 (A) unionised in the small intestine and in the stomach
 (B) completely ionised in stomach and small intestine
 (C) ionised in stomach and almost unionised in small intestine
 (D) none of these
15. The Henderson equation for an acidic buffer is : [NSEC-2002]
 (A) $\text{pH} = \text{pK}_a + \log \frac{[\text{acid}]}{[\text{salt}]}$ (B) $\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$
 (C) $\text{pK}_w = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$ (D) $\text{pH} = \text{pK}_a + \log (\text{salt}) \cdot (\text{acid})$



16. For the salt of strong acid and weak base, the hydrolysis constant is given by : [NSEC-2002]
 (A) $K_h = \frac{K_w}{K_b}$ (B) $K_h = \frac{K_w}{K_a}$ (C) $K_h = \frac{K_w}{K_a \cdot K_b}$ (D) $K_h = \sqrt{\frac{K_w}{K_a \cdot K_b}}$
17. When a salt 'X' is dissolved in water at pH = 7, the resulting solution becomes alkaline in nature. Salt 'X' is made up of : [NSEC-2002]
 (A) strong acid and strong base (B) weak acid and weak base
 (C) weak acid and strong base (D) strong acid and weak base
18. The pH of 10^{-6} M KOH is [NSEC-2002]
 (A) 6.0 (B) 7.0 (C) 8.0 (D) 9.0
19. If H_2S gas is passed in a acidified $ZnSO_4$, the precipitation of ZnS does not take place because : [NSEC-2002]
 (A) K_{sp} of ZnS is not exceeded (B) K_{sp} of ZnS is decreases
 (C) K_{sp} of ZnS is increases (D) none of these
20. pH of circulating blood is maintained around 7.4 by the action of buffer system of : [NSEC-2002]
 (A) $H_2PO_4^-/HPO_4^{2-}$ (B) CO_2 / HCO_3^- (C) NH_4Cl/NH_4OH (D) CH_3COO^-/CH_3COONa .
21. Lead chloride has a solubility product of 1.7×10^{-5} at 300 K. Its solubility will be [NSEC-2002]
 (A) $1.62 \times 10^{-2} \text{ mol dm}^{-3}$ (B) $4.123 \times 10^{-3} \text{ mol dm}^{-3}$
 (C) $4.123 \times 10^{-6} \text{ mol dm}^{-3}$ (D) $5.1 \times 10^{-3} \text{ mol dm}^{-3}$
22. The pK_a of acetic acid is 4.74, which implies that : [NSEC-2003]
 (A) pH of 1 N acetic acid is 4.74.
 (B) at pH 4.74, the dissociation of acetic acid is maximum.
 (C) at pH 4.74, half of the acetic acid molecules are dissociated in the solution.
 (D) at pH 4.74, the dissociation of acetic acid is minimum.
23. Sodium hydroxide cannot be used as a primary standard for acid base titration because : [NSEC-2003]
 (A) it is corrosive and reacts with glass.
 (B) the dissolution of sodium hydroxide in water is highly exothermic, hence its concentration changes on dissolution.
 (C) it is hygroscopic and also reacts with atmospheric carbon dioxide.
 (D) hydroxides cannot be used as primary standards.
24. The molar concentration of pure water at 4°C and 1 atm pressure is [NSEC-2003]
 (A) 1 (B) 5.5 (C) 18 (D) 55.5
25. The solubility product of $AgCl$ is 1.8×10^{-10} . The minimum volume (in L) of water, required to dissolve 1.9 mg of $AgCl$ is approximately [NSEC-2003]
 (A) 10 (B) 2 (C) 1 (D) 20
26. In a titration experiment, 50.0 mL of 0.1 N HCl is being titrated against 0.1 N $NaOH$. The pH of the solution on addition of 49.9 mL of $NaOH$ is approximately : [NSEC-2003]
 (A) 7.0 (B) 6.0 (C) 4.0 (D) 3.0
27. The solubility product of Ag_2CrO_4 is 1.9×10^{-12} . The volume of water in mL that can just dissolve 4 mg of Ag_2CrO_4 is about : [NSEC-2004]
 (A) 150 mL (B) 1000 mL (C) 250 mL (D) 500 mL
28. The precipitate of $AgCl$ dissolves in : [NSEC-2004]
 (A) conc. HNO_3 (B) excess of HCl (C) dilute H_2SO_4 (D) aq. ammonia.
- 29.* In its 0.20 M solution, methanoic acid has degree of dissociation 0.032. Hence, its dissociation constant would be : [NSEC-2004]
 (A) 2.1×10^{-2} (B) 2.1×10^{-4} (C) 1.1×10^{-6} (D) 1.6×10^{-8}
30. A pair that forms a buffer system is : [NSEC-2004]
 (A) $BaCl_2 + Ba(NO_3)_2$ (B) $NaCl + HCl$
 (C) $Na_3PO_4 + Na_2HPO_4$ (D) $AgCN + KCN$.

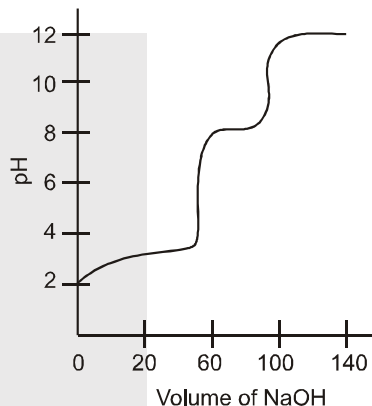
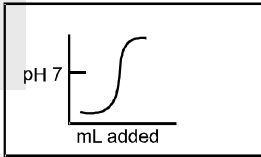


31. A buffer solution is prepared by mixing 0.050 moles of a weak acid HA and 0.20 moles of NaA in sufficient amount of water to give 500 mL of solution (K_a for HA is 4.5×10^{-4}). The pH of this solution is: [NSEC-2004]
 (A) 1.97 (B) 2.17 (C) 2.74 (D) 3.95
- 32.* Consider various species generated when H_3PO_4 dissolved in water. Among these, the conjugate acid of HPO_4^{2-} is : [NSEC-2004]
 (A) H_3PO_4 (B) $H_2PO_4^-$ (C) PO_4^{3-} (D) H_3O^+
33. If a weak base has the dissociation constant K_b , then the value of the dissociation constant, K_a of its conjugate acid is given by : [NSEC-2004]
 (A) $1/K_b$ (B) K_w/K_b (C) K_b/K_w (D) $K_w K_b$
34. The salt that when added to water will not change its pH is : [NSEC-2004]
 (A) Na_2CO_3 (B) NaCl (C) KCN (D) NH_4Cl
35. What is the pH of 10^{-3} (N) CH_3COOH solution? (Dissociation constant of CH_3COOH is 1.74×10^{-5}) [NSEC-2005]
 (A) 3.88 (B) 4.76 (C) 5.55 (D) 3.0
36. The ionic product of water at $25^\circ C$ is 10^{-14} . The corresponding dissociation constant is : [NSEC-2005]
 (A) 2.0×10^{-16} (B) 3.6×10^{-16} (C) 1.5×10^{-16} (D) 1.8×10^{-16}
37. The solubility product of $Mg(OH)_2$ is $3.4 \times 10^{-11} \text{ mol}^3 \text{ L}^{-3}$. What will be its solubility in g L^{-1} (molecular weight of $Mg(OH)_2$ is 58.3) [NSEC-2005]
 (A) 2.0×10^{-4} (B) 1.2×10^{-2} (C) 1.6×10^{-3} (D) 3.5×10^{-4}
38. pH of the blood in the body is maintained by : [NSEC-2005]
 (A) glucose and salt concentration (B) protein and salt concentration
 (C) carbonate and bicarbonate concentration (D) salt and carbonate
39. Which indicator will be suitable for the titration of acetic acid vs NaOH ? [NSEC-2005]
 (A) methyl orange [$pK_{in} = 3.7$] (B) bromocresol green [$pK_{in} = 4.7$]
 (C) chlorophenol red [$pK_{in} = 6.1$] (D) phenolphthalein [$pK_{in} = 9.6$]
40. The compound that may be added to water to increase its pH from 7 to a higher value is : [NSEC-2006]
 (A) Na_2CO_3 (B) $AlCl_3$ (C) NaCl (D) HCl
41. The most acidic aqueous solution is : [NSEC-2006]
 (A) CH_3COONa (B) Na_2CO_3 (C) NH_4Cl (D) Na_2HPO_4
42. The slightly soluble Ag_2CrO_4 is expected to be more soluble in : [NSEC-2006]
 (A) 0.1 M HNO_3 (B) 0.01 M K_2CrO_4 (C) pure water (D) 0.1 M $AgNO_3$.
43. Pure water is saturated with a sparingly soluble salt, PbI_2 which has a solubility product K_{sp} . In this saturated solution : [NSEC-2006]
 (A) $[Pb^{2+}] = [I^-]$ (B) $[Pb^{2+}] = K_{sp}$ (C) $[Pb^{2+}] = (K_{sp})^{1/2}$ (D) $2[Pb^{2+}] = [I^-]$
44. Which of the following metal sulphides has maximum solubility in water? (K_{sp} values are indicated in parentheses). [NSEC-2006]
 (A) HgS (10^{-54}) (B) CdS (10^{-30}) (C) FeS (10^{-20}) (D) ZnS (10^{-22}) .
45. If 25 mL of 0.01 N NaOH is added to 50 mL of 0.01 N acetic acid, the pH of the resultant solution will be: (pK_a of $CH_3COOH = 4.76$) [NSEC-2006]
 (A) 5.5 (B) 4.5 (C) 4.76 (D) 2.48
46. Calculate the pH of a solution made by mixing 150 cm^3 of 0.10 M CH_3COONa and 250 cm^3 of 0.10 M CH_3COOH . [K_a of $CH_3COOH = 1.8 \times 10^{-5}$] [NSEC-2007]
 (A) 237 (B) 4.52 (C) 4.74 (D) 4.97
47. The solubility product constant, K_{sp} of Ag_3PO_4 is 1.8×10^{-18} . What is the molar solubility of Ag_3PO_4 in water? (Neglect any hydrolysis) [NSEC-2007]
 (A) 1.6×10^{-5} (B) 8.4×10^{-7} (C) 1.3×10^{-9} (D) 4.5×10^{-19}



48. If each of the following salts has $K_{sp} = 1 \times 10^{-9}$, which of them is the least soluble in water? [NSEC-2007]
 (A) XY (B) XY_2 (C) X_2Y (D) X_3Y
49. The pH of a 0.01 M solution of ammonium acetate can be changed by changing : [NSEC-2007]
 (A) the temperature (B) the volume of solution
 (C) the concentration (D) the pressure on solution
50. The pH of 0.1 M NH_4OH ($K_b = 1.8 \times 10^{-5}$) is : [NSEC-2008]
 (A) 1 (B) 13 (C) 5.7 (D) 11.1
51. The pH of mixture of 100 mL of 0.25 N HCl and 100 mL of 0.125 N NaOH is : [NSEC-2008]
 (A) 1.20 (B) 0.90 (C) 0.70 (D) 0.42
52. CaF_2 (Mol. Wt. = 78) has a solubility of 1.6×10^{-2} g/dm³ at 293 K. The solubility product of CaF_2 is : [NSEC-2008]
 (A) 3.45×10^{-11} (B) 4.20×10^{-8} (C) 2.56×10^{-4} (D) 4.69×10^{-9}
53. Which of these mixtures constitute buffer solutions ? [NSEC-2008]
Mixture 1 : 25 mL of 0.10 M HNO_3 and 25 mL of 0.10 M $NaNO_3$
Mixture 2 : 25 mL of 0.10 M $HC_2H_3O_2$ and 25 mL of 0.10 M NaOH
 (A) 1 only (B) 2 only (C) both 1 and 2 (D) neither 1 nor 2
54. A 0.200 g sample of benzoic acid, C_6H_5COOH , is titrated with a 0.120 M $Ba(OH)_2$ solution. What volume of the $Ba(OH)_2$ solution is required to reach the equivalence point ? [NSEC-2008]
 $[C_6H_5COOH, \text{Molar Mass} = 122.1 \text{ g mol}^{-1}]$
 (A) 6.82 mL (B) 13.6 mL (C) 17.6 mL (D) 35.2 mL
55. The amount of sodium hydrogen carbonate, $NaHCO_3$, in an antacid tablet is to be determined by dissolving the tablet in water and then titrating the resulting solution with hydrochloric acid. Which indicator is the most appropriate for this titration ? [NSEC-2008]
- | Acid | K_a |
|-----------|----------------------|
| H_2CO_3 | 2.5×10^{-4} |
| HCO_3^- | 2.4×10^{-8} |
- (A) methyl orange, $pK_{in} = 3.7$ (B) bromothymol blue, $pK_{in} = 7.0$
 (C) phenolphthalein, $pK_{in} = 9.3$ (D) alizarin yellow, $pK_{in} = 12.5$
56. The hydrolysis constant of 0.5 M ammonium benzoate is 6.25×10^{-6} . The percentage hydrolysis of the salt is : [NSEC-2009]
 (A) 0.25 (B) 0.177 (C) 0.125 (D) 0.50
57. When CO_2 dissolves in water, the following equilibrium is established – [NSEC-2009]
 $CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^{1-}$
- If the equilibrium constant for the reaction is 3.8×10^{-7} and pH = 6, the ratio $\frac{[HCO_3^-]}{[CO_2]}$ will be:
 (A) 3.8×10^{-13} (B) 3.8×10^{-1} (C) 6.0 (D) 3.0
58. Carbonic acid, H_2CO_3 , is a diprotic acid for which $K_1 = 4.2 \times 10^{-7}$, and $K_2 = 4.7 \times 10^{-11}$. The solution which will have a pH closest to 9 is : [NSEC-2009]
 (A) 0.1 M H_2CO_3 (B) 0.1 M Na_2CO_3
 (C) 0.1 M $NaHCO_3$ (D) 0.1 M $NaHCO_3$ + 0.1 M Na_2CO_3
59. The strongest Bronsted base among the following compounds is – [NSEC-2009]
 (A) $NaCH_3$ (B) NaOH (C) NaF (D) $NaNH_2$
60. The ionization of benzoic acid is represented by the equation : [NSEC-2009]
 $C_6H_5COOH(aq) \rightleftharpoons H^+(aq) + C_6H_5COO^-(aq)$
 If a 0.045 M solution of benzoic acid has an $[H^+] = 1.7 \times 10^{-3}$ M, then, the K_a of benzoic acid is :
 (A) 7.7×10^{-5} (B) 6.4×10^{-5} (C) 3.8×10^{-2} (D) 8.4×10^{-1}
61. If the K_{sp} of CaF_2 at 25°C is 1.6×10^{-10} , then the number of moles of the salt must be dissolved in 2.0 L of water at 25°C to form a saturated solution is : [NSEC-2009]
 (A) 2.6×10^{-2} mol (B) 1.3×10^{-3} mol (C) 6.8×10^{-4} mol (D) 3.4×10^{-4} mol



62. A solution of 0.10 M NaZ has pH = 8.90. The K_a of HZ is : [NSEC-2010]
 (A) 1.6×10^{-4} (B) 1.6×10^{-5} (C) 6.3×10^{-10} (D) 6.3×10^{-11}
63. The pH of a solution made of mixing 200 mL of 0.0657 M NaOH, 140 mL of 0.107 M HCl and 150 mL of water would be : [NSEC-2010]
 (A) 3.04 (B) 2.43 (C) 2.74 (D) 2.27
64. The molarity of NH_3 solution of pH 12.0 at 25°C is (K_b of NH_3 at 25°C is 1.8×10^{-5}) [NSEC-2010]
 (A) 0.55 M (B) 5.5 M (C) 0.01 M (D) 11.7 M
65. A weak acid HA has $K_a = 1.00 \times 10^{-5}$ at 25°C . If 0.100 mole of this acid are dissolved in 1 L water, the percentage of the acid dissociated at equilibrium will be closest to : [NSEC-2010]
 (A) 0.100% (B) 99.0% (C) 1.00% (D) 99.9%
66. The solubility of a salt MX is $3.60 \times 10^{-5} \text{ mol dm}^{-3}$. The K_{sp} MX is: [NSEC-2010]
 (A) 3.60×10^{-5} (B) 12.96×10^{-10} (C) 7.20×10^{-5} (D) 1.80×10^{-10}
67. The following curve represents the titration of : [NSEC-2010]
 (A) a diprotic acid
 (B) two monoprotic acids with the same K_a , but different concentrations
 (C) two monoprotic acids with the different K_a , but the same concentration
 (D) two monoprotic acids with the different K_a , and different concentrations
- 
68. The solubility of calcium phosphate is $S \text{ mol dm}^{-3}$. Hence, the solubility product is : [NSEC-2011]
 (A) S^5 (B) $27S^3$ (C) $54S^4$ (D) $108S^5$
69. 0.1 dm³ of 0.1 M acetic acid is titrated against 0.1 M NaOH. When 50 cm³ of 0.1 M NaOH are added, the pH of the solution will be : ($\text{p}K_a = 4.74$) [NSEC-2011]
 (A) 2.37 (B) 4.74 (C) 1.34 (D) 5.74
70. A 0.056 M solution of benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, is titrated with a strong base. $[\text{H}^+]$ of the solution when half of the solution is titrated before the equivalence point is : (K_a of benzoic acid = 6.3×10^{-5})
 (A) $6.3 \times 10^{-5} \text{ M}$ (B) $1.8 \times 10^{-3} \text{ M}$ (C) $7.9 \times 10^{-3} \text{ M}$ (D) $2.6 \times 10^{-2} \text{ M}$
71. The following titration curve represents the titration of a _____ acid with a _____ base. [NSEC-2011]
 (A) strong, strong
 (B) weak, strong
 (C) strong, weak
 (D) weak, weak
- 
72. Hypochlorous acid ionizes as
 $\text{HOCl(aq)} \rightleftharpoons \text{H}^+(\text{aq}) + \text{OCl}^-(\text{aq})$, $\text{OCl}^-(\text{aq}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{HOCl(aq)} + \text{OH}^-(\text{aq})$
 K_a for this reaction at 25°C is 3.0×10^{-8} ($K_w = 1.0 \times 10^{-14}$ at 25°C)
 Hence, K_b for OCl^- is : [NSEC-2011]
 (A) 3.3×10^{-7} (B) 3.0×10^{-8} (C) 3.0×10^6 (D) 3.3×10^7
73. When equal volumes of the following solutions are mixed, precipitation of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) will occur only with : [NSEC-2012]
 (A) 10^{-4} M Ag^+ and 10^{-4} M Cl^- (B) 10^{-5} M Ag^+ and 10^{-5} M Cl^-
 (C) 10^{-6} M Ag^+ and 10^{-6} M Cl^- (D) 10^{-10} M Ag^+ and 10^{-10} M Cl^-

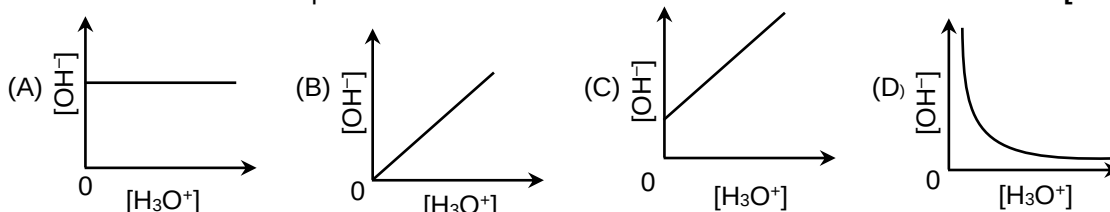


74. Equal volumes of two solutions of pH = 2 and pH = 4 are mixed together. The pH of the resulting solution will be : [NSEC-2012]
(A) 2.0 (B) 3.1 (C) 4.2 (D) 2.3
75. The solubility of SrF_2 in water at 303 K is $9.55 \times 10^{-5} \text{ mol dm}^{-3}$. The solubility product of the salt is : [NSEC-2012]
(A) 8.7×10^{-17} (B) 9.1×10^{-11} (C) 9.55×10^{-5} (D) 3.48×10^{-12}
76. The pK_a value of picric acid, acetic acid and phenol are in the order : [NSEC-2012]
(A) Picric acid 0.4, acetic acid 4.75, phenol 10.0
(B) Acetic acid 0.4, picric acid 4.75, phenol 10.0
(C) picric acid 0.4, phenol 4.75, acetic acid 10.0
(D) phenol 0.4, acetic acid 4.75, picric acid 10.0
77. If each of the following salts has solubility product $K_{\text{sp}} = 1 \times 10^{-9}$, which of them is least soluble in water? [NSEC-2013]
(A) XY (B) X_2Y (C) XY_2 (D) X_3Y
78. Which of the following salts produces the most basic solution ? [NSEC-2013]
(A) $\text{Al}(\text{CN})_3$ (B) $\text{KC}_2\text{H}_3\text{O}_2$ (C) FeCl_3 (D) KCl
79. The pH of 0.1 M NH_4OH , ($K_b = 1.8 \times 10^{-5}$, $K_w = 10^{-14}$), is : [NSEC-2013]
(A) 1.0 (B) 5.7 (C) 11.1 (D) 13.0
80. Which solution has the highest pH ? [NSEC-2013]
(A) 0.01 M CaCl_2 (B) 0.01 M KNO_2 (C) 0.01 M CH_3COOH (D) 0.01 M CH_3COCH_3
81. pH of a saturated solution of magnesium hydroxide in water at 298 K is 10.5. The solubility of the hydroxide in water at 298 K is [NSEC-2014]
(A) $1.58 \times 10^{-4} \text{ mol L}^{-1}$ (B) $1.58 \times 10^{-11} \text{ mol L}^{-1}$ (C) $3.16 \times 10^{-4} \text{ mol L}^{-1}$ (D) $9.98 \times 10^{-8} \text{ mol L}^{-1}$
82. Solubility product of silver chloride and silver thiocyanate are 1.2×10^{-10} and 7.1×10^{-13} respectively. The equilibrium constant for the reaction $\text{AgCl}(\text{s}) + \text{CNS}^{-1}(\text{aq}) \rightleftharpoons \text{AgCNS}(\text{s}) + \text{Cl}^{-1}(\text{aq})$ is [NSEC-2014]
(A) 0.0625 (B) 169 (C) 13 (D) 1.40×10^{-4}
83. If a weak base has the dissociation constant, K_b , then the value of the dissociation constant, K_a , of its conjugate acid is given by [NSEC-2014]
(A) $1/K_b$ (B) K_w/K_b (C) K_b/K_w (D) $K_w K_b$
84. When 1L of 0.1 M sulphuric acid solution is allowed to react with 1L of 0.1 M sodium hydroxide solution, the amount of sodium sulphate (anhydrous) that can be obtained from the solution formed and the concentration of H^+ in the solution respectively are [NSEC-2015]
(A) 3.55 g, 0.1 M (B) 7.10 g, 0.025M (C) 3.55g, 0.025M. (D) 7.10g, 0.05 M
85. A water sample from a municipal water supply was found to have a pH = 7.0. On evaporating 2 L of this water, 2.016 g of white solid was left behind in the evaporation vessel, i.e., the total dissolved solid (TDS) content of this water was 1008 mg L^{-1} . However, addition of soap to a bucket of this water did not produce any visible scum. Based on these findings, one can conclude that [NSEC-2015]
(A) There are no Ca^{2+} or Mg^{2+} ion in the water
(B) There are no CO_3^{2-} or HCO_3^{-} ion in the water
(C) concentration of any ion in the water is lower than 0.038M
(D) water may be containing Na^+ ions in concentration $> 0.04\text{M}$
86. A 50 mL solution of pH = 1 is mixed with a 50 mL solution of pH = 2. The pH of the mixture is : [NSEC-2015]
(A) 0.86 (B) 1.26 (C) 1.76 (D) 2.26
87. The solubility products (K_{sp}) of three salts MX , MY_2 and MZ_3 are 1×10^{-8} , 4×10^{-9} and 27×10^{-8} , respectively. The correct order for solubilities of these salts is [NSEC-2015]
(A) $\text{Mx} > \text{MY}_2 > \text{MZ}_3$ (B) $\text{MZ}_3 > \text{MY}_2 > \text{MX}$ (C) $\text{MZ}_3 > \text{MX} > \text{MY}_2$ (D) $\text{MY}_2 > \text{MX} > \text{MZ}_3$





88. If the solubility product of iron(III) hydroxide is 1.8×10^{-37} , the pH of a saturated solution of iron(III) hydroxide in distilled water is close to : [NSEC-2016]
 (A) 4 (B) 5 (C) 7 (D) 9
89. The pH of a $1.0 \times 10^{-3} \text{ mol L}^{-1}$ solution of weak acid HA is 3.60. The dissociation constant of the acid is: [NSEC-2016]
 (A) 8.4×10^{-8} (B) 8.4×10^{-6} (C) 8.4×10^{-5} (D) 8.4×10^{-2}
90. The colour changes of an indicator HIn in acid base titrations is given below
 $\text{HIn (aq)} \rightleftharpoons \text{H}^+ \text{ (aq)} + \text{In}^- \text{ (aq)}$
 Colour X Colour Y
 Which of the following statements is correct? [NSEC-2017]
 (A) In a strong alkaline solution colour Y will be observed
 (B) In a strong acidic solution colour Y will be observed
 (C) Concentration of In^- is higher than that of HIn at the equivalence point
 (D) In a strong alkaline solution colour X is observed
91. The table below gives the results of three titrations carried out with 0.200 M HCl to determine the molarity of a given NaOH solution using phenolphthalein as indicator. NaOH was taken in the burette and HCl was taken in a conical flask for the titrations [NSEC-2017]
- | Titration No. | V _{HCl} (mL) | V _{NaOH} (mL) | M _{NaOH} mol dm ⁻³ |
|---------------|-----------------------|------------------------|--|
| I | 21.4 | 19.3 | 0.222 |
| II | 18.6 | 16.8 | 0.221 |
| III | 22.2 | 21.1 | 0.210 |
- The actual molarity of the prepared NaOH solution was $0.220 \text{ mol dm}^{-3}$. Which among the following could be the reason for the wrong value obtained in titration III?
 (A) Number of drops of phenolphthalein added to the titration flask was more in this titration
 (B) The concentration of HCl was wrongly used as 0.250 M for the calculation of M_{NaOH}
 (C) A few drops of NaOH solution were spilled outside the titration flask during titration
 (D) A few drops of the neutralized solution from titration II were left behind in the flask
92. The solution with pH value close to 1 is [NSEC-2017]
 (A) 10 mL, 0.1 M HCl + 90 mL, 0.1 M NaOH (B) 55 mL, 0.1 M HCl + 45 mL, 0.1 M NaOH
 (C) 75 mL, 0.2 M HCl + 25 mL, 0.2 M NaOH (D) 75 mL, 0.2 M HCl + 25 mL, 0.1 M NaOH
93. Which of the following is not a pair of a Lewis acid and a Lewis base ? [NSEC-2017]
 (A) H^+ , $(\text{C}_2\text{H}_5)_2\text{O}$ (B) H_2O , AlCl_3 (C) Fe^{3+} , CO (D) SiF_4 , BF_3
94. Which of the following statement/s is/are correct about weak acids in aqueous solutions? [NSEC-2017]
 I. When $\text{pH} = \text{pK}_a$ of a monoprotic acid, 50% of the acid is ionised
 II. If $\text{pH} = \text{pK}_{a2}$ of a diprotic acid, the average charge of all the ionised species is 0.5
 III. When $\text{pH} = \text{pK}_a + 1$, 10% of the acid is ionised
 IV. When $\text{pH} = 7$, 50% of a monibasic acid is ionized.
 (A) I and IV (B) I, II and IV (C) I, II and IV (D) I only
95. Solubility product of AgCl is 1.8×10^{-10} . The minimum volume (in L) of water required to dissolve 1 mg of AgCl is close to [NSEC-2018]
 (A) 0.5 (B) 7.5 (C) 50 (D) 0.75
96. If for an aqueous solution of a weak acid, $\text{pH} = \text{pK}_a + 2$ at 25°C , the approximate fraction of the acid in the dissociated form is [NSEC-2018]
 (A) 1.1 % (B) 0.99 % (C) 99.0 % (D) 9.9 %
97. Which of the following graphs describes the relationship between $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ in an aqueous solution at a constant temperature ? [NSEC-2018]





PART - III : HIGH LEVEL PROBLEMS (HLP)

THEORY

Levelling / Differentiating Effect of Solvent :

A solvent is said to be showing levelling effect if various electrolytes (generally acids and bases) get completely dissociated in it and thus have the same strength. For example, in water solvent, all strong acids (HClO_4 , H_2SO_4 , HI , HBr and HCl) are completely dissociated and are said to have the same strength. Similarly, all strong bases (CsOH , RbOH , KOH , NaOH and Ba(OH)_2) are also completely dissociated in water solvent. So, their relative strengths cannot be compared in water solvent.

However, a solvent is said to be showing differentiating effect if various acids and bases dissociate to different degrees in it and thus their relative strengths can be compared. For example, in water solvent, weak acids (like CH_3COOH , HCOOH etc.) dissociate to different extents and on comparing their degree of dissociation, their relative strengths can be compared. Same is the case with weak bases (like $\text{NH}_3 \cdot \text{H}_2\text{O}$, Aniline etc.). So, it can be concluded that water in general behaves as a levelling solvent for strong acids as well as strong bases, but it behaves as a differentiating solvent for weak acids as well as weak bases.

Considering other solvents, in glacial acetic acid solvent, strong acids dissociate to different degrees. So, strengths of strong acids can be compared in this solvent and in general, it behaves as differentiating solvent for strong acids. However, due to acidic nature of solvent, weak bases dissociate completely in it. So, in general, it behaves as a levelling solvent for weak bases.

Similarly, behaviour of ammonia solvent (basic in nature) can be thought of.

Note : In anhydrous HF solvent, only HClO_4 behaves like an acid because HF does not have a tendency to accept H^+ from other acids and form an unstable species H_2F^+ (positive charge on most electronegative atom F). So, rest all acids behave as bases in HF solvent.

INDICATORS :

Indicators are the substances, which indicate the end-point of a titration generally by changing their colour.

They are, in general, either weak organic acids or weak organic bases, having characteristically different colours in the ionised and unionised forms. For example, methyl orange is a weak base (having red colour in ionised form and yellow colour in the unionised form) and phenolphthalein is a weak acid (having pink colour in the ionized form while its unionised form is colourless)

Let us consider the equilibrium between the ionised and unionised form of an acid indicator (HIn)



$$\therefore K_{\text{In}} = \frac{[\text{H}^+][\text{In}^-]}{[\text{HIn}]} \quad [K_{\text{In}} = \text{indicator constant or dissociation constant of indicator}]$$

$$\text{or} \quad [\text{H}^+] = K_{\text{In}} \times \frac{[\text{HIn}]}{[\text{In}^-]}$$

Taking negative logarithm of both sides,

$$-\log [\text{H}^+] = -\log K_{\text{In}} - \log \frac{[\text{HIn}]}{[\text{In}^-]}$$

$$\therefore \text{pH} = \text{p}K_{\text{In}} + \log \frac{[\text{In}^-]}{[\text{HIn}]} \quad \Rightarrow \quad \text{pH} = \text{p}K_{\text{In}} + \log \frac{[\text{Ionised form}]}{[\text{Unionised form}]}$$

In general, the intensity of a coloured solution depends on the concentration of the colour imparting species. If the solution contains two coloured species, then the colour of the solution depends on the relative concentration of the colour imparting species. When the solution contains two coloured species as HIn and In^- , then it is a medical fact that the solution acquires a distinct colour of In^- for visibility by human eyes only when the concentration of In^- is approximately atleast 10 times greater than the concentration of HIn and vice versa. Actually, this numerical factor has different value for different colours, but if not specified, we take this factor as 10 in general.



Let us consider two cases :

Case-I : For the solution to show colour due to In^- , the minimum ratio $\frac{[\text{In}^-]}{[\text{HIn}]}$ should be 10.

$$\therefore \text{pH} = \text{pK}_{\text{In}} + \log 10 = \text{pK}_{\text{In}} + 1$$

At this stage, the percentage of the indicator in In^- form would be given by :

$$\frac{[\text{In}^-]}{[\text{In}^-] + [\text{HIn}]} \times 100 = \frac{10}{10 + 1} \times 100 \approx 91\%$$

Thus, nearly 91% of the indicator has been present in the ionised form (In^-). In fact, $\text{pH} = \text{pK}_{\text{In}} + 1$ is the minimum pH upto which the solution has a distinct colour characteristic of In^- . At pH greater than this value, some more indicator will be present in the ionised form. Thus at $\text{pH} \geq \text{pK}_{\text{In}} + 1$, the solution has a colour characteristic of In^- .

Case – II : For the solution to show colour due to HIn , the minimum ratio of $\frac{[\text{In}^-]}{[\text{HIn}]}$ should be $\frac{1}{10}$.

$$\therefore \text{pH} = \text{pK}_{\text{In}} + \log \frac{1}{10} = \text{pK}_{\text{In}} - 1.$$

At this stage, the percentage of the indicator in In^- form would be given by :

$$\frac{[\text{In}^-]}{[\text{In}^-] + [\text{HIn}]} \times 100 = \frac{1}{10 + 1} \times 100 \approx 9\%$$

Thus, nearly 9% of the indicator has been present in the ionised form (In^-) and $\approx 91\%$ of the indicator is present in the unionized form (HIn). In fact, $\text{pH} = \text{pK}_{\text{In}} - 1$ is the maximum pH upto which the solution has a distinct colour characteristic of HIn . At pH smaller than this value, some more indicator will be present in the unionised form. Thus, at $\text{pH} \leq \text{pK}_{\text{In}} - 1$, the solution has a colour characteristic of HIn .

$\therefore$ General pH range of an indicator : $\text{pK}_{\text{In}} - 1$ to $\text{pK}_{\text{In}} + 1$.

SIGNIFICANCE OF INDICATORS :

Indicators work in specific pH range i.e. they change their colour distinctly once a specific pH range is crossed. For example, phenolphthalein works in pH range 8.3–10 while methyl orange in the range 3.1 to 4.4. Range of some common indicators is given below :

Table-1

Indicator	pH range	Colour	
		If $\text{pH}_{\text{solution}} < \text{Minimum value of pH range}$	If $\text{pH}_{\text{solution}} > \text{Maximum value of pH range}$
Methyl orange	3.1 – 4.4	Red	Yellow
Methyl red	4.2 – 6.3	Red	Yellow
Litmus	5.0 – 8.0	Red	Blue
Phenol red	6.4 – 8.2	Yellow	Red
Phenolphthalein	8.3 – 10.0	Colourless	Pink



Extent of reaction of different bases with acid (HCl) using two indicators Phenolphthalein & Methyl orange is summarised below :

	Phenolphthalein	Methyl Orange
NaOH	100% reaction is indicated	100% reaction is indicated
	$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
Na_2CO_3	50% reaction upto NaHCO_3 stage is indicated	100% reaction is indicated
	$\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaHCO}_3 + \text{NaCl}$	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$
NaHCO_3	No reaction is indicated	100% reaction is indicated
		$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$



Solved Examples

Ex-1. pH of 0.1 M solution of NaA (sodium salt of a weak acid HA) is 8.92. Calculate pK_a of HA. If a drop of HPh ($pK_{In} = 9.52$) be added to the above solution, predict whether the pink colour will be visible or not, under the medical fact that our eyes can see the pink color if the mole % of ionised form of indicator is 25% or more.

Sol.
$$pH = \frac{1}{2} (pK_a + pK_w + \log c)$$

$$8.92 = 7 + \frac{1}{2} pK_a + \frac{1}{2} (-1)$$

$$\frac{1}{2} pK_a = 8.92 - 6.5 = 2.42$$

$\therefore pK_a \text{ of HA} = 4.84$

$$pH = pK_{In} + \log \frac{[In^-]}{[HIn]}$$

$$8.92 = 9.52 + \log \frac{[In^-]}{[HIn]}$$

$$\frac{[In^-]}{[HIn]} = \frac{\text{ionised}}{\text{unionised}} = \frac{1}{4}$$

Since ionised form % = $\frac{1}{1+4} \times 100 = 20\%$, pink colour will not be visible.

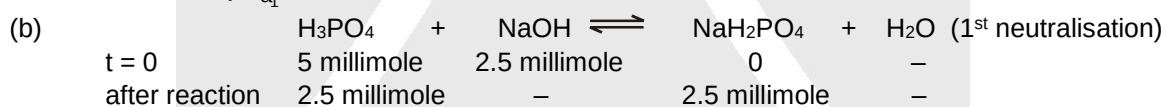
Titration of Poly protic acid with SB :

Ex. 50 mL of 0.1 M H_3PO_4 with 0.1 M NaOH. For plotting curve, calculate pH when volume of NaOH added is :

- (a) 0 mL (b) 25 mL (c) 50 mL (d) 75 mL (e) 100 mL
(f) 125 mL (g) 150 mL (h) 200 mL (i) 90 mL

For H_3PO_4 : $pK_{a_1} = 3$; $pK_{a_2} = 7$; $pK_{a_3} = 11$

(a) $pH = 1/2 (pK_{a_1} - \log C) = 1/2 (3 + 1) = 2$



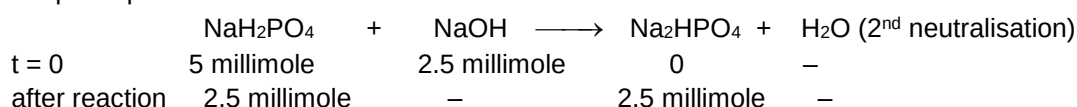
Buffer with equal amounts of H_3PO_4 & $H_2PO_4^- \Rightarrow$ Half equivalence point with respect to 1st neutralisation.

$$pH = pK_{a_1} = 3$$

(c) Solution of $H_2PO_4^-$ (amphiprotic species) will be obtained as amount of NaOH added is just sufficient to complete 1st neutralisation of H_3PO_4 .

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2} = \frac{3+7}{2} = 5 \text{ (1st eq pt)}$$

(d) 50 ml NaOH solution will be consumed in 1st neutralisation & remaining 25 mL (2.5 millimole) will participate in 2nd neutralisation.



Buffer with equal amounts of $H_2PO_4^-$ & $HPO_4^{2-} \Rightarrow$ Half equivalence point with respect to 2nd neutralisation.

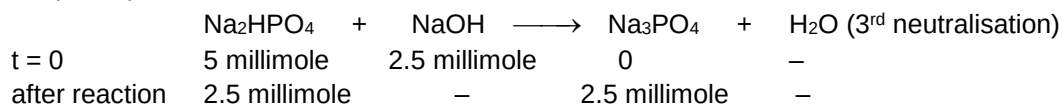
$$pH = pK_{a_2} = 7$$

(e) Solution of HPO_4^{2-} solution (amphiprotic species) will be obtained as amount of NaOH added is just sufficient to complete 2nd neutralisation of H_3PO_4 .



$$\text{pH} = \frac{\text{pK}_{a_2} + \text{pK}_{a_3}}{2} = 9 \text{ (2nd eq pt)}$$

- (f) 100 mL NaOH solution will be consumed in 2 neutralisations & remaining 25 mL (2.5 millimole) will participate in 3rd neutralisation.



Buffer with equal amounts of HPO_4^{2-} & $\text{PO}_4^{3-} \Rightarrow$ Half equivalence point with respect to 3rd neutralisation.

$$\text{pH} = \text{pK}_{a_3} = 11$$

- (g) Solution of PO_4^{3-} will be obtained as amount of NaOH added is just sufficient to complete 3rd neutralisation of H_3PO_4 .

$$[\text{Na}_3\text{PO}_4] = \frac{\text{Millimole}}{\text{Total volume in mL}} = \frac{5}{200} = 1/40 \text{ M}$$

Assuming h to be negligible

$$h = \sqrt{\frac{K_w / K_{a_3}}{C}} = \sqrt{\frac{10^{-14} / 10^{-11}}{1/40}} = 0.2 \text{ (not negligible)}$$

$$\therefore \frac{1}{40} = \frac{h^2}{1-h} = \frac{10^{-14}}{10^{-11}}$$

On solving this quadratic, $h = 0.18$

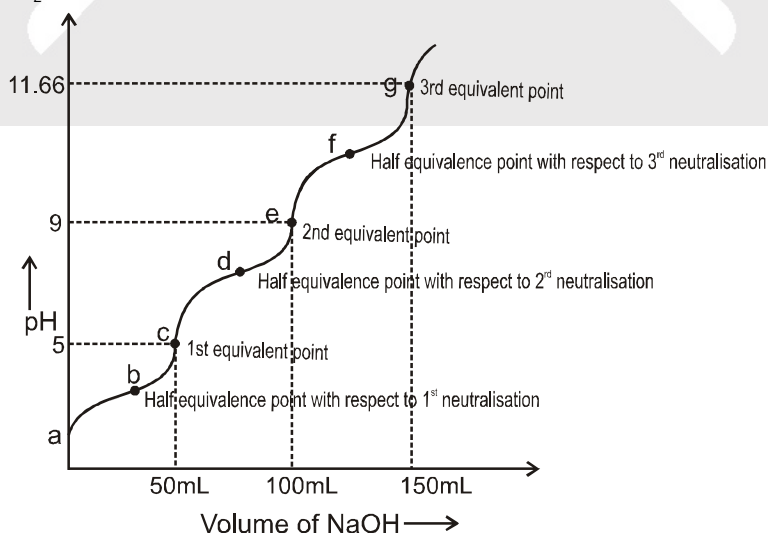
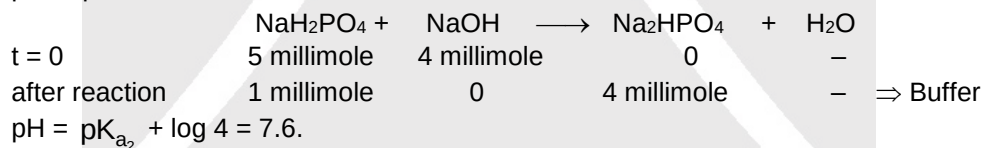
$$\therefore [\text{OH}^-] = ch = 4.5 \times 10^{-3} \text{ M}$$

$$\therefore \text{pH} = 11.66$$

- (h) 150 mL NaOH will be consumed in three neutralisations & 50 mL will be left unreacted. Also, hydrolysis of Na_3PO_4 can be neglected in presence of NaOH left.

$$[\text{OH}^-] = \frac{\text{Millimole}}{\text{Total volume in mL}} = \frac{5}{250} = 1/50 \text{ M} \quad \therefore \text{pOH} = 1.7 \text{ \& } \text{pH} = 12.3$$

- (i) 50 mL NaOH will be consumed in 1st neutralisation & remaining 40 mL (4 millimole) will participate in 2nd neutralisation





Solved Examples

Ex-2. Find the pH at equivalence point when a solution of 0.1 M acetic acid is titrated with a solution of 0.3 M NaOH. K_a for acetic acid = 7.5×10^{-6} .

Sol. At equivalence point CH_3COONa will be formed.

$$M_{\text{base}} V_{\text{base}} = M_{\text{acid}} V_{\text{acid}} \Rightarrow V_{\text{acid}} = \frac{0.3}{0.1} V_{\text{base}} = 3 V_{\text{base}}$$

$$[\text{CH}_3\text{COONa}]_{\text{Equivalence point}} = \frac{\text{Total moles}}{\text{Total volume}} = \frac{0.3 \times V_{\text{base}}}{V_{\text{acid}} + V_{\text{base}}} = \frac{0.3}{4} \frac{V_{\text{base}}}{V_{\text{base}}} = 0.075$$

$$[\text{OH}^-] = \sqrt{\frac{K_w}{K_a} \times C} = \sqrt{\frac{10^{-14}}{7.5 \times 10^{-6}} \times 0.075} = 10^{-5} \Rightarrow \text{pOH} = 5$$

$$\therefore \text{pH} = 9$$

Ex-3. Predict titration curve for titration of Na_2CO_3 & HCl .

Sol. $\text{Na}_2\text{CO}_3 + \text{HCl} \longrightarrow \text{NaHCO}_3 + \text{NaCl}$

pH_1 = Half equivalence point with respect to 1st neutralisation = pK_{a_2} ($\because$ solution will be a buffer consisting of equal amount of CO_3^{2-} & HCO_3^-)

$$\text{pH}_{1^{\text{st eq pt}}} = \frac{\text{pK}_{a_1} + \text{pK}_{a_2}}{2} (\because \text{amphiprotic anion})$$



pH_2 = Half equivalence point with respect to 2nd neutralisation = pK_{a_1} ($\because$ solution will be a buffer consisting of equal amount of HCO_3^- & H_2CO_3)

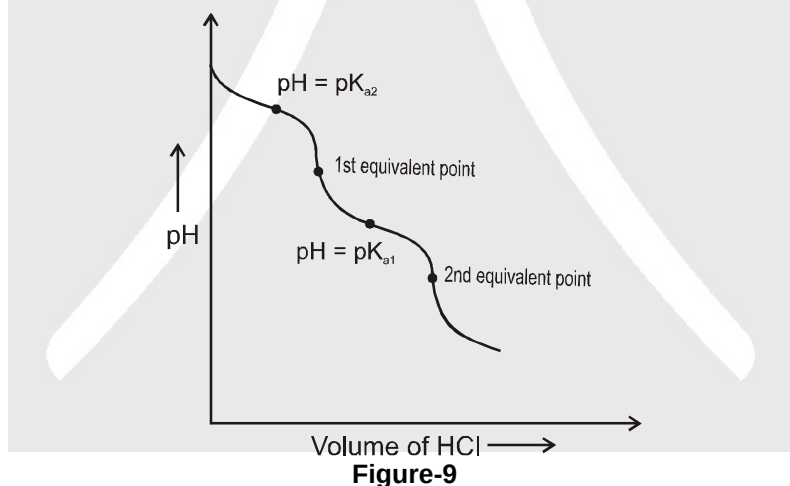


Figure-9

$$\text{pH}_{2^{\text{nd eq pt}}} = 1/2 (\text{pK}_{a_1} - \log C) (\because \text{weak acid})$$

Ionic Equilibrium-III

SOLUBILITY(s) AND SOLUBILITY PRODUCT (K_{sp})

Some complicated cases for solubility of sparingly soluble salts could be as follows :

(A) **Effect on solubility because of complex formation**

○ When ions of a salt form complex in the solution, then concentration of that ion decreases, so solubility equilibrium is shifted forward & solubility increases.



Solved Examples

Ex-4. What must be the concentration of aqueous NH_3 which must be added to a solution containing $4 \times 10^{-3} \text{ M Ag}^+$ and 0.001 M NaCl , to prevent the precipitation of AgCl .

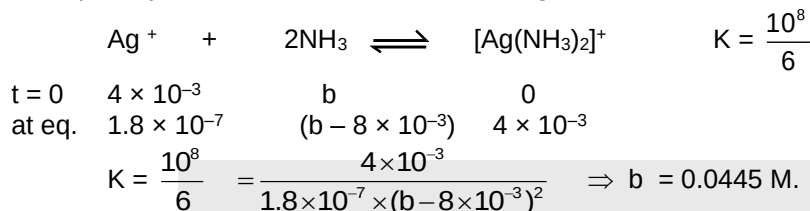
Given that $K_{\text{sp}}(\text{AgCl}) = 1.8 \times 10^{-10}$ and the formation constant of $[\text{Ag}(\text{NH}_3)_2]^+$ is $K_{\text{formation}} = \frac{10^8}{6}$.

Sol. Silver ion concentration which can be allowed to remain in the solution,

$$1.8 \times 10^{-10} = [\text{Ag}^+][\text{Cl}^-]$$

$$[\text{Ag}^+] = \frac{1.8 \times 10^{-10}}{0.001} = 1.8 \times 10^{-7} \text{ M},$$

This quantity is so small that almost all the Ag^+ ion will be consumed by NH_3 in complex formation.



(B) Solubility in appropriate buffer solutions

Appropriate buffer means that the components of buffer should not interfere with the salt or only H^+ or OH^- ions should be interacting with the ions of the salt.

Solved Examples

Ex-5. The solubility of $\text{Pb}(\text{OH})_2$ in water is $6.7 \times 10^{-6} \text{ M}$. Calculate the solubility of $\text{Pb}(\text{OH})_2$ in a buffer solution of $\text{pH} = 8$. Take $(6.7)^3 = 301$

Sol. $K_{\text{sp}} = 4 \times (6.7 \times 10^{-6})^3 = 1.204 \times 10^{-15}$

$$s \times (10^{-6})^2 = 1.204 \times 10^{-15} \Rightarrow s = 1.204 \times 10^{-3} \text{ M}$$

Ex-6. Calculate solubility of MnS in a buffer solution of given pH . K_{sp} of MnS and K_{a_1} & K_{a_2} for H_2S are given.

Sol. Let the new solubility of $\text{MnS} = x$

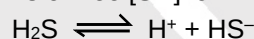
$\therefore [\text{Mn}^{2+}] = x = \text{Initial concentration of } \text{S}^{2-} \text{ ions, but free } \text{S}^{2-} \text{ ions will be less because some of the } \text{S}^{2-} \text{ ions will react with } \text{H}^+ \text{ from buffer to form } \text{HS}^- \text{ and } \text{H}_2\text{S}.$

$$[\text{Mn}^{2+}] = x = [\text{S}^{2-}] + [\text{HS}^-] + [\text{H}_2\text{S}] \quad \text{.....(1)}$$

Let us calculate the value of $[\text{HS}^-]$ & $[\text{H}_2\text{S}]$, in terms of free $[\text{S}^{2-}]$ ion. For that, consider :



$$K_2 = \frac{[\text{H}^+][\text{S}^{2-}]}{[\text{HS}^-]}$$



$$K_1 = \frac{[\text{H}^+][\text{HS}^-]}{[\text{H}_2\text{S}]}$$

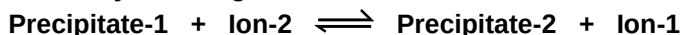
$$\therefore [\text{HS}^-] = \frac{[\text{H}^+][\text{S}^{2-}]}{K_2} \quad \text{....(2)}$$

$$\text{and } [\text{H}_2\text{S}] = \frac{[\text{H}^+][\text{HS}^-]}{K_1} = \frac{[\text{H}^+]^2[\text{S}^{2-}]}{K_1 K_2} \quad \text{....(3)}$$

Put (2) & (3) in (1)

$$x = [\text{S}^{2-}] \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} \right)$$

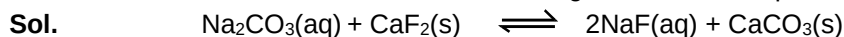
$$x = \frac{K_{\text{sp}}}{[\text{Mn}^{2+}]} \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} \right) \Rightarrow x = \sqrt{K_{\text{sp}} \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} \right)}$$

**(C) Solubility Exchange :**

One precipitate is mixed with solution of an ion to get another precipitate and another ion in solution.

Solved Examples

Ex-7. 2M solution of Na_2CO_3 is boiled in a closed container with excess of CaF_2 . Very little amount of CaCO_3 and NaF are formed. If the solubility product (K_{sp}) of CaCO_3 is x and molar solubility of CaF_2 is y , find the molar concentration of F^- in resulting solution after equilibrium is attained.



$$t = 0 \quad 2 \quad - \quad 0 \quad -$$

$$t = \text{eq} \quad 2 - a \quad - \quad 2a \quad -$$

where a is very small

For CaCO_3 , $K_{sp} = x = [\text{Ca}^{2+}][\text{CO}_3^{2-}] = [\text{Ca}^{2+}] \times 2$ ($\because \text{CO}_3^{2-}$ mainly coming from Na_2CO_3)

$$\therefore [\text{Ca}^{2+}] = \frac{x}{2}$$

$$\text{For } \text{CaF}_2, K_{sp} = 4y^3 = \left(\frac{x}{2}\right) [\text{F}^-]^2 \Rightarrow [\text{F}^-] = \sqrt{\frac{8y^3}{x}}$$

(D) Solubility related to H_2S

The sulphide ion concentration in a solution that has been saturated with H_2S is given by :

$$[\text{S}^{2-}] = \frac{K_{a1} K_{a2} [\text{H}_2\text{S}]}{[\text{H}^+]^2} = \frac{K_{sp(\text{H}_2\text{S})}}{[\text{H}^+]^2}$$

Solved Examples

Ex-8. What $[\text{H}_3\text{O}^+]$ must be maintained in a saturated H_2S solution to precipitate Pb^{2+} , but not Zn^{2+} from a solution in which each ion is present at a concentration of 0.01 M ?

(K_{sp} of $\text{H}_2\text{S} = 1.1 \times 10^{-22}$; K_{sp} of $\text{ZnS} = 1.0 \times 10^{-21}$; $\sqrt{11} = 3.3$)

Sol. For ZnS not to be precipitated from a solution of Zn^{2+} and Pb^{2+} , $IP_{\text{ZnS}} < K_{sp}(\text{ZnS})$

$$[\text{Zn}^{2+}][\text{S}^{2-}] < K_{sp} \text{ of } \text{ZnS}$$

$$[10^{-2}][\text{S}^{2-}] < 1.0 \times 10^{-21} \therefore [\text{S}^{2-}] < 10^{-19} \text{ M}$$

So, at $[\text{S}^{2-}] = 10^{-19} \text{ M}$ or less, no precipitation of ZnS will occur.



$$\therefore [\text{H}^+]^2 [\text{S}^{2-}] = K_{sp(\text{H}_2\text{S})} = 1.1 \times 10^{-22} \quad \therefore [\text{H}^+]_{\min}^2 [10^{-19}] = 1.1 \times 10^{-22}$$

$$[\text{H}^+]_{\min}^2 = 11 \times 10^{-4}$$

$$\therefore [\text{H}^+]_{\min} = 3.3 \times 10^{-2} \text{ M}$$

Thus, if $[\text{H}^+] = 3.3 \times 10^{-2} \text{ M}$ or more, the precipitation of ZnS will not take place and only PbS will precipitate.

SUBJECTIVE QUESTIONS

- (a) For which of the following, water would behave as a levelling solvent and for which, it would behave as a differentiating solvent ?
Strong acids, weak acids, strong bases, weak bases.

(b) For which of the following, glacial acetic acid would behave as a levelling solvent and for which, it would behave as a differentiating solvent ?
Strong acids, weak acids, strong bases, weak bases.

(c) For which of the following, liquid ammonia would behave as a levelling solvent and for which, it would behave as a differentiating solvent ?
Strong acids, weak acids, strong bases, weak bases.
- Calculate pOH , $[\text{CH}_3\text{NH}_3^+]$, $[\text{C}_4\text{H}_{13}\text{N}_2^+]$ & $[\text{C}_4\text{H}_{14}\text{N}_2^{2+}]$ in an aqueous solution consisting of 0.2 M Methyl amine ($K_b = 4.1 \times 10^{-4}$) & 0.1 M Butane-1, 4-diamine ($K_{b1} = 6.2 \times 10^{-4} \text{ M}$; $K_{b2} = 2.25 \times 10^{-5}$).



3. Upon adding 50 mL of 0.2 M KOH solution to 50 mL of 0.5 M solution of Ethylene diamine ($K_{b1} = 8 \times 10^{-5}$; $K_{b2} = 2.7 \times 10^{-8}$), calculate pH of final solution and $[C_2N_2H_{10}^{2+}]$ in resulting solution.
4. Determine the pH of 0.5 M BH_2Cl_2 solution (salt of a diacidic base B). Also calculate $[BH^+]$ & $[B]$. Given : K_{b1} & K_{b2} for B are 10^{-6} & 2×10^{-10}
5. Prove that buffer capacity of 0.2 M CH_3COOH –0.2M CH_3COONa buffer is less than the 0.4 M CH_3COOH –0.4 M CH_3COONa .
6. A small quantity of phenolphthalein (an acid indicator) is added to a decimolar solution of Sodium butyrate. Calculate the ratio of the coloured to the colourless form of the indicator. K_a for butyric acid 1.5×10^{-5} , K for the indicator $= 3.075 \times 10^{-10}$ and $K_w = 10^{-14}$. Take $\log 1.23 = 0.09$
7. At pH = 2, half of the indicator, thymol blue (an acid type indicator) is in unionised form. Find the % of indicator in unionised form in the solution with $[H^+] = 4 \times 10^{-3}$ M.
8. A base type indicator B differs in colour from its conjugate acid (BH^+). Acidic form is red in colour while basic form is blue in colour. Human eye can sense blue colour distinctly when ratio of blue form concentration to red form concentration is $\frac{a}{b}$ or more. However, red colour can be sensed by human eye distinctly when ratio of red form concentration to blue form concentration is $\frac{c}{d}$ or more. Determine the pH range of solution in which human eyes will be unable to observe distinct red or distinct blue colour. Take ionisation constant of B as K_{eq} .
9. Determine the maximum number of moles of MgF_2 that can dissolve in 1000 L of a buffer solution of pH = 4. Given : K_{sp} of $MgF_2 = \frac{9}{14} \times 10^{-8}$ and K_a of HF $= 3.5 \times 10^{-4}$. Take $(2.17)^3 = 10$.
- 10(a). Determine the concentration of NH_3 solution whose 1 L can dissolve 0.1 mole $CuCO_3$.
Given : K_{sp} of $CuCO_3 = 1.4 \times 10^{-10}$ and K_f of $[Cu(NH_3)_4]^{2+} = 2 \times 10^{13}$. Take $\sqrt[4]{\frac{1000}{2.8}} = 4.4$. Assume no other reaction to take place.
- 10(b). An aqueous solution of a metal bromide, MBr_2 (0.05 M) is saturated with H_2S . Calculate the minimum pH at which the metal sulphide will be precipitated. Concentration of H_2S in a saturated solution = 0.1M. [K_{a1} of $(H_2S) = 10^{-7}$, K_{a2} of $(HS^-) = 1.2 \times 10^{-13}$, K_{sp} for MS $= 6 \times 10^{-21}$]

ONLY ONE OPTION CORRECT TYPE

11. In the reaction : $Ni^{2+} + 6H_2O \rightarrow [Ni(H_2O)_6]^{2+}$,
(A) Ni^{2+} behaves like a Lewis acid. (B) H_2O behaves like a Lewis base.
(C) Above cannot be called an acid-base reaction. (D) Both (A) and (B).
12. In water, the bases NaOH, KOH, RbOH, CsOH and $Ba(OH)_2$ exhibit the same strength as they are completely ionised in water (an acid). This is called of the solvent water.
(A) Strength (B) Capacity (C) Buffer effect (D) Levelling effect
13. The ratio of dissociation constant of two weak acids HA and HB is 4:1. At what initial molar concentration ratio of two acid solution, the two will have same pH in separate solutions ? Assume negligible dissociation of both acids in their solutions.
(A) 2 : 1 (B) 1 : 2 (C) 4 : 1 (D) 1 : 4
14. For 10^{-4} M BOH (weak base ; $K_b = 5 \times 10^{-5}$) :
(A) $\alpha = 0.707$ (B) $[OH^-] = 5 \times 10^{-5}$ M (C) pH = 9.85 (D) All of these
15. In a 0.2 M aqueous solution of Ethylene diamine ($H_2NCH_2CH_2NH_2$) :
Given : $K_{b1} = 8 \times 10^{-5}$ & $K_{b2} = 2.7 \times 10^{-8}$
(A) $[OH^-] = 2 \times 10^{-2}$ M (B) $[C_2N_2H_{10}^{2+}] = 2.7 \times 10^{-8}$ M
(C) Both (A) and (B) (D) none of these



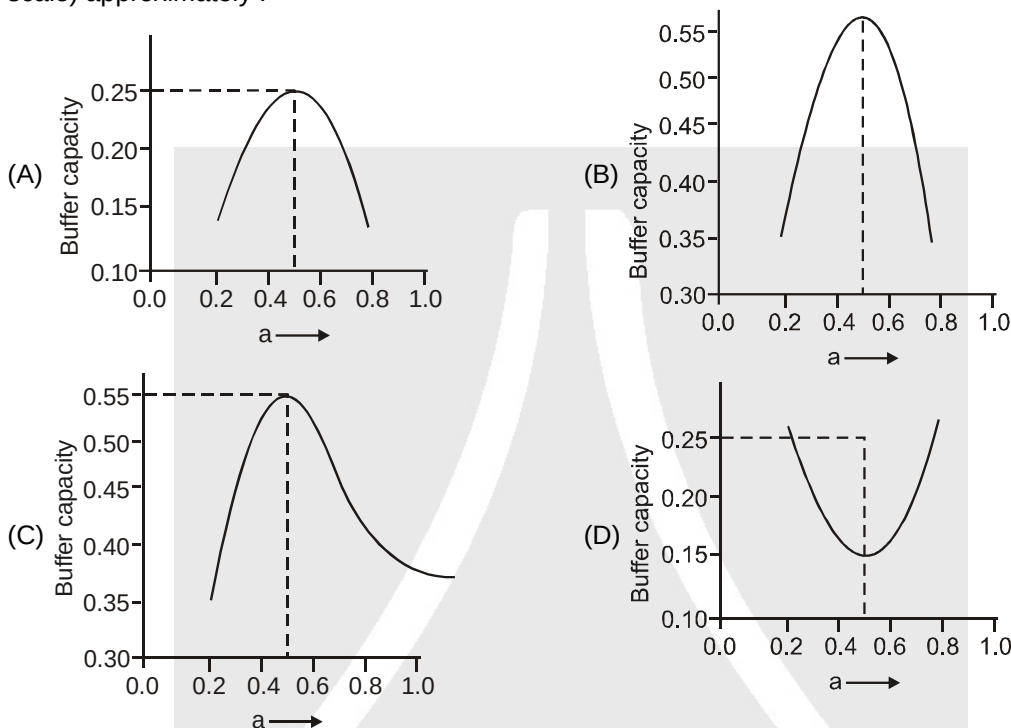
16. (a) In a solution containing 0.1 M HCl & 0.1 M H_3PO_4 (having dissociation constants K_{a_1} , K_{a_2} & K_{a_3}), assuming $\alpha \ll 1$ for H_3PO_4 :
- (A) $\text{pH} = 1$ (B) $\alpha_{\text{H}_3\text{PO}_4} = 10 K_{a_1}$ (C) $[\text{HPO}_4^{2-}] = 10 K_{a_1} K_{a_2}$ (D) All of these
16. (b) In above question
- (A) $[\text{PO}_4^{3-}] = 100 K_{a_1} K_{a_2} K_{a_3}$ (B) $[\text{PO}_4^{3-}] = 10 K_{a_1} K_{a_2} K_{a_3}$
 (C) $[\text{PO}_4^{3-}] = 1000 K_{a_1} K_{a_2} K_{a_3}$ (D) $[\text{PO}_4^{3-}] = K_{a_1} K_{a_2} K_{a_3}$
17. A pair of salts are given in a solution each in 0.1M concentration. Which solution has highest pH ?
 Given : Order of acidic strength of acids is : $\text{HF} > \text{HOCl} > \text{HOBr} > \text{HCN}$
 (A) NaCN and NaOBr (B) NaF and NaOCl (C) NaF and NaOBr (D) NaCN and NaOCl
18. (a) For a c M K_3A (potassium salt of a tribasic acid H_3A) solution :
 (Dissociation constants of acid are K_{a_1} , K_{a_2} & K_{a_3} ; $h \ll 1$)
- (A) $\text{pH} = \frac{1}{2} (\text{p}K_w + \text{p}K_{a_3} + \log c)$ (B) $\text{pH} = \frac{1}{2} (\text{p}K_w + \text{p}K_{a_2} + \log c)$
 (C) $\text{pH} = \frac{1}{2} (\text{p}K_w + \text{p}K_{a_1} + \log c)$ (D) $\text{pH} = \frac{1}{2} (\text{p}K_w - \text{p}K_{a_3} - \log c)$
18. (b) In above solution,
- (A) $[\text{HA}^{2-}] = \sqrt{\frac{cK_w}{K_{a_3}}}$ (B) $[\text{H}_2\text{A}^-] = \frac{K_w}{K_{a_2}}$
 (C) $[\text{H}_3\text{A}] = \frac{K_w}{K_{a_1} K_{a_2}} \sqrt{\frac{K_w K_{a_3}}{c}}$ (D) All of these
19. Identify acids in the following reactions according to Bronsted - Lowry concept :
- (I) $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+} (\text{aq.}) + [\text{Al}(\text{H}_2\text{O})_6]^{3+} (\text{aq.}) \longrightarrow [\text{Fe}(\text{H}_2\text{O})_6]^{3+} (\text{aq.}) + [\text{Al}(\text{H}_2\text{O})_5(\text{OH})]^{2+} (\text{aq.})$
 (II) $\text{O}^{2-} (\text{aq.}) + \text{H}_2\text{O} (\text{aq.}) \longrightarrow 2\text{OH}^- (\text{aq.})$
 (III) $\text{CH}_3\text{OH} (\text{aq.}) + \text{H}^- (\text{aq.}) \longrightarrow \text{CH}_3\text{O}^- (\text{aq.}) + \text{H}_2 (\text{g})$
- (A) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$, H_2O , H^- (B) $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$, H_2O , CH_3OH
 (C) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$, H_2O , CH_3OH (D) $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$, O^{2-} , H^-
20. In aqueous solution, boric acid H_3BO_3 is a :
 (A) Arrhenius acid (B) Bronsted acid (C) Lewis acid (D) Both (B) & (C)
21. pOH of a 0.004 M aqueous solution of a diacidic weak base B ($\text{p}K_{b1} = 3.7$ & $\text{p}K_{b2} = 7.3$) is :
 (A) 3.05 (B) 3.1 (C) 0.65 (D) 3.15
22. What concentration of OH^- ions will reduce NH_4^+ ion to 2×10^{-5} M in 0.4 M solution of NH_4OH ? K_b (NH_4OH) = 1.8×10^{-5}
 (A) 0.36 M (B) 0.036 M (C) 2×10^{-5} M (D) None of these
23. Calculate the pH of a 0.15 M aqueous solution of AlCl_3 .
 Given : $[\text{Al}(\text{H}_2\text{O})_6]^{3+} (\text{aq.}) + \text{H}_2\text{O} (\ell) \rightleftharpoons [\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+} (\text{aq.}) + \text{H}_3\text{O}^+ (\text{aq.})$; $K_a = 1.5 \times 10^{-5}$
 (A) 2.82 (B) 5 (C) 9 (D) 11.18
24. Calculate $[\text{OH}^-]$ & $[\text{H}_2\text{C}_2\text{O}_4]$ in a 0.005 M $\text{Na}_2\text{C}_2\text{O}_4$ solution. Given : K_{a_1} & K_{a_2} for oxalic acid are 5.6×10^{-2} & 5.4×10^{-5} . Take $\sqrt{\frac{1}{108}} = 0.096$
- (A) $[\text{OH}^-] = 9.6 \times 10^{-7}$ M (B) $[\text{H}_2\text{C}_2\text{O}_4] = \frac{5}{28} \times 10^{-12}$ M
 (C) Both (A) & (B) (D) None of these



25. Species acting as both bronsted acid and base is :
 (A) HSO_4^- (B) Na_2CO_3 (C) NH_3 (D) OH^-
26. Which one of the following statements is not true ?
 (A) The conjugate base of H_2PO_4^- is HPO_4^{2-} .
 (B) $\text{pH} + \text{pOH} = 14$ for all aqueous solutions at 25°C .
 (C) The pH of 1×10^{-8} M HCl is 8.
 (D) 96, 500 coulombs of electricity when passed through a CuSO_4 solution deposits 1 gram equivalent of copper at the cathode.
27. When rain is accompanied by a thunderstorm, the collected rain water will have a pH value :
 (A) slightly lower than that of rain water without thunderstorm.
 (B) slightly higher than that when the thunderstorm is not there.
 (C) uninfluenced by occurrence of thunderstorm.
 (D) which depends on the amount of dust in air.
28. 100 mL of 0.05 M $\text{C}_3\text{N}_2\text{H}_{12}\text{SO}_4$ (salt of Propane-1,2-diamine & H_2SO_4) and some volume of 0.1 M $\text{C}_3\text{N}_2\text{H}_{11}\text{Cl}$ (salt of propane-1,2-diamine & HCl) are available. What volume (in mL) of 2nd salt solution should be taken to prepare a buffer solution of $\text{pH} = 6.31$? Given : pK_{b1} & pK_{b2} of propane-1, 2-diamine are 4.18 & 7.39 respectively.
 (A) 50 mL (B) 25 mL (C) 12.5 mL (D) 0.015 mL
29. Select the correct set of statements for an acid type indicator used in a titration :
 (i) In general, pH range of an indicator is $(\text{pK}_a - 1)$ to $(\text{pK}_a + 1)$ and indicator shows its characteristic colours in this range.
 (ii) In general, pH range of an indicator is $(\text{pK}_a - 1)$ to $(\text{pK}_a + 1)$ and indicator does not show its characteristic colours in this range.
 (iii) In the relation : $\text{pH} = \text{pK}_{\text{In}} + \log_{10} \frac{[\text{In}^-]}{[\text{HIn}]}$, pH value represents pH of indicator solution.
 (iv) In the relation : $\text{pH} = \text{pK}_{\text{In}} + \log_{10} \frac{[\text{In}^-]}{[\text{HIn}]}$, pH value represents pH of resulting solution containing indicator.
 (A) i & iii (B) ii & iv (C) ii & iii (D) i & iv
30. Methyl orange indicator (pH range = 3.1 – 4.4) cannot be used for end point detection of following titration(s) :
 (A) CH_3COOH vs NaOH (B) H_2O_2 vs KMnO_4 (in acidic medium)
 (C) Both (A) & (B) (D) None of these
31. A certain indicator (an organic dye) has $\text{pK}_a = 5$. For which of the following titrations may it be suitable ?
 (A) acetic acid against NaOH (B) aniline hydrochloride against NaOH
 (C) sodium carbonate against HCl (D) barium hydroxide against oxalic acid
32. The best indicator (with their pH range given) for the detection of end point in titration of a weak acid and a strong base is :
 (A) Methyl orange (3 to 4) (B) Methyl red (5 to 6)
 (C) Bromothymol blue (6 to 7.5) (D) Phenolphthalein (8 to 9.6)
33. At $\text{pH} < 3.1$, the indicator methyl red is coloured red, at $\text{pH} > 6.3$, it is yellow and at the intermediate values of the pH, it is orange. What will the colour of indicator be in a 0.1 M solution of NH_4Br ? Take $\text{pK}_b(\text{NH}_4\text{OH}) = 4.74$.
 (A) Red (B) Yellow
 (C) Orange (D) $\text{pH} = 3.1$ so colour cannot be predicted



34. What fraction of an indicator HIn is in basic form at a pH of 6 if the pK_a of the indicator is 5 ?
 (A) $\frac{9}{10}$ (B) $\frac{1}{11}$ (C) $\frac{10}{11}$ (D) $\frac{1}{10}$
35. 50 mL of 0.1 M NaOH is added to 60 mL of 0.15 M H_3PO_4 solution (K_1 , K_2 and K_3 for H_3PO_4 are 10^{-3} , 10^{-8} and 10^{-13} respectively). The pH of the mixture would be about :
 (A) 3.1 (B) 5.5 (C) 4.1 (D) 6.5
36. A buffer solution is prepared by mixing 'a' moles of CH_3COONa and 'b' moles of CH_3COOH such that **(a + b) = 1**, into water to make 1L buffer solution. If the instantaneous (differential) buffer capacity of this buffer solution is plotted against moles of salt CH_3COONa (a), then the plot obtained will be (to the scale) approximately :



37. An acid-base indicator which is a weak acid has a pK_{In} value = 5.45. At what concentration ratio of sodium acetate to acetic acid would the indicator show a colour half-way between those of its acid and conjugate base forms? [pK_a of acetic acid = 4.75]
 (A) 4 : 1 (B) 6 : 1 (C) 5 : 1 (D) 3 : 1
38. If the indicator is a weak acid with $K_{In} = 4 \times 10^{-4}$, for indication of equivalence point of which of the following neutralisation(s), would the indicator be useful ?
 (A) $HCl + NaOH$ (B) $HCl + NH_3$ (C) $NaOH + CH_3COOH$ (D) Both (A) and (B)
39. 50 mL of 0.05 M Na_2CO_3 is titrated against 0.1 M HCl. On adding 40 mL of HCl, pH of the solution will be : [Given : For H_2CO_3 , $pK_{a_1} = 6.35$, $pK_{a_2} = 10.33$]
 (A) 6.35 (B) 6.526 (C) 8.34 (D) 6.173
40. One mole of a sparingly soluble salt MX are added in 1 L saturated solution of another sparingly soluble salt NX. If $K_{sp}(MX) \gg K_{sp}(NX)$, then : (Assume no chemical reaction to be taking place).
 (A) Some MX & NX solid will get collected at the bottom of vessel.
 (B) $[X^-] \approx \sqrt{K_{sp}(MX)}$ (C) $[N^+] \approx \frac{K_{sp}(NX)}{\sqrt{K_{sp}(MX)}} (\ll [M^+])$ (D) All of these

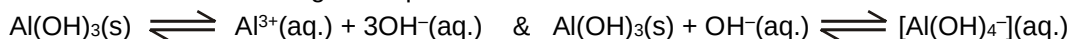


41. The solubility of CaCO_3 is 7 mg / litre. Calculate the solubility of BaCO_3 (in mol/L) from this information and from the fact that when Na_2CO_3 is added slowly to a solution containing equimolar concentration of Ca^{+2} and Ba^{+2} , no precipitate of CaCO_3 is formed until 90% of Ba^{+2} has been precipitated as BaCO_3 . (Assume no hydrolysis of CO_3^{2-} ion).
 (A) 4.9×10^{-10} (B) 7×10^{-6} (C) 2.21×10^{-5} (D) 4.9×10^{-11}
42. To a solution formed by adding n mole of sparingly soluble salt MX in 1 L water, NaY solid (a readily soluble salt) is added until x% of MX is converted to MY(s) (another sparingly soluble salt). Then :
 (Assume no hydrolysis of any ion).
 (A) Resulting solution can be called saturated with respect to both MX & MY.
 (B) $[X^-] = \frac{nx}{100}$
 (C) $[Y^-] = \frac{K_{sp}(\text{MY})}{K_{sp}(\text{MX})} \times \frac{nx}{100}$
 (D) All of these
43. A sample of AgCl was treated with 5 mL of 1.6 M Na_2CO_3 solution to give Ag_2CO_3 . The resulting solution contained 0.00284 g of Cl^- per litre. Determine $K_{sp}(\text{AgCl})$, if $K_{sp}(\text{Ag}_2\text{CO}_3) = 8.1 \times 10^{-12}$. (Assume no hydrolysis of any ion).
 (A) 1.8×10^{-10} (B) 10^{-10} (C) 10^{-5} (D) 1.34×10^{-5}
44. To a solution of 0.1 M Mg^{2+} and 0.8 M NH_4Cl , an equal volume of NH_3 is added which just gives precipitate. Calculate $[\text{NH}_3]$ in its solution. K_{sp} of $\text{Mg}(\text{OH})_2 = 1.62 \times 10^{-11}$ and K_b of $\text{NH}_4\text{OH} = 1.8 \times 10^{-5}$.
 (A) 0.4 M (B) 0.8 M (C) 1.6 M (D) 0.08 M
45. Solubility of $\text{Fe}(\text{OH})_3$ ($K_{sp} = 10^{-38}$) :
 (A) In a buffer solution of pH = 2 is 10^{-2} mol/L (B) In a buffer solution of pH = 11 is 10^{-29} mol/L
 (C) Both (A) & (B) (D) None of these
46. What maximum pH must be maintained in a saturated H_2S solution (0.1 M) to avoid precipitation of both Mn^{2+} & Fe^{2+} from a solution, in which each ion is present at a concentration of 0.01 M ?
 (K_a of $\text{H}_2\text{S} = 9.6 \times 10^{-21}$; K_{sp} of $\text{MnS} = 2.5 \times 10^{-13}$; K_{sp} of $\text{FeS} = 6.4 \times 10^{-18}$)
 (A) 5.2 (B) 2.91 (C) 4.7 (D) 2.41
47. What minimum pH must be maintained in a saturated H_2S solution (0.1 M) to cause precipitation of both Mn^{2+} & Fe^{2+} from a solution, in which each ion is present at a concentration of 0.01 M ?
 (K_a of $\text{H}_2\text{S} = 9.6 \times 10^{-21}$; K_{sp} of $\text{MnS} = 2.5 \times 10^{-13}$; K_{sp} of $\text{FeS} = 6.4 \times 10^{-18}$)
 (A) 5.21 (B) 2.91 (C) 4.7 (D) 2.41
48. A decimolar solution of HCl is saturated with H_2S gas. Calculate the minimum $[\text{Cu}^{2+}]$ ion that will saturate the above solution with CuS. Given K_{sp} of CuS = 6.4×10^{-36} , K_a for $\text{H}_2\text{S} = 9.6 \times 10^{-21}$ and solubility of $\text{H}_2\text{S} = 0.1$ mole litre $^{-1}$.
 (A) $\frac{2}{3} \times 10^{-16}$ M (B) $\frac{2}{3} \times 10^{-14}$ M (C) $\frac{2}{3} \times 10^{-18}$ M (D) $\frac{3}{2} \times 10^{-16}$ M
- 49(a). The salt $\text{Al}(\text{OH})_3$ is involved in the following two equilibria :
 $\text{Al}(\text{OH})_3(\text{s}) \rightleftharpoons \text{Al}^{3+}(\text{aq}) + 3\text{OH}^-(\text{aq}) ; K_{sp}$
 $\text{Al}(\text{OH})_3(\text{s}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{Al}(\text{OH})_4^-(\text{aq}) ; K_c$
 Which of the following relationship is correct when the solubility is minimum ?
 (A) $[\text{OH}^-] = \left(\frac{K_{sp}}{K_c} \right)^{1/3}$ (B) $[\text{OH}^-] = \left(\frac{K_c}{K_{sp}} \right)^{1/4}$
 (C) $[\text{OH}^-] = \sqrt{\left(\frac{K_{sp}}{K_c} \right)^{1/4}}$ (D) None of these



- 49(b). Arrange solubility of AgBr in the given solutions in increasing order : (consider no hydrolysis of Ag^+ ion)
- (i) 0.1 M NH_3 (ii) 0.1 M AgNO_3 (iii) 0.2 M NaBr (iv) pure water
- (A) (iii) < (ii) < (iv) < (i) (B) (iii) < (ii) < (i) < (iv) (C) (iii) < (ii) = (i) < (iv) (D) (ii) < (iii) < (iv) < (i)

- 49(c). Al(OH)_3 is involved in the following two equilibria :



Then :

- (A) Solubility of Al(OH)_3 will be more in acidic solutions & less in basic solution, both as compared to in water.
- (B) Solubility of Al(OH)_3 will be less in acidic solutions & more in basic solution, both as compared to in water.
- (C) Solubility of Al(OH)_3 will be more in acidic solutions & more in basic solution, both as compared to in water.
- (D) Solubility of Al(OH)_3 will be less in acidic solutions & less in basic solution, both as compared to in water.

MATCH THE COLUMN

50. Match the entries of column II for which the equality or inequality given in the column I are satisfied. (Use $\log 1.8 = 0.26$, K_a of formic acid = 1.8×10^{-4} , K_a of acetic acid = 1.8×10^{-5} , K_b of ammonia = 1.8×10^{-5} , K_{a1} of H_2S = 10^{-7} and K_{a2} of H_2S = 10^{-14} , for the following matchings)

	Column I		Column II
(A)	10^{-5} M HCl solution > 0.1 M H_2S solution	(p)	α_{water} (degree of dissociation of water)
(B)	CH_3COOH solution of pH equal to 4.74 = $\text{NH}_3 \cdot \text{H}_2\text{O}$ solution of pH equal to 9.26	(q)	$[\text{OH}^-]$
(C)	0.1 M CH_3COOH solution = 1.0 M HCOOH solution	(r)	α (degree of dissociation)
(D)	0.1 M solution of a weak acid HA_1 ($K_a = 10^{-5}$) < 0.01 M solution of a weak acid HA_2 ($K_a = 10^{-6}$)	(s)	pH

51. Match the column:

	Column-I		Column-II
	Mixture of various solutions		pH of resulting solution
(A)	10 L of 0.015 M Ba(OH)_2 + 5 L of 0.08 M HCl + 485 L of 0.05 M BaCl_2	(p)	5.5
(B)	100 mL of 0.15 M CH_3COOH ($\text{p}K_a = 4.74$) + 50 mL of 0.3 M NaOH	(q)	11.2
(C)	20 mL of 0.5 M RNH_3Cl ($K_b = 10^{-9}$) + 20 mL of 0.5 M KOH	(r)	8.87
(D)	10 mL of 0.1 M H_3A ($\text{p}K_{a1} = 4$, $\text{p}K_{a2} = 7$, $\text{p}K_{a3} = 11$) + 10 mL of 0.1 M NaOH	(s)	3.7

52. Match the effect of addition of 1 M NaOH to 50 mL of 1 M $\text{H}_2\text{C}_2\text{O}_4$ (diprotic acid) in column I with column II (Given : $K_{a1} = 10^{-4}$, $K_{a2} = 10^{-9}$)

	Column-I		Column-II
(A)	25 mL of NaOH	(p)	Buffer solution
(B)	50 mL of NaOH	(q)	pH is independent of concentration of species present in the solution.
(C)	75 mL of NaOH	(r)	anionic hydrolysis
(D)	100 mL of NaOH	(s)	pH > 7



53. Match the column:

	Column-I		Column-II
	Titration		Best Indicator (with its pH range) to detect end point
(A)	Titration of 0.1 M H_2SO_4 by adding 0.1 M KOH	(p)	Methyl red (4.2 – 6.3)
(B)	Titration of 0.5 M HCOOH ($\text{pK}_a = 3.74$) by adding 0.125 M NaOH	(q)	Phenol red (6.8 – 8.2)
(C)	Titration of 0.2 M NH_4OH ($\text{pK}_b = 4.74$) by adding 0.1 M H_2SO_4	(r)	Methyl orange (3.1 – 4.4)
(D)	Titration of 0.3 M Na_2CO_3 by adding 0.3 M HCl (K_{a_1} & K_{a_2} of H_2CO_3 are 4×10^{-7} & 5.6×10^{-11})	(s)	Phenolphthalein (8.3 – 10)

54. Match the column:

	Column-I		Column-II
	(Concentration/solubility)		(Solutions)
(A)	8×10^{-5} M	(p)	$[\text{Mg}^{2+}]_{\text{min}}$ required for precipitation of $\text{Mg}(\text{OH})_2$ ($\text{K}_{\text{sp}} = 1.794 \times 10^{-11}$) from a solution containing 0.36 M NaOH.
(B)	1.38×10^{-10} M	(q)	Solubility of Hg_2Cl_2 ($\text{K}_{\text{sp}} = 1.296 \times 10^{-18}$) in 0.18 M BaCl_2 solution.
(C)	10^{-17} M	(r)	$[\text{Ca}^{2+}]$ in a solution saturated with respect to both CaCO_3 ($\text{K}_{\text{sp}} = 2.6 \times 10^{-9}$) & CaC_2O_4 ($\text{K}_{\text{sp}} = 3.8 \times 10^{-9}$).
(D)	7.5×10^{-18} M	(s)	Solubility of $\text{Fe}(\text{OH})_3$ ($\text{K}_{\text{sp}} = 10^{-38}$) in pure water.
		(t)	$[\text{Ag}^+]$ in resulting solution obtained by mixing equal volumes of 0.06 M AgNO_3 solution & 0.2 M KCN solution (K_f of $[\text{Ag}(\text{CN})_2]^- = 2.5 \times 10^{18}$).

SINGLE AND DOUBLE VALUE INTEGER TYPE

55. What is the sum of basicity of H_3PO_4 , H_3PO_3 , H_3PO_2 , H_3BO_3 , $(\text{COOH})_2$, $\text{H}_2\text{S}_2\text{O}_8$, H_2SO_4 , H_2SO_3 , H_2CO_3 , H_2S and $\text{H}_4\text{P}_2\text{O}_5$?
56. 'A' solution of strong acid/strong base of $\text{pH} = x$ is mixed with equal volume of 'B' solution of strong acid/strong base of $\text{pH} = (14 - x)$, such that the resulting mixture has a $\text{pH} = 7$. If only natural number values of x are allowed, excluding $x = 7$, then how many possible values of x would be there? (Negative values of pH are not to be considered)
57. An $\text{NH}_4^+ - \text{NH}_3$ buffer is supposed to keep the pH of the solution constant within 0.3 pH unit during the reaction. $\text{CH}_3\text{COOCH}_3(\text{aq}) + 2\text{H}_2\text{O}(\text{aq}) \longrightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq}) + \text{CH}_3\text{OH}(\text{aq})$
If this solution had initial concentrations : $[\text{NH}_4^+]_0 = 0.1$ M, $[\text{NH}_3]_0 = 0.06$ M, $[\text{CH}_3\text{COOCH}_3]_0 = 0.02$ M, determine the magnitude of pH change as a result of reaction.
Multiply the magnitude by 10 & add 1 if it is a satisfactory buffer, otherwise subtract 1. Report the answer rounding it off to the nearest whole number.
 $[\text{K}_b(\text{NH}_3) = 1.8 \times 10^{-5}]$
58. 0.98 g H_3PO_4 is dissolved in 100 mL of a buffer solution ($\text{pH} = 5$). If the equilibrium concentrations of H_2PO_4^- , HPO_4^{2-} , PO_4^{3-} and H_3PO_4 are C_1 , C_2 , C_3 and C_4 respectively. Find $\text{pC}_1 + \text{pC}_2 + \text{pC}_3 + \text{pC}_4$.
($\text{pC}_i = -\log \text{C}_i$; where $i = 1, 2, 3, 4$) Given : For H_3PO_4 , $\text{K}_{a_1} = 10^{-3}$, $\text{K}_{a_2} = 10^{-8}$, $\text{K}_{a_3} = 10^{-12}$.



59. End point/equivalence point of how many of the following titrations cannot be detected with the indicators given against them ?

	Titration	Indicator with its pH range
(i)	KOH + HCN	Methyl orange (3.1 - 4.4)
(ii)	NaOH + HF	HIn ($K_{In} = 3 \times 10^{-4}$)
(iii)	HNO ₃ + Sr(OH) ₂	Phenol red (6.8 - 8.2)
(iv)	HClO ₄ + Aniline	Methyl red (4.2 - 6.3)
(v)	HCl + Dimethyl amine	HIn ($K_{In} = 5 \times 10^{-5}$)
(vi)	Ba(OH) ₂ + HNO ₂	Phenolphthalein (8.3 - 10)
(vii)	NaH ₂ PO ₂ + H ₂ SO ₄	InOH ($K_{In} = 3 \times 10^{-5}$)
(viii)	Pyridine + Benzoic acid	Phenol red (6.8 - 8.2)
(ix)	KH ₂ BO ₃ + HI	Methyl red

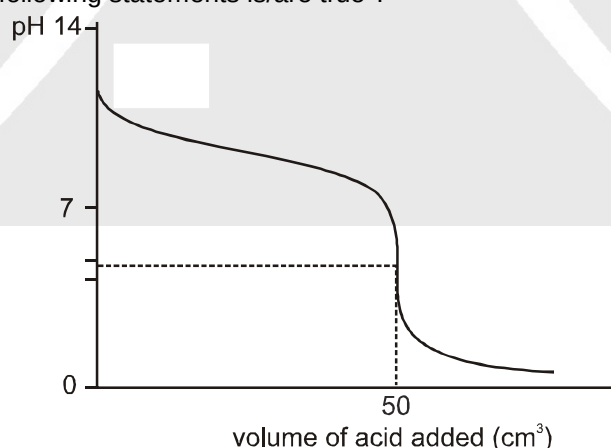
60. Number of compounds among following which give two end points when titrated with HCl is
 Na_2HPO_4 , Na_2S , $\text{K}_2\text{C}_2\text{O}_4$, KHC_2O_4 , H_3PO_3 , $\begin{matrix} \text{COOH} \\ | \\ \text{COOH} \end{matrix}$, K_2CO_3 , NaHCO_3 , Ethylene diamine
61. On addition of V mL volume of 0.1 M KOH, buffer capacity is found maximum at $\text{pH}_{\text{solution}} = x$ (> 7), when it is titrated against 50 mL, 0.2 M H₂CO₃. (For H₂CO₃: $K_{a1} = 4.3 \times 10^{-6}$, $K_{a2} = 5.6 \times 10^{-11}$)
 Report your answer as $(V + x) \times 0.4$, rounding it off to the nearest whole number.
62. The solubility product of Ag₂C₂O₄ at 25°C is $1.29 \times 10^{-11} \text{ mol}^3 \text{ L}^{-3}$. A solution of K₂C₂O₄ containing 0.152 mole in 500 mL water is shaken with excess of Ag₂CO₃ till the following equilibrium is reached :
 $\text{Ag}_2\text{CO}_3(\text{s}) + \text{K}_2\text{C}_2\text{O}_4(\text{aq}) \rightleftharpoons \text{Ag}_2\text{C}_2\text{O}_4(\text{s}) + \text{K}_2\text{CO}_3(\text{aq})$
 At equilibrium, the solution contains 0.0358 mole of K₂CO₃. Assuming the degree of dissociation of K₂C₂O₄ and K₂CO₃ to be equal, if the solubility product of Ag₂CO₃ is $3.794 \times 10^{-x} \text{ mol}^3 \text{ L}^{-3}$, find x.
63. Find out maximum number of moles of PbCO₃ soluble in 100 L buffer solution of pH = 3, rounding it off to the nearest whole number.
 $K_{a1}(\text{H}_2\text{CO}_3) = \frac{50}{11} \times 10^{-7}$
 $K_{a2}(\text{H}_2\text{CO}_3) = 5.5 \times 10^{-11}$
 $K_{sp}(\text{PbCO}_3) = 7.5 \times 10^{-14}$; (Take $\sqrt{0.3} = 0.55$)
64. 0.1 millimole of CdSO₄ are present in 10 mL acid solution of 0.08 M HCl. Now H₂S is passed to precipitate all the Cd²⁺ ions. Find the pOH of the solution after filtering off precipitate, boiling off H₂S and making the solution 100 mL by adding H₂O.

ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

65. Select correct statement(s) :
 (A) CH₃COOH (a weak acid) behaves as a strong acid in NH₃.
 (B) HNO₃ (strong acid) behaves as a base in HF (anhydrous).
 (C) H₂SO₄ dissociates to a very small extent in glacial CH₃COOH.
 (D) HI is a stronger acid than HCl in H₂O.
66. C M fluoroacetic acid solution was found to contain $[\text{H}^+] = 1.5 \times 10^{-3} \text{ M}$. K_a of fluoroacetic acid = 2.5×10^{-3} . Then :
 (A) C = $9 \times 10^{-4} \text{ M}$ (B) $[\text{FCH}_2\text{COOH}] = 9 \times 10^{-4} \text{ M}$
 (C) C = $2.4 \times 10^{-3} \text{ M}$ (D) $\alpha = 0.625$



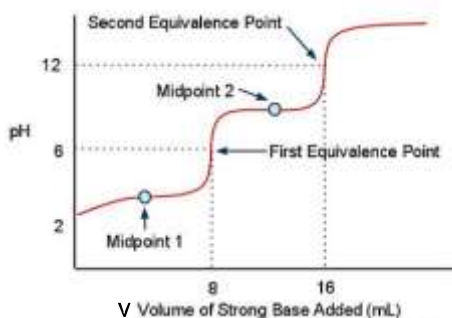
67. To 1 L of a 1.6×10^{-3} M aqueous solutions of ethylene diamine ($K_{b_1} = 8 \times 10^{-5}$, $K_{b_2} = 2.7 \times 10^{-8}$), 5×10^{-4} mole of $\text{Ba}(\text{OH})_2$ is added. Then :
- (A) $\text{pH}_f - \text{pH}_i = 0.5$ (B) $\alpha_i - \alpha_f = 0.12$ (C) $\frac{[\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_i}{[\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_f} = 0.128$ (D) $\frac{[\text{C}_2\text{N}_2\text{H}_9^+]_i}{[\text{C}_2\text{N}_2\text{H}_9^+]_f} = 2.5$
68. The correct relationship(s) between the pH of isomolar solutions of Na_2O (pH_1), Na_2S (pH_2), Na_2Se (pH_3) and Na_2Te (pH_4) is/are :
- (A) $\text{pH}_1 > \text{pH}_2$ (B) $\text{pH}_2 < \text{pH}_4$ (C) $\text{pH}_2 < \text{pH}_3$ (D) $\text{pH}_3 > \text{pH}_4$
69. For a 0.072 M NaHSO_4 solution, select the incorrect option(s) : (K_{a_1} and K_{a_2} of $\text{H}_2\text{SO}_4 = \infty$ & 1.2×10^{-2})
- (A) $\text{pH} = 1.62$ (B) $\text{pH} = 7.39$
 (C) Degree of hydrolysis, $h = \frac{1}{\sqrt{6}}$ (D) $h = 3.4 \times 10^{-6}$
70. Choose the correct statement(s) about buffer capacity during titration of NH_4OH with HCl :
- (A) Buffer capacity during titration first increases, then decreases.
 (B) Buffer capacity reaches to maximum at equivalence point.
 (C) Buffer capacity will increase if concentration of NH_4OH increases.
 (D) Buffer capacity value will remain same if strong acid HCl is replaced by H_2SO_4 of equal $[\text{H}_3\text{O}^+]$.
71. Which of the following statements is/are true for indicators ?
- (A) Indicators catalyse the acid base reactions by releasing or accepting H^+ ions.
 (B) Indicators do not significantly affect the pH of the solution to which they are added.
 (C) Acid-base reactions do not occur in absence of indicators.
 (D) Indicators have different colours in dissociated and undissociated forms.
72. Let the colour of the indicator HIn (colourless) will be visible only when its ionised form (pink) is 25% or more in a solution. Suppose HIn ($\text{pK}_{\text{in}} = 9$) is added to a solution of $\text{pH} = 9.6$. Predict what will happen :
- (A) pink colour will be visible. (B) pink colour will not be visible.
 (C) % of ionised form will be less than 25%. (D) % of ionised form will be more than 25%.
73. When a weak base solution (50 mL of 0.1 M NH_4OH) is titrated with a strong acid (0.1 M HCl), the pH of solution initially decreases fast and then decreases slowly till near equivalence point (as shown in the figure). Which of the following statements is/are true ?



- (A) The initial fast decrement in pH is due to fast consumption of free OH^- ions by HCl .
 (B) The slow decrease of pH is due to formation of an acidic buffer solution after addition of some HCl .
 (C) The slope of shown pH graph (magnitude only) will be minimum when 25 mL of 0.1 M HCl is added.
 (D) The slow decrease of pH is due to formation of a basic buffer solution after addition of some HCl .

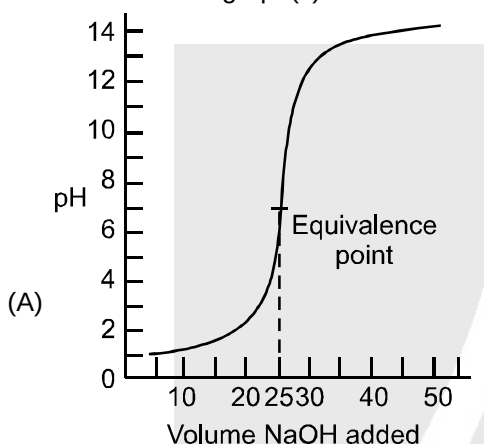


74. A solution of a substance is titrated against a strong base (or acid), volume V of strong base (or acid) is plotted against pH of the solution (as shown in figure). The substance could be :

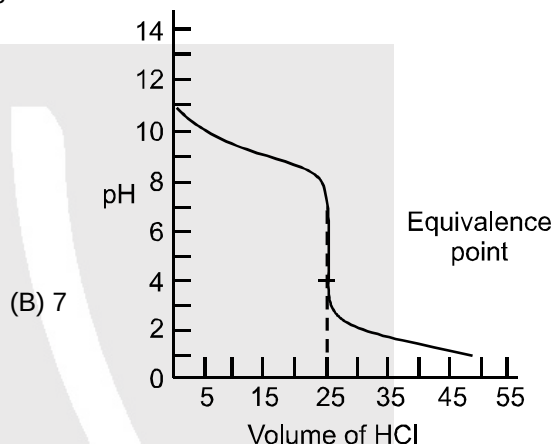


- (A) Na_2CO_3 (B) Ethylene diamine (C) $\text{H}_2\text{C}_2\text{O}_4$ (D) $\text{CH}_2(\text{COOH})_2$

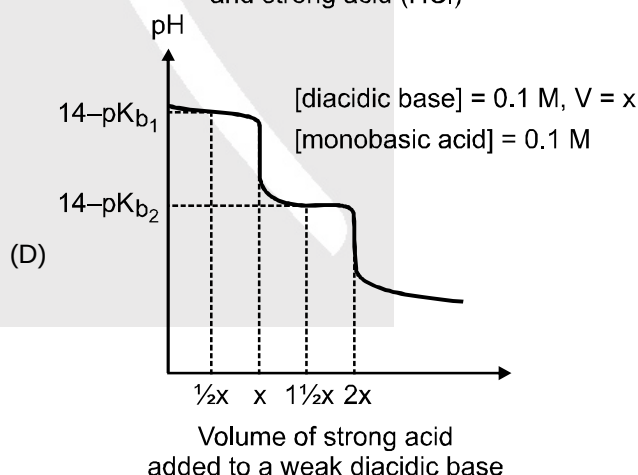
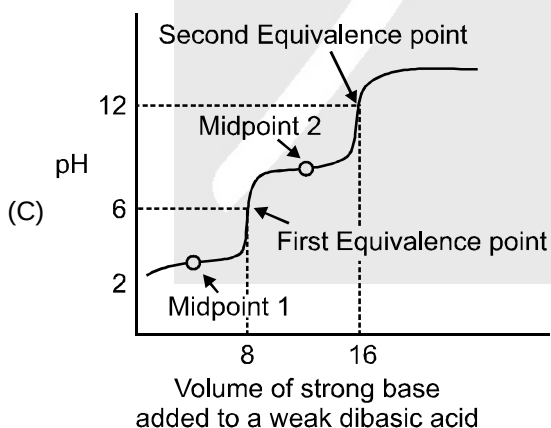
75. Select the correct graph(s) for the corresponding acid-base titration :



Titration curve of strong acid (HCl) with a strong base (NaOH)



The pH titration curve of weak base (NH_4OH) and strong acid (HCl)



76. 100 mL of a clear saturated solution of Ag_2SO_4 is added to 250 mL of a clear saturated solution of PbCrO_4 . Then select the incorrect option(s). Given K_{sp} values for Ag_2SO_4 , Ag_2CrO_4 , PbCrO_4 and PbSO_4 are 1.4×10^{-5} , 2.4×10^{-12} , 2.8×10^{-13} and 1.6×10^{-8} respectively.
- (A) $[\text{Ag}^+]_{\text{Just after mixing}} = 3.038 \times 10^{-2} \text{ M}$.
 (B) $[\text{Pb}^{2+}]_{\text{Just after mixing}} = 3.78 \times 10^{-7} \text{ M}$
 (C) Only PbSO_4 will get precipitated.
 (D) Both Ag_2CrO_4 and PbSO_4 will get precipitated.



77. Calculate $[Ag^+]$, $[CO_3^{2-}]$ & $[CrO_4^{2-}]$ in a solution saturated with respect to both Ag_2CO_3 & Ag_2CrO_4 .
 Given: $K_{sp}(Ag_2CO_3) = 4 \times 10^{-12}$; $K_{sp}(Ag_2CrO_4) = 2.4 \times 10^{-12}$ & $\frac{1}{4\sqrt{5}} = 0.146$
- (A) $[Ag^+] = 2.336 \times 10^{-4} M$ (B) $[CO_3^{2-}] = 7.3 \times 10^{-5} M$
 (C) $[CrO_4^{2-}] = 4.35 \times 10^{-5} M$ (D) $2[Ag^+] = [CO_3^{2-}] + [CrO_4^{2-}]$
78. A sample of $AgCl$ was treated with 5 mL of 1.5 M Na_2CO_3 solution to give Ag_2CO_3 . The remaining solution contained 0.0026 g of Cl^- per litre. Then : ($K_{sp}(Ag_2CO_3) = 8.2 \times 10^{-12}$)
- (A) $K_{sp}(AgCl) = 1.17 \times 10^{-10}$ (B) $K_{sp}(AgCl) = 1.17 \times 10^{-9}$
 (C) $[Ag^+] = 2.34 \times 10^{-7} M$ (D) $[Ag^+] = 2.34 \times 10^{-6} M$
79. In which of the following solutions, the solubility of $AgCN$ will be greater than that in pure water ?
 Given : $K_{sp}(AgCN) = 4 \times 10^{-16}$, $K_a(HCN) = 5 \times 10^{-10}$
- (A) 0.01 M $AgNO_3$ solution (B) A buffer solution of pH = 12
 (C) 0.2 M NH_3 solution (D) A buffer solution of pH = 5
80. The simultaneous solubility of $AgCN$ ($K_{sp} = 2.5 \times 10^{-16}$) and $AgCl$ ($K_{sp} = 1.6 \times 10^{-10}$) in 1 M $NH_3(aq)$ are s_1 and s_2 respectively. Then : [Given : $K_f[Ag(NH_3)_2^+] = 10^7$]
- (A) $s_1 = \frac{1}{27} \text{ mol/L}$ (B) $s_2 = \frac{10^{-4}}{1728} \text{ mol/L}$ (C) $\frac{[Cl^-]}{[CN^-]} = 6.4 \times 10^5$
 (D) Separate values of s_1 and s_2 cannot be determined, just their ratio $\frac{s_2}{s_1} = 1.5625 \times 10^{-6}$ can be known.

PART - IV : PRACTICE TEST-2 (IIT-JEE (ADVANCED Pattern))

Max. Time : 1 Hr.

Max. Marks : 66

Important Instructions

A. General :

- The test is of 1 hour duration.
- The Test Booklet consists of 22 questions. The maximum marks are 66.

B. Question Paper Format :

- Each part consists of five sections.
- Section 1 contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE is correct.
- Section 2 contains 5 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE OR MORE THAN ONE are correct.
- Section 3 contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9 (both inclusive).
- Section 4 contains 1 paragraph, describing theory, experiment and data etc. 3 questions relate to paragraph. Each question pertaining to a particular passage should have only one correct answer among the four given choices (A), (B), (C) and (D).
- Section 5 contains 1 multiple choice questions. Question has two lists (list-1 : P, Q, R and S; List-2 : 1, 2, 3 and 4). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

C. Marking Scheme :

- For each question in Section 1, 4 and 5 you will be awarded 3 marks if you darken the bubble corresponding to the correct answer and zero mark if no bubble is darkened. In all other cases, minus one (– 1) mark will be awarded.
- For each question in Section 2, you will be awarded 3 marks. If you darken all the bubble(s) corresponding to the correct answer(s) and zero mark. If no bubbles are darkened. No negative marks will be answered for incorrect answer in this section.



11. For each question in Section 3, you will be awarded 3 marks if you darken only the bubble corresponding to the correct answer and zero mark if no bubble is darkened. No negative marks will be awarded for incorrect answer in this section.

Note : Take water as solvent and temperature as 25°C, if not specified.
Take $\log 2 = 0.3$, $\log 3 = 0.48$, $\log 5 = 0.7$, $\log 7 = 0.845$, if not specified.

SECTION-1 : (Only One option correct Type)

This section contains 7 multiple choice questions. Each questions has four choices (A), (B), (C) and (D) out of which Only ONE option is correct.

1. Azhar prepared three different solutions of HCl, NaOH and brine (NaCl) but forgot to label them. In order to identify he kept them on table and named them as solution I, solution II and solution III. When he added a sample from solution II or solution III to a sample from solution I, the pH of the solution I increased. However, when a sample from solution III was added to a sample from solution II, the pH of the solution II decreased. Identify the incorrect option(s) :

(A) Solution I, II and III contain HCl, NaOH and NaCl respectively.
(B) Solutions I and solution III can be mixed to get the pH of solution II.
(C) $\text{pH}_I < \text{pH}_{II} < \text{pH}_{III}$
(D) Both (B) & (C)
2. An aqueous solution contains 0.01 M RNH_2 ($K_b = 2 \times 10^{-6}$) & 10^{-4} M NaOH. The concentration of OH^- is nearly :

(A) 1.414×10^{-4} M (B) 10^{-4} M (C) 3×10^{-4} M (D) 2×10^{-4} M
3. A 0.01 M solution of $\text{PuO}_2(\text{NO}_3)_2$ was found to have a pH of 4. Then select the INCORRECT option :

(A) % hydrolysis of $\text{PuO}_2^{2+} = 1\%$. (B) K_b for $\text{PuO}_2(\text{OH})^+ = 10^{-8}$.
(C) Hydrolysis is expected to be endothermic. (D) None of these
4. 10 mL of 0.1 M HCl solution is added in 90 mL of a buffer solution having 0.1 M NH_4OH and 0.1 M NH_4Cl . The percentage change in pH of solution is : (K_a of $\text{NH}_4^+ = 5 \times 10^{-10}$)

(A) $\frac{10}{4.7}$ % increase (B) $\frac{10}{4.7}$ % decrease (C) $\frac{10}{9.3}$ % increase (D) $\frac{10}{9.3}$ % decrease
5. Select INCORRECT statement :

(A) Phenolphthalein is not a suitable indicator for the titration of $\text{HCl}(\text{aq})$ with $\text{NH}_4\text{OH}(\text{aq})$.
(B) An acid-type indicator in a buffer solution of $\text{pH} = \text{pK}_{\text{in}} + 1$ is ionized to the extent of $\frac{1000}{11}\%$.
(C) In the titration of a monoacidic weak base with a strong acid, the pH at the equivalent point is always calculated by $\text{pH} = \frac{1}{2} [\text{pK}_w - \text{pK}_b - \log C]$.
(D) When $\text{Na}_3\text{PO}_4(\text{aq})$ is titrated with $\text{HCl}(\text{aq})$, the pH of solution at second equivalent point is calculated by $\frac{1}{2} [\text{pK}_{a_1} + \text{pK}_{a_2}]$.
6. The solubility of SrF_2 ($K_{\text{sp}} = 2.5 \times 10^{-9}$) in 0.1 M SrCl_2 solution is : (neglect hydrolysis of F^-)

(A) $\sqrt{\frac{5}{2}} \times 10^{-4}$ mol/L (B) 5×10^{-3} mol/L (C) $5\sqrt{\frac{5}{2}} \times 10^{-5}$ mol/L (D) $\frac{5}{2} \times 10^{-3}$ mol/L
7. pH of a saturated solution of silver salt of monobasic acid HA is found to be 9. Find the K_{sp} of sparingly soluble salt $\text{AgA}(\text{s})$.
Given : $K_a(\text{HA}) = 10^{-10}$

(A) 1.1×10^{-11} (B) 1.1×10^{-10} (C) 10^{-12} (D) Data insufficient



Section-2 : (One or More than one options correct Type)

This section contains 5 multipole choice questions. Each questions has four choices (A), (B), (C) and (D) out of which ONE or MORE THAN ONE are correct.

8. Which of the following options is/are true for alkaline aqueous solution ?
 (A) $\text{pH} > \frac{\text{pK}_w}{2}$ (B) $\text{pH} > \text{pOH}$ (C) $\text{pOH} < \frac{\text{pK}_w}{2}$ (D) $\text{pH} < \text{pOH}$
9. pH of aqueous solution of which of the following salt(s) is/are independent of its concentration ?
 (A) CH_3COONa (B) $\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$ (C) KClO_4 (D) $\text{CH}_3\text{COONH}_4$
10. A base type indicator is half in ionised form when pH is 7.2. If the ratio of unionised form to ionised form is 1 : 5, let pH of the solution is pH_1 . With the same pH of solution, indicator is altered (not its type) such that the ratio of unionised form to ionised form is 1 : 4. Let pH_2 be the pH of solution when 50% of new indicator is in ionised form. Then :
 (A) $\text{pH}_1 = 7.9$ (B) $\text{pH}_1 = 6.5$ (C) $\text{pH}_2 = 7.3$ (D) $\text{pH}_2 = 7.1$
11. A 2.5 g impure sample containing weak monoacidic base (Mol. wt. = 45 u) is dissolved in 100 mL water and titrated with 0.5 M HCl. When $\left(\frac{1}{5}\right)^{\text{th}}$ of the base was neutralised, the pH was found to be 9 and at equivalence point, pH of solution is 4.5. Select correct statement(s) :
 (A) K_b of base is less than 10^{-6} .
 (B) Concentration of salt (C) at equivalent point is 0.25 M.
 (C) Volume of HCl is used at equivalent point is 100 mL.
 (D) Weight percentage of a base in given sample is 80%.
12. The solubility of metal sulphides in saturated solution of H_2S $\{[\text{H}_2\text{S}] = 0.1 \text{ M}\}$ can be represented by :

$$\text{MS} + 2\text{H}^+ \rightleftharpoons \text{M}^{2+} + \text{H}_2\text{S} ; K_{\text{eq}} = \frac{[\text{M}^{2+}][\text{H}_2\text{S}]}{[\text{H}^+]^2}$$

 The value of K_{eq} is given for few metal sulphides. If concentration of each metal ion in solution is 0.01 M, which metal sulphide(s) is/are selectively precipitated at total $[\text{H}^+] = 1 \text{ M}$ in saturated H_2S solution ?

Metal sulphide :	MnS	ZnS	CoS	PbS
K_{eq} :	3×10^{10}	3×10^{-2}	3	3×10^{-7}
(A) MnS	(B) ZnS	(C) CoS	(D) PbS	

Section-3 : (One Integer Value Correct Type.)

This section contains 6 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive)

13. Value of K_{eq} at 25°C for reaction : $\text{NH}_3 + \text{H}_3\text{O}^+ \rightleftharpoons \text{NH}_4^+ + \text{H}_2\text{O}$ is 2×10^9 . Determine pK_b of NH_4OH , rounding it off to nearest whole number. Report your answer as '0' if value cannot be determined.
14. In a 3.24 L closed cylindrical container, the following equilibrium is established at 0°C :
 $2\text{Ag}(\text{s}) + 2\text{I}^-(\text{aq}) + 2\text{H}_2\text{O}(\ell) \rightleftharpoons 2\text{AgI}(\text{s}) + \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$
 The equilibrium constant for the above reaction is 4×10^{-9} . At equilibrium, solution occupies $\left(\frac{1}{3.24}\right)^{\text{th}}$ height of the container and contains 0.5 M I^- and other species. If gaseous matter weighs 2 g, find pOH of solution. Neglect aqueous tension.
15. Calculate $[\text{A}^{2-}]/[\text{H}^+]$ in an aqueous solution of 0.1 M H_2A (dibasic weak acid).
 Given : For H_2A , $K_{a1} = 0.05$, $K_{a2} = 2 \times 10^{-6}$.
 Report your answer after multiplying by 1,00,000.



16. To a 200 mL of 0.1 M weak base BOH solution, 46 mL of 0.2 M solution of HCl are added. Now, what volume (in mL) of 0.1 M H_2SO_4 should be added into above solution, so that pH of resulting solution becomes 9? [$K_b(\text{BOH}) = 10^{-5}$]
17. 3.29×10^{-3} mg AgBr(s) ($K_{sp} = 4.9 \times 10^{-13}$) is added to 250 mL water. Determine the % saturation of solution. Report your answer after dividing by 10. (Report your answer as '0', if the given amount of salt cannot dissolve/just dissolves).
18. A buffer solution has 0.25 M CH_3COOH , 0.15 M CH_3COONa , $[\text{Mn}^{2+}] = 0.015$ M and is saturated with H_2S (0.1 M). Given: $K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}$, $K_a(\text{H}_2\text{S}) = 9 \times 10^{-21}$, $K_{sp}(\text{MnS}) = 2.4 \times 10^{-13}$. Concentration of a component of the buffer may have to be increased to start the precipitation of MnS. What would be its new concentration (in mole per litre)? Report your answer after multiplying by 10. (Report your answer as '0', if the concentration of any component need not be increased to start the precipitation).

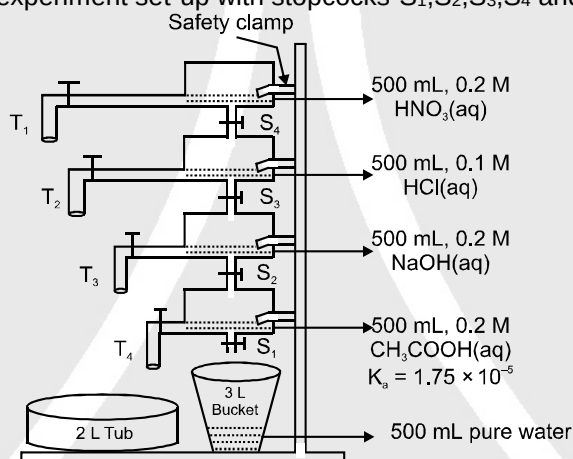
SECTION-4 : Comprehension Type (Only One options correct)

This section contains 1 paragraph, describing theory, experiments, data etc. 3 questions relate to the paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D)

Paragraph for Questions 19 to 21

Read the following passage carefully and answer the questions.

Consider the following experiment set-up with stopcocks- S_1, S_2, S_3, S_4 and taps- T_1, T_2, T_3, T_4 .



19. In which of the following cases will the solutions in tub have a pH different from others?
 (A) T_1 and T_4 are opened. (B) Only T_2 is opened.
 (C) T_1, T_2, T_3 and T_4 are all opened. (D) T_1 is opened and the content of bucket is poured in the tub.
20. A pH meter is immersed in the bucket of the original set-up shown. It shows a reading of 7. Now, S_1 is first opened and all content is allowed to drain out into the bucket. pH of the solution in the bucket is measured again. Similarly S_2, S_3 and S_4 are sequentially opened and the pH of the solution in the bucket is measured each time after complete drainage. Opening which stop cock would cause maximum pH change (increase/decrease) compared to the previous pH in bucket?
 (A) S_1 (B) S_2 (C) S_3 (D) S_4
21. The above apparatus was setup to study the functioning of the human stomach, where a low pH is maintained at a nearly constant value even when reactions take place in it. Following the same procedure as in the previous question, can a solution imitating stomach-like pH properties be generated in the bucket, and if so, by opening which stopcock? Would this solution have the lowest pH possible from this setup? If your answer is "No" for any of these questions, then after answering "No", choose the stopcock, opening of which will lead to solution in bucket with minimum pH, following the same procedure as above.
 (A) S_4 ; Yes (B) No; S_4 (C) S_3 ; No; S_1 (D) S_3 ; No; S_4



SECTION-5 : Matching List Type (Only One options correct)

This section contains 1 question, having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D) out of which one is correct

22. 0.01 mole of AgNO_3 are gradually added to 1 litre of a solution, which is 0.1 M in Na_2CrO_4 and 0.005 M in NaIO_3 . (K_{sp} values of Ag_2CrO_4 and AgIO_3 are 10^{-8} and 10^{-13} respectively)
Now match the values in list II with list I.

	List-I		List-II
(I)	Mole of precipitate formed	(P)	0.0975
(II)	Equilibrium concentration of Ag^+ (in mol/L)	(Q)	0.00032
(III)	Equilibrium concentration of IO_3^- (in mol/L)	(R)	0.0075
(IV)	Equilibrium concentration of CrO_4^{2-} (in mol/L)	(S)	3.125×10^{-10}

Code :

- (A) I-Q, II-R, III-S, IV-P
(C) I-R, II-S, III-Q, IV-P

- (B) I-Q, II-P, III-S, IV-R
(D) I-R, II-Q, III-S, IV-P

Practice Test-2 (IIT-JEE (ADVANCED Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22								
Ans.										



APSP Answers

PART - I

1.	(4)	2.	(3)	3.	(2)	4.	(2)	5.	(3)
6.	(4)	7.	(2)	8.	(4)	9.	(2)	10.	(4)
11.	(2)	12.	(3)	13.	(4)	14.	(3)	15.	(1)
16.	(4)	17.	(2)	18.	(1)	19.	(2)	20.	(1)
21.	(2)	22.	(4)	23.	(1)	24.	(4)	25.	(1)
26.	(3)	27.	(3)	28.	(4)	29.	(3)	30.	(3)

PART - II

1.	(B)	2.	(A)	3.	(D)	4.	(B)	5.	(D)
6.	(B)	7.	(B)	8.	(B)	9.	(D)	10.	(B)
11.	(D)	12.	(A)	13.	(B)	14.	(C)	15.	(B)
16.	(A)	17.	(C)	18.	(C)	19.	(A)	20.	(B)
21.	(A)	22.	(C)	23.	(C)	24.	(D)	25.	(C)
26.	(C)	27.	(C)	28.	(D)	29.	(B)	30.	(C)
31.	(D)	32.	(B)	33.	(C)	34.	(B)	35.	(A)
36.	(D)	37.	(B)	38.	(C)	39.	(D)	40.	(A)
41.	(C)	42.	(A)	43.	(D)	44.	(C)	45.	(C)
46.	(B)	47.	(A)	48.	(A)	49.	(A)	50.	(D)
51.	(A)	52.	(A)	53.	(D)	54.	(A)	55.	(A)
56.	(A)	57.	(B)	58.	(C)	59.	(A)	60.	(B)
61.	(C)	62.	(B)	63.	(B)	64.	(A)	65.	(C)
66.	(B)	67.	(D)	68.	(D)	69.	(B)	70.	(A)
71.	(A)	72.	(A)	73.	(A)	74.	(D)	75.	(D)
76.	(A)	77.	(A)	78.	(B)	79.	(C)	80.	(B)
81.	(A)	82.	(B)	83.	(B)	84.	(D)	85.	(C)
86.	(B)	87.	(B)	88.	(C)	89.	(C)	90.	(A)
91.	(C)	92.	(C)	93.	(D)	94.	(D)	95.	(A)
96.	(C)	97.	(D)						





PART - III

1. (a) As levelling solvent : Strong acids, strong bases. As differentiating solvent : Weak acids, weak bases.
(b) As levelling solvent : Strong bases, weak bases. As differentiating solvent : Strong acids, weak acids.
(c) As levelling solvent : Strong acids, weak acids, As differentiating solvent : Strong bases, weak bases.
2. $1.92 ; \frac{41}{60} \times 10^{-2} \text{ M} ; \frac{31}{60} \times 10^{-2} \text{ M} ; \frac{1395}{144} \times 10^{-6} \text{ M}$
3. $13, 5.4 \times 10^{-11} \text{ M}$
4. $2.3 ; 5 \times 10^{-3} \text{ M} ; 10^{-8} \text{ M}$
5. 0.23 for first solution and 0.46 for second solution
6. 0.25
7. 28.57%
8. $14 - \text{pK}_{\text{eq}} - \log \frac{b}{a}$ to $14 - \text{pK}_{\text{eq}} - \log \frac{c}{d}$
9. 1.395
- 10(a). 0.444 M
- 10(b). 1
11. (D)
12. (D)
13. (D)
14. (B)
15. (B)
16. (a) (D)
16. (b) (A)
17. (A)
18. (a) (A)
18. (b) (D)
19. (C)
20. (C)
21. (B)
22. (A)
23. (A)
24. (C)
25. (A)
26. (C)
27. (A)
28. (B)
29. (B)
30. (C)
31. (C)
32. (D)
33. (C)
34. (C)
35. (A)
36. (B)
37. (C)
38. (D)
39. (D)
40. (D)
41. (C)
42. (D)
43. (A)
44. (B)
45. (C)
46. (B)
47. (A)
48. (A)
- 49(a). (D)
- 49(b). (A)
- 49(c). (C)
50. (A – p, q, r, s); (B – p, r); (C – r); (D – p, q, s)
51. (A – s); (B – r); (C – q); (D – p)
52. (A – p, q); (B – q, r); (C – p, q, s); (D – r, s)
53. (A – q); (B – s); (C – p); (D – r)
54. (A → r); (B → p); (C → q, s); (D → t)
55. 21
56. 13
57. 4
58. 19
59. 3 [(i), (ii) & (vii)]
60. 5 [Na_2HPO_4 , Na_2S , $\text{K}_2\text{C}_2\text{O}_4$, K_2CO_3 , Ethylene diamine]
61. 64
62. 12
63. 6 (actual answer = 5.5)
64. 12
65. (ABC)
66. (BCD)
67. (ABD)
68. (AD)
69. (BCD)
70. (ACD)
71. (BD)
72. (AD)
73. (ACD)
74. (CD)
75. (ABCD)
76. (ACD)
77. (ABC)
78. (AD)
79. (CD)
80. (ABC)

PART - IV

1. (D)
2. (D)
3. (D)
4. (D)
5. (C)
6. (C)
7. (A)
8. (ABC)
9. (CD)
10. (BD)
11. (BC)
12. (D)
13. 5 (exact answer = 4.7)
14. 5
15. 4
16. 4
17. 1
18. 6
19. (C)
20. (B)
21. (D)
22. (D)



APSP Solutions

PART - I

- CO doesn't have a vacant orbital. Infact, due to the presence of lone pair on carbon, it behaves like a Lewis base.
- $$\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$$

$$t = \text{eq} \quad \text{C}(1 - \alpha) \quad \text{C}\alpha \quad \text{C}\alpha \quad (\text{C} = 55.55 \text{ M} ; \alpha = 1.8 \times 10^{-9})$$

$$K_a = \frac{\text{C}\alpha^2}{1 - \alpha} \approx \text{C}\alpha^2 = 1.8 \times 10^{-16}$$
- pH = 0 means $[\text{H}^+] = 10^0 = 1\text{M}$. Hence solution is strongly acidic.
- Na_2SO_4 will not exert any common ion effect on H_2SO_4 (since strong electrolyte).
 So, $[\text{H}^+] = 2 \times [\text{H}_2\text{SO}_4] = 2 \times 10^{-2} \text{ M}$
 $\therefore \text{pH} = 2 - \log 2 = 1.7$.
- $$[\text{H}^+] = \frac{0.1 \times 100 + 0.01 \times 100}{200} = \frac{11}{200} \text{ M}$$
 So, $\text{pH} = 2 - 0.74 = 1.26$.
- Millimole of $\text{H}^+ = V_1 \times 0.1 \times 1$
 & Millimole of $\text{OH}^- = V_2 \times 0.1 \times 2$
 If we put $V_1 = 2V_2$, the final solution will be neutral i.e.
 Millimole of $\text{H}^+ = 2V_2 \times 0.1 = \text{Millimole of OH}^-$
- $$\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_{b1}}{K_{b2}}} = \sqrt{\frac{2.56 \times 10^{-4}}{1.6 \times 10^{-5}}} = 4 : 1$$
- $[\text{H}^+]$ from $\text{HCl} = 10^{-3} \text{ M}$ & $[\text{H}^+]$ from $\text{HA} = \sqrt{\text{CK}_a} = 10^{-3} \text{ M}$.
 So, addition of HCl will not exert common ion effect on HA .
 So, neither pH nor degree of dissociation of HA will change.
- $$[\text{H}^+] = \sqrt{K_{a1}C_1 + K_{a2}C_2} = \sqrt{10} \times 10^{-3} \text{ M}$$

$$\therefore \text{pH} = 3 - \frac{1}{2} \log 10 = 2.5$$
- α is negligible w.r.t. 1.
 $[\text{H}^+]$ mainly comes from first step ionisation.

$$[\text{H}^+] = \sqrt{K_{a1} \times C} = \sqrt{10^{-5} \times 0.1} = 10^{-3} \text{ M}$$

$$\therefore K_{a3} = \frac{[\text{H}^+][\text{A}^{3-}]}{[\text{HA}^{2-}]} \Rightarrow 10^{-13} = \frac{10^{-3}[\text{A}^{3-}]}{[\text{HA}^{2-}]}$$

$$X = \frac{[\text{A}^{3-}]}{[\text{HA}^{2-}]} = 10^{-10} \quad \therefore \text{pX} = 10$$
- Hydrolysis of Fe^{3+} takes place :

$$\text{Fe}^{3+} + 3 \text{H}_2\text{O} \longrightarrow \text{Fe}(\text{OH})_3 + 3 \text{H}^+$$



$$12. \quad h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{K_w}{K_a \cdot C}} = \sqrt{\frac{10^{-14}}{K_a \times (0.01)}}$$

$$\Rightarrow 0.01 = \sqrt{\frac{10^{-14}}{K_a \times (0.01)}} \Rightarrow 10^{-14} = \frac{10^{-14}}{K_a \times 10^{-2}} \Rightarrow K_a = 10^{-8}.$$

Also, hydrolysis product of $A^- = [OH^-] = Ch = 10^{-4} \text{ M}$.

$$\therefore [H^+] = 10^{-10} \text{ M}$$

$$\therefore \text{pH} = 10.$$

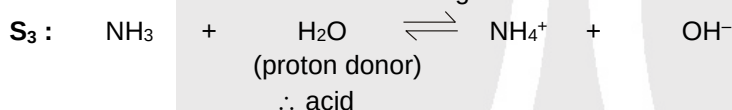
$$13. \quad \text{pH of a salt solution made from weak acid and weak base} = \frac{1}{2} (\text{pK}_w + \text{pK}_a - \text{pK}_b)$$

= independent of concentration of salt solution.

$$14. \quad \text{pH} = \frac{\text{pK}_{a_1} + \text{pK}_{a_2}}{2} = 9 \quad (\text{amphoteric species } HA^-)$$

15. **S₁** : Final solution will contain NaCl (salt of SA and SB) and $\text{CH}_3\text{COONH}_4$ (salt of WA and WB with $K_a = K_b$). So, final solution will be neutral ($\text{pH} = 7$ at 25°C).

S₂ : Equivalence point pH will be greater than 7, but MeOH indicator has pH range = 3.1 – 4.4. So, MeOH cannot be used as indicator in given titration.



16. (1) Salt of WA and WB.
 (2) Weak acid (H_2CO_3) + its conjugate base (HCO_3^-).
 (3) Weak acid (H_2PO_4^-) + its conjugate base (HPO_4^{2-}).

$$17. \quad \text{Buffer solution} \Rightarrow \text{pH} = \text{pK}_a + \log \frac{(0.02 \times 500)}{(0.01 \times 500)}$$

$$\Rightarrow 5.3 = \text{pK}_a + \log 2$$

$$\Rightarrow \text{pK}_a = 5 \quad \therefore \text{pK}_b = 14 - \text{pK}_a = 9$$

$$18. \quad \text{pH} = 9.7 = 10 - \log 2$$

$$\therefore [H^+] = 2 \times 10^{-10} \text{ M}$$

$$\text{PrNH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{PrNH}_3^+ + \text{OH}^- \quad ; \quad K_b = 5 \times 10^{-4}$$

$$\text{and } H^+ + \text{OH}^- \rightleftharpoons \text{H}_2\text{O} \quad ; \quad K_{eq} = \frac{1}{K_w} = 10^{14}$$

$$\therefore \text{PrNH}_2 + \text{H}^+ \rightleftharpoons \text{PrNH}_3^+ \quad ; \quad K_{eq} = \frac{K_b}{K_w} = 5 \times 10^{10}$$

$$5 \times 10^{10} = \frac{[\text{PrNH}_3^+]}{[\text{PrNH}_2] \times 2 \times 10^{-10}} \quad \therefore \quad \frac{[\text{PrNH}_2]}{[\text{PrNH}_3^+]} = \frac{1}{10} = 0.1.$$

$$19. \quad \text{pH} = \text{pK}_{in} + \log \frac{[In^-]}{[HIn]} \Rightarrow 5 = 4.7 + \log \frac{[In^-]}{[HIn]}$$

$$\Rightarrow 0.3 = \log \frac{[In^-]}{[HIn]} \Rightarrow \log 2 = \log \frac{[In^-]}{[HIn]}$$

$$\Rightarrow 2 = \frac{[In^-]}{[HIn]}$$

$$\text{Fraction} = \frac{HIn}{In^- + HIn} = \frac{1}{3}$$



20. $B + H^+ \longrightarrow BH^+$
At the half equivalent point, $[B] = [BH^+] \Rightarrow pOH = pK_b$ (from buffer formula)
21. $n_{HX} = n_{NaOH}$
 $\therefore \frac{8}{80} = 0.25 \times V_L$
 $\therefore V_L = 0.4 \text{ L}$
 $[NaX] = \frac{1}{5} = 0.2 \text{ M}$ (Since volume of solution becomes 5 times, so concentration of salt will become 1/5 times the concentration of HX).
 $pH = \frac{1}{2} (pK_w + pK_a + \log C) = \frac{1}{2} (14 + 4 - 0.7) = 8.65.$
22. $[CH_3COOH] = 0.1 \text{ M}$
 $\therefore pH = \frac{1}{2} (pK_a - \log C) = \frac{1}{2} (4.76 + 1) = 2.88.$
23. $s^2 = 2.5 \times 10^{-9}$
 $\therefore s = 5 \times 10^{-5} \text{ mol/L} = 5 \times 10^{-5} \times 100 \times 1000 \text{ mg/L} = 5 \text{ mg/L}.$
So, for dissolving 10 mg $CaCO_3$, $(V_{min})_{H_2O}$ required = 2 L.
24. For sparingly soluble salts having same K_{sp} values, the one producing maximum number of ions per formula-unit will be most soluble.
25. $4s_1^3 = 8.64 \times 10^{-13}$
 $\therefore s_1 = 6 \times 10^{-5} \text{ mol/L (in water)}$
 $Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{2-}$
Let solubility $2s_2 \quad s_2 + 0.6$
 $= s_2 \text{ mol/L} \quad \approx 0.6$
 $(2s_2)^2 \times 0.6 = 8.64 \times 10^{-13}$
 $\therefore s_2 = 6 \times 10^{-7} \text{ mol/L (in } Na_2CrO_4 \text{ solution)}$
 $\therefore \frac{s_1}{s_2} = 100.$
26. $[Ag^+][Cl^-] = K_{sp} = \text{constant}$
 $\Rightarrow xy = \text{constant}$
So, shape of graph should be a rectangular hyperbola.
27. $SrCO_3 \rightleftharpoons Sr^{2+} + CO_3^{2-}$
 $\quad \quad \quad S_1 \quad \quad \quad S_1 + S_2$
 $ZnCO_3 \rightleftharpoons Zn^{2+} + CO_3^{2-}$
 $\quad \quad \quad S_2 \quad \quad \quad S_1 + S_2$
 $S_1 \times (S_1 + S_2) = 10^{-10}$
 $S_2 \times (S_1 + S_2) = 1.5 \times 10^{-11}$
 $\Rightarrow \frac{S_1}{S_2} = \frac{[Sr^{2+}]}{[Zn^{2+}]} = \frac{10^{-10}}{1.5 \times 10^{-11}} = \frac{10}{1.5} = \frac{100}{15} = \frac{20}{3}$
28. Higher the concentration of H^+ , higher is the solubility of $Fe(OH)_3$. (H^+ ions will shift its solubility equilibrium in forward direction by consuming the OH^- ions produced). So, solubility of $Fe(OH)_3$ is maximum in 0.1 M H_2SO_4 .





29. Due to complex formation, solubility of sparingly soluble salt increases because of solubility equilibrium shifting in forward direction.
30. As pH decreases, $[H^+]$ increases. So, CN^- will combine with H^+ forming HCN making the solubility equilibrium shift in forward direction.

PART - III

1. Refer class notes / sheet theory.
2. Assuming $\alpha_2 \ll \alpha_1$ & $\alpha_2 \ll \alpha$ due to common ion effect of OH^- from first dissociation of same base (with α_1) & dissociation of CH_3NH_2 (with α) on second dissociation of base (with α_2), we have :

$$[OH^-] = \sqrt{0.2 \times (4.1 \times 10^{-4}) + 0.1 \times (6.2 \times 10^{-4})} = \sqrt{(8.2 + 6.2) \times 10^{-5}} = \sqrt{144 \times 10^{-6}}$$

$$\therefore pOH = -\log 0.012 \approx 1.92$$

$$\text{Now, } K_b = \frac{[OH^-][CH_3NH_3^+]}{[CH_3NH_2]} \Rightarrow 4.1 \times 10^{-4} = \frac{0.012}{0.2} [CH_3NH_3^+] \Rightarrow [CH_3NH_3^+] = \frac{41}{60} \times 10^{-2} \text{ M}$$

$$\text{Again } \therefore, K_{b_1} = \frac{[OH^-][C_4H_{13}N_2^+]}{[C_4H_{12}N_2]} \Rightarrow 6.2 \times 10^{-4} = \frac{0.012}{0.1} [C_4H_{13}N_2^+] \Rightarrow [C_4H_{13}N_2^+] = \frac{31}{60} \times 10^{-2} \text{ M}$$

$$\text{And } K_{b_2} = \frac{[OH^-][C_4H_{14}N_2^{2+}]}{[C_4H_{13}N_2^+]} \Rightarrow 2.25 \times 10^{-5} = \frac{0.012[C_4H_{14}N_2^{2+}]}{\frac{31}{60} \times 10^{-2}} \Rightarrow [C_4H_{14}N_2^{2+}] = \frac{1395}{144} \times 10^{-6} \text{ M}$$

3. $[OH^-] = \frac{0.2}{2} = 0.1 \text{ M} \Rightarrow pH = 14 - (-\log 0.1) = 13.$

$$K_{b_1} \times K_{b_2} = \frac{[OH^-]^2 [C_2N_2H_{10}^{2+}]}{[C_2N_2H_8]} \Rightarrow 8 \times 2.7 \times 10^{-13} = \frac{(0.1)^2 [C_2N_2H_{10}^{2+}]}{(0.5/2)}$$

$$\Rightarrow [C_2N_2H_{10}^{2+}] = \frac{8 \times 2.7 \times 10^{-13}}{4 \times 10^{-2}} \text{ M} = 5.4 \times 10^{-11} \text{ M}$$

4. $pH = \frac{1}{2}(14 - pK_{b_2} - \log 0.5) = 7 - 4.7 = 2.3.$

$$[BH^+] = [H^+] = 10^{-2.3} = 5 \times 10^{-3} \text{ M}.$$

$$\frac{10^{-14} \times 10^{-14}}{2 \times 10^{-6} \times 10^{-10}} = \frac{(5 \times 10^{-3})^2 [B]}{0.5} \Rightarrow [B] = 10^{-8} \text{ M}.$$

5. (i) $\frac{dx}{d(\Delta pH)} = \frac{2.303 (a-b)}{(a+b)} = \frac{2.303 \times 0.2 \times 0.2}{0.4} = 0.23$

(ii) $\frac{dx}{d(\Delta pH)} = \frac{2.303(0.4 \times 0.4)}{0.8} = 0.46$

6. For sodium butyrate

$$pH = 7 + \frac{1}{2} pK_a + \frac{1}{2} \log C = 7 + \left(\frac{5 - \log 1.5}{2} \right) + \frac{1}{2} \log (0.1) = 7 + 2.41 - 0.5 = 8.91$$

For Indicator

$$K_{In} = \frac{[H^+][In^-]}{[HIn]} \Rightarrow \frac{[In^-]}{[HIn]} = \frac{K_{In}}{[H^+]} = \frac{3.075 \times 10^{-10}}{10^{-8.91}} = 0.25$$



7. For this indicator, $\text{pH} = \text{pK}_{\text{In}} = 2$

Now, at $[\text{H}^+] = 4 \times 10^{-3} \text{ M}$,

$$K_{\text{In}} = \frac{(4 \times 10^{-3}) [\text{In}^-]}{[\text{HIn}]} \Rightarrow \frac{[\text{HIn}]}{[\text{In}^-]} = 0.4$$

$$\therefore \% \text{ of HIn} = \frac{[\text{HIn}]}{[\text{In}^-] + [\text{HIn}]} \times 100 = \frac{0.4}{1.4} \times 100 = \mathbf{28.57 \%}$$

8. $\therefore \text{pOH} = \text{pK}_{\text{eq}} + \log \frac{[\text{Acidic form}]}{[\text{Basic form}]} = \text{pK}_{\text{eq}} + \log \frac{[\text{Red}]}{[\text{Blue}]}$

$$(\text{pOH})_{\text{blue}} = \text{pK}_{\text{eq}} + \log \frac{b}{a} \quad \text{and} \quad (\text{pOH})_{\text{red}} = \text{pK}_{\text{eq}} + \log \frac{c}{d}$$

$$\text{So, range} = \text{pH}_{\text{blue}} \text{ to } \text{pH}_{\text{red}} = 14 - \text{pK}_{\text{eq}} - \log \frac{b}{a} \text{ to } 14 - \text{pK}_{\text{eq}} - \log \frac{c}{d}$$

9. $K_{\text{sp}} = [\text{Mg}^{2+}][\text{F}^-]^2$ & $K_{\text{a}} = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]} \Rightarrow 3.5[\text{HF}] = [\text{F}^-]$

$$\text{Now, } \therefore [\text{F}^-] + [\text{HF}] = 2s \Rightarrow \frac{4.5}{3.5} [\text{F}^-] = 2s \quad \dots(1)$$

Since, $K_{\text{sp}} = s[\text{F}^-]^2$

$$\therefore \frac{9}{14} \times 10^{-8} = 4s^3 \times \left(\frac{3.5}{4.5}\right)^2 \quad \text{from (1)}$$

$$\left(\frac{9}{14}\right)^3 \times 10^{-9} \times 10 = s^3$$

$$\therefore s = 1.395 \times 10^{-3} \text{ mole/L}$$

$$\text{So, for 1000 L : Moles} = s \times 10^3 = 1.395.$$

- 10(a). $\text{CuCO}_3 (\text{s}) + 4\text{NH}_3 \rightleftharpoons \text{CO}_3^{2-} + [\text{Cu}(\text{NH}_3)_4]^{2+}$; $K_{\text{f}} \times K_{\text{sp}} = 2.8 \times 10^3$

$$t = 0 \quad 0.1 \quad c \quad 0 \quad 0$$

$$t = \text{eq} \quad 0 \quad c-0.4 \quad 0.1 \quad 0.1$$

$$2.8 \times 10^3 = \frac{0.1 \times 0.1}{(c-0.4)^4} \Rightarrow c-0.4 = \sqrt[4]{\frac{1000}{2.8 \times 10^8}} = 0.044 \Rightarrow c = 0.444 \text{ M.}$$

- 10(b). For precipitation of MS, $K_{\text{sp}} \text{ of (MS)} = [\text{M}^{+2}] [\text{S}^{2-}] = 6 \times 10^{-21}$

$$\Rightarrow [\text{S}^{2-}]_{\text{req}} = \frac{100 \times 6 \times 10^{-21}}{5} = 120 \times 10^{-21} = 1.2 \times 10^{-19} \text{ M}$$

Precipitation of MS will start when H_2S will provide atleast $1.2 \times 10^{-19} \text{ M S}^{2-}$.

$$\text{For } \text{H}_2\text{S}, \quad \text{H}_2\text{S} \rightleftharpoons 2 \text{H}^+ + \text{S}^{2-} \quad ; \quad K = K_{\text{a1}}.K_{\text{a2}} = \frac{[\text{H}^+]^2 [\text{S}^{2-}]}{[\text{H}_2\text{S}]}$$

$$10^{-7} \times 1.2 \times 10^{-19} = \frac{[\text{H}^+]^2 \times 1.2 \times 10^{-19}}{0.1} \quad ; \quad [\text{H}^+]^2 = 10^{-2}$$

$$\text{So, } [\text{H}^+] = 0.1 \text{ M} \quad \therefore \text{pH} = 1$$

11. Ni^{2+} accepts a pair of electron and H_2O donates it. This an acid-base reaction.

12. It is definition of Levelling effect.



$$13. \quad K_a = \frac{[H^+][A^-]}{[HA]} = \frac{[H^+]^2}{[HA]} \quad (\because [H^+] = [A^-]) \quad \text{and} \quad K_b = \frac{[H^+][B^-]}{[HB]} = \frac{[H^+]^2}{[HB]} \quad (\because [H^+] = [B^-])$$

$$\text{Also } H^+ \text{ are same} \quad \therefore \frac{K_a}{K_b} = \frac{[HB]}{[HA]} = \frac{4}{1} \quad \frac{[HA]}{[HB]} = \frac{1}{4}$$

$$14. \quad \alpha = \sqrt{\frac{5 \times 10^{-5}}{10^{-4}}} > 0.1 \quad \text{So, } \frac{C\alpha^2}{1-\alpha} = K_b \Rightarrow 5 \times 10^{-5} = \frac{10^{-4}\alpha^2}{1-\alpha} \Rightarrow 2\alpha^2 + \alpha - 1 = 0 \Rightarrow \alpha = 0.5$$

$$\text{Hence, } [OH^-] = C\alpha = 5 \times 10^{-5} \quad \& \text{ (तथा) } pH = 9.7$$

$$15. \quad [OH^-] = \sqrt{K_{b1} \times C} = \sqrt{1.6 \times 10^{-5}} = 4 \times 10^{-3} M$$

$$K_{b1} \times K_{b2} = \frac{[OH^-]^2 [C_2H_{10}N_2^{2+}]}{[C_2N_2H_8]} \Rightarrow 8 \times 2.7 \times 10^{-13} = \frac{16 \times 10^{-3} [C_2H_{10}N_2^{2+}]}{0.2}$$

$$\therefore [C_2H_{10}N_2^{2+}] = 2.7 \times 10^{-8} M$$

$$16. (a) \quad [H^+] = 0.1 M \& pH = 1$$

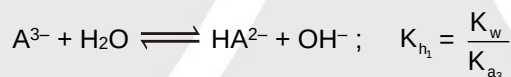
$$K_{a1} = \frac{0.1 [H_2PO_4^-]}{0.1} \Rightarrow [H_2PO_4^-] = K_{a1} \Rightarrow C\alpha = K_{a1} \Rightarrow \alpha = \frac{K_{a1}}{0.1} = 10 K_{a1}$$

$$K_{a2} = \frac{0.1 [HPO_4^{2-}]}{K_{a1}} \Rightarrow [HPO_4^{2-}] = K_{a1} \times K_{a2} \times 10$$

$$16. (b) \quad K_{a3} = \frac{0.1 [PO_4^{3-}]}{10 K_{a1} K_{a2} K_{a3}} \Rightarrow [PO_4^{3-}] = 100 K_{a1} K_{a2} K_{a3}$$

17. Factual

18.(a) & 18(b) :

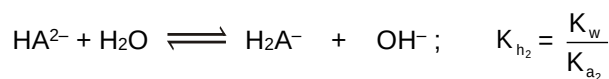


$$t = eq \quad c(1-h) \quad ch \quad ch$$

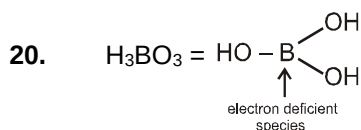
$$\frac{K_w}{K_{a3}} = \frac{ch^2}{1-h} \approx ch^2 \therefore h = \sqrt{\frac{K_w}{cK_{a3}}}$$

$$\therefore [OH^-] = [HA^{2-}] = ch = \sqrt{\frac{cK_w}{K_{a3}}}$$

$$\therefore [H^+] = \frac{K_w}{[OH^-]} = \sqrt{\frac{K_w K_{a3}}{c}} \& pH = \frac{1}{2} (pK_w + pK_{a3} + \log c)$$



19. According to Bronsted-Lowry concept, acid gives H^+ ion and base accept H^+ ion.

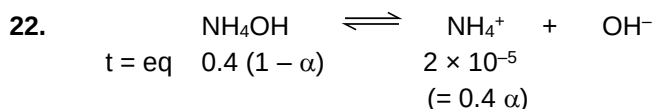


So it will behave as Lewis acid.



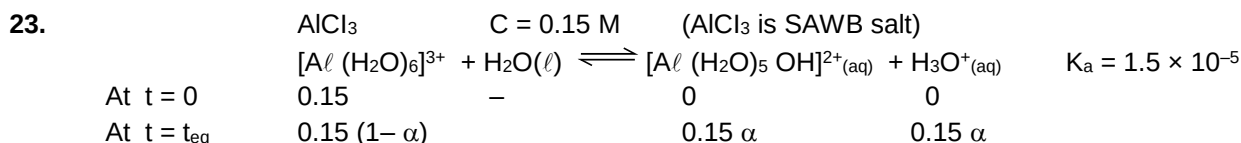
$$21. \quad 2 \times 10^{-4} = \frac{4 \times 10^{-3} \alpha^2}{1 - \alpha} \Rightarrow 20\alpha^2 + \alpha - 1 = 0 \Rightarrow \alpha = 0.2$$

$$\therefore [\text{OH}^-] = C\alpha = 8 \times 10^{-4} \text{ M} \Rightarrow \text{pOH} = 3.1$$



$$\text{So, } \alpha = 5 \times 10^{-5} (<< 1) \therefore 0.4(1 - \alpha) \approx 0.4$$

$$K_b = \frac{2 \times 10^{-5} \times [\text{OH}^-]}{0.4} = 1.8 \times 10^{-5} \Rightarrow [\text{OH}^-] = 0.36 \text{ M}$$

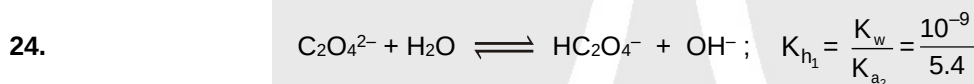


$$K_a = \frac{0.15\alpha \times 0.15\alpha}{0.15[1 - \alpha]} = 1.5 \times 10^{-5}$$

$$\Rightarrow 1.5 \times 10^{-1} \alpha^2 = 1.5 \times 10^{-5} \Rightarrow \alpha^2 = 10^{-4} \Rightarrow \alpha = 10^{-2}$$

$$[\text{H}_3\text{O}^+] = 0.15 \times 10^{-2} = 1.5 \times 10^{-3} \text{ M}$$

$$\Rightarrow \text{pH} = 3 - \log 1.5 = 3 - 0.18 = 2.82$$

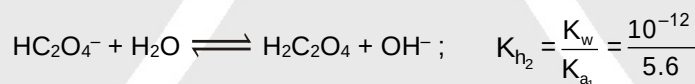


$$t = \text{eq} \quad 0.005(1 - h) \quad 0.005h \quad 0.005h$$

$$(\approx 0.005)$$

$$\frac{(0.005h)(0.005h)}{(0.005)} = \frac{10^{-9}}{5.4} \therefore h = 1.92 \times 10^{-4}$$

$$[\text{OH}^-] = 0.005h = 9.6 \times 10^{-7} \text{ M}$$



$$t = \text{eq} \quad \approx 9.6 \times 10^{-7} \quad \approx 9.6 \times 10^{-7}$$

$$\therefore [\text{H}_2\text{C}_2\text{O}_4] = K_{h_2} = \frac{5}{28} \times 10^{-12} \text{ M}$$



Note : Answer could also be NH_3 or OH^- .

26. We know that for acids at 25°C , pH must be less than 7.

27. When rain is accompanied by a thunderstorm,



$$28. \quad 6.31 = 6.61 + \log\left(\frac{M \times V}{100 \times 0.05}\right) \Rightarrow -0.3 = \log\left(\frac{MV}{5}\right)$$

$$\Rightarrow \frac{1}{2} = \frac{MV}{5} \Rightarrow 0.1V = \frac{5}{2} \Rightarrow V = 25 \text{ mL}$$

29. It is factual.



30. End points of both titrations will lie in acidic range.
31. $pK_{HIn} = 5$
 (a) $CH_3COOH + NaOH$, end point $pH > 7$ (b) Aniline hydrochloride + $NaOH$, end point $pH > 7$
 (c) $NaHCO_3 + HCl$, end point $pH < 7$ (d) $Ba(OH)_2 + H_2C_2O_4$, end point $pH > 7$
32. WA Vs SB end point > 7 Phenolphthalein
33. $NH_4Br \longrightarrow$ Salt of SA WB

$$pH = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C = 7 - \frac{4.74}{2} - \frac{1}{2}\log 10^{-1} = 7 - 2.37 + \frac{1}{2} = 5.13$$
34. $pH = pK_a + \log \frac{[Ionised]}{[un\ Ionised]} \Rightarrow 6 = 5 + \log \frac{[Ionised]}{[un\ Ionised]} \Rightarrow 1 = \log \frac{[Ionised]}{[un\ Ionised]}$

$$\frac{[Ionised]}{[un\ Ionised]} = 10 \Rightarrow \frac{[Ionised]}{[Ionised] + [un\ Ionised]} = \left[\frac{10}{11} \right]$$
35. $H_3PO_4 + NaOH \longrightarrow NaH_2PO_4$

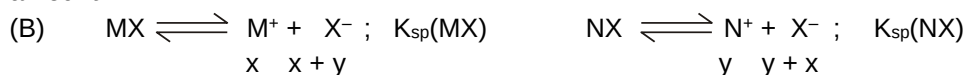
9	5	0
4	0	5

$$pH = pK_a + \log \frac{5}{4} = 3 + 0.1 = 3.1$$
36. Maximum buffer capacity of a solution is given by, buffer capacity = $2.303 \left(\frac{ab}{a+b} \right)$. Hence the result.
 and $a = b = 0.5$, $BC = 2.303 \times \frac{(0.5)^2}{1} = 0.57$
37. $pK_a = 5.45$
 $pH = pK_{HIn} + \log \frac{[Base\ form]}{[Acid\ form]} \Rightarrow pH = pK_{HIn} = 5.45$
 For a Buffer solution
 $pH = pK_a + \log \frac{[CH_3COONa]}{[CH_3COOH]} \Rightarrow 5.45 = 4.75 + \log \frac{[CH_3COONa]}{[CH_3COOH]}$
 $0.7 = \log \frac{[CH_3COONa]}{[CH_3COOH]} \Rightarrow \frac{5}{1} = \frac{[CH_3COONa]}{[CH_3COOH]}$
38. The end points of all the titrations will lie in acidic range.
- 39.
- | | | | | | |
|---------------------|-------------|---|--------|-------------------|-----------|
| | CO_3^{2-} | + | H^+ | $\longrightarrow$ | HCO_3^- |
| Initial milli-moles | 50×0.05 | | 40×0.1 | | — |
| Final milli-moles | — | | 1.5 | | 2.5 |
-
- | | | | | | |
|---------------------|-----------|---|-------|-------------------|-----------|
| | HCO_3^- | + | H^+ | $\longrightarrow$ | H_2CO_3 |
| Initial milli-moles | 2.5 | | 1.5 | | — |
| Final milli-moles | 1 | | — | | 1.5 |
- $$pH = pK_{a_1} + \log \frac{[HCO_3^-]}{[H_2CO_3]} = 6.173$$





40. (A) As both salts are sparingly soluble, due to common ion effect, both will be precipitated in some amount.



$$K_{sp}(MX) + K_{sp}(NX) = (x+y)^2$$

$$\Rightarrow \sqrt{K_{sp}(MX)} \approx x+y = [X^-] \dots (1) \quad (\because K_{sp}(MX) \gg K_{sp}(NX))$$

$$(C) \quad (y+x)y = K_{sp}(NX) \Rightarrow y = [N^+] \approx \frac{K_{sp}(NX)}{\sqrt{K_{sp}(MX)}} \dots (2)$$

$$\text{Similarly, } [M^+] = x \approx \frac{K_{sp}(MX)}{\sqrt{K_{sp}(MX)}} \dots (3)$$

From (2) & (3) : $[M^+] \gg [N^+]$

41. Solubility of $\text{CaCO}_3 = \left[\frac{7 \times 10^{-3}}{100} \right] \text{ mole/L} = 7 \times 10^{-5} \text{ mole/L} = s$

$$K_{sp} \text{ of } \text{CaCO}_3 = s^2 = 49 \times 10^{-10}.$$

When $[\text{Ba}^{+2}]$ is 90% precipitated, then only CaCO_3 starts precipitation.

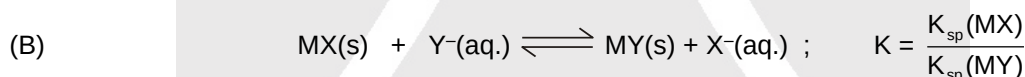
If original solution contain 'a' mole/L of Ca^{+2} & Ba^{+2} , then for CaCO_3 precipitation,

$$[\text{Ca}^{+2}][\text{CO}_3^{-2}] = 49 \times 10^{-10}, \quad [\text{CO}_3^{-2}] = \left[\frac{49 \times 10^{-10}}{a} \right] \text{ M.}$$

$$\text{Now for } \text{BaCO}_3, K_{sp} = [\text{Ba}^{+2}][\text{CO}_3^{-2}] = \frac{a \times 10}{100} \times \frac{49 \times 10^{-10}}{a} = 4.9 \times 10^{-10}. (\because 90\% \text{ Ba}^{+2} \text{ has precipitated,})$$

$$\text{so } [\text{Ba}^{+2}] = \frac{a \times 10}{100} \text{ M)}$$

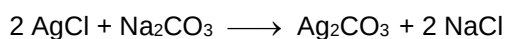
42. (A) As long as a solid-aqueous equilibrium exists, a salt solution is saturated.



$$\begin{array}{ccccc} t=0 & n & \frac{nx}{100} & 0 & 0 \\ t=\text{eq} & n - \frac{nx}{100} & y & \frac{nx}{100} & \frac{nx}{100} = [X^-] \end{array}$$

$$(C) \quad \frac{K_{sp}(\text{MX})}{K_{sp}(\text{MY})} = \frac{[X^-][M^+]}{[Y^-][M^+]} \Rightarrow [Y^-] = \frac{K_{sp}(\text{MY})}{K_{sp}(\text{MX})} \times [X^-] = \frac{K_{sp}(\text{MY})}{K_{sp}(\text{MX})} \times \frac{nx}{100}.$$

43. The concerned chemical reaction is :



Calculation of $[\text{Ag}^+]$ left in the solution :

$$K_{sp}(\text{Ag}_2\text{CO}_3) = [\text{Ag}^+][\text{CO}_3^{2-}]$$

$$[\text{Ag}^+] = \sqrt{\frac{8.1 \times 10^{-12}}{1.6}} = 2.25 \times 10^{-6} \text{ M}$$

$$\text{Concentration of Cl}^- \text{ left} = 0.00284 \text{ g/L} = \frac{0.00284}{35.5} \text{ mol/L} = 8 \times 10^{-5} \text{ M}$$

$$\therefore K_{sp}(\text{AgCl}) = [\text{Ag}^+][\text{Cl}^-] = (2.25 \times 10^{-6})(8 \times 10^{-5}) = 1.8 \times 10^{-10}.$$



44. Suppose V mL of solution contains 0.1 M Mg^{2+} and 0.8 M NH_4Cl
 Now, V mL of 'a' M NH_3 is added, which just gives precipitate of $\text{Mg}(\text{OH})_2$. Then :
 $[\text{Mg}^{2+}][\text{OH}^-]^2 = K_{\text{sp}} \text{Mg}(\text{OH})_2$

$$\left[\frac{0.1V}{2V} \right] [\text{OH}^-]^2 = 1.62 \times 10^{-11} \quad \left([\text{Mg}^{2+}] = \frac{\text{Millimole}}{\text{Total volume in mL}} \right)$$

 $\therefore [\text{OH}^-] = 1.8 \times 10^{-5} \text{ M}$
 Now, if the $[\text{OH}^-] = 1.8 \times 10^{-5} \text{ M}$, on addition of NH_3 in NH_4Cl , then $\text{Mg}(\text{OH})_2$ will precipitate. For buffer solution of NH_3 and NH_4Cl :

$$[\text{OH}^-] = K_b \times \frac{[\text{NH}_3]}{[\text{NH}_4^+]}$$

$$1.8 \times 10^{-5} = 1.8 \times 10^{-5} \times \frac{(0.8 \times V)/2V}{(a \times V)/2V}$$

 $\therefore a = 0.8 \text{ M}.$
45. (A) $S_{\text{pH}=2} = \frac{10^{-38}}{(10^{-12})^3} = 10^{-2} \text{ M}$ (B) $S_{\text{pH}=11} = \frac{10^{-38}}{(10^{-3})^3} = 10^{-29} \text{ M}$
46. For FeS and MnS not to be precipitated from a solution of Fe^{2+} and Mn^{2+} , $\text{IP}_{\text{FeS}} < K_{\text{sp}}(\text{FeS})$
 $[\text{Fe}^{2+}][\text{S}^{2-}] < K_{\text{sp}} \text{ of FeS}$
 $[10^{-2}][\text{S}^{2-}] < 6.4 \times 10^{-18} \therefore [\text{S}^{2-}] < 6.4 \times 10^{-16} \text{ M}$
 So, at $[\text{S}^{2-}] = 6.4 \times 10^{-16} \text{ M}$ or less, no precipitation of FeS and MnS will occur.
 $\text{H}_2\text{S} \rightleftharpoons 2\text{H}^+ + \text{S}^{2-}$
 $\therefore [\text{H}^+]^2 [\text{S}^{2-}] = K_{\text{a}(\text{H}_2\text{S})} \times [\text{H}_2\text{S}] = 9.6 \times 10^{-22}$
 $\therefore [\text{H}^+]_{\text{min}}^2 [6.4 \times 10^{-16}] = 9.6 \times 10^{-22}$

$$[\text{H}^+]_{\text{min}}^2 = 3/2 \times 10^{-6} \quad \therefore [\text{H}^+]_{\text{min}} = \sqrt{\frac{3}{2}} \times 10^{-3} \text{ M}$$

 Thus, if $[\text{H}^+] = \sqrt{\frac{3}{2}} \times 10^{-3} \text{ M}$ or more, the precipitation of FeS and MnS will not take place.
 Therefore, $\text{pH}_{(\text{max.})} = 3 - \frac{1}{2} \log \frac{3}{2} = 2.91 \text{ Ans.}$
47. For FeS and MnS to be precipitated from a solution of Fe^{2+} and Mn^{2+} , $\text{IP}_{\text{MnS}} > K_{\text{sp}}(\text{MnS})$
 $[\text{Mn}^{2+}][\text{S}^{2-}] > K_{\text{sp}} \text{ of MnS}$
 $[10^{-2}][\text{S}^{2-}] > 2.5 \times 10^{-13} \therefore [\text{S}^{2-}] > 2.5 \times 10^{-11} \text{ M}$
 So, at $[\text{S}^{2-}] = 2.5 \times 10^{-11} \text{ M}$ or more, precipitation of FeS and MnS will occur.
 $\text{H}_2\text{S} \rightleftharpoons 2\text{H}^+ + \text{S}^{2-}$
 $\therefore [\text{H}^+]^2 [\text{S}^{2-}] = K_{\text{a}(\text{H}_2\text{S})} \times [\text{H}_2\text{S}] = 9.6 \times 10^{-22}$
 $\therefore [\text{H}^+]_{\text{max}}^2 [2.5 \times 10^{-11}] = 9.6 \times 10^{-22}$

$$[\text{H}^+]_{\text{max}}^2 = 3.84 \times 10^{-11} \quad \therefore [\text{H}^+]_{\text{max}} = 8 \sqrt{\frac{3}{5}} \times 10^{-6} \text{ M}$$

 Thus, if $[\text{H}^+] = 8 \sqrt{\frac{3}{5}} \times 10^{-6} \text{ M}$ or less, the precipitation of FeS and MnS will take place.
 Therefore, $\text{pH}_{(\text{min.})} = 6 - \frac{1}{2} \log \frac{192}{5} = 5.21 \text{ Ans.}$



$$48. \quad 9.6 \times 10^{-21} = \frac{(0.1)^2 [S^{2-}]}{(0.1)} \Rightarrow [S^{2-}] = 9.6 \times 10^{-20} \text{ M}$$

$$9.6 \times 10^{-20} [Cu^{2+}]_{\min} = 6.4 \times 10^{-36} \Rightarrow [Cu^{2+}]_{\min} = \frac{2}{3} \times 10^{-16}$$

49(a). Dissolved $Al(OH)_3$ present in solution as $Al^{3+}(aq)$ as well as $Al(OH)_4^- (aq)$

$$\therefore S = [Al^{3+}(aq)] + [Al(OH)_4^-(aq)]$$

$$S = \frac{K_{sp}}{[OH^-]^3} + K_c [OH^-]$$

$$\text{for minimum solubility : } \frac{dS}{d(OH^-)} = 0$$

$$\text{or } -\frac{K_{sp} \times 3}{[OH^-]^4} + K_c = 0$$

$$[OH^-] = \left(\frac{3K_{sp}}{K_c} \right)^{1/4}$$

49(b). Order of solubility : Complex formation > Pure water > Common ion effect.

AgBr form complex $[Ag(NH_3)_2]^+$ in NH_3 , so solubility is maximum in $NH_3(aq)$.

49(c). In acidic medium, it behaves as a base & in basic medium, it behaves as an acid (accepting OH^-), driving the equilibrium forward in both cases.

50. (A) $[H^+]$ in HCl solution = 10^{-5} M

$$[H^+] \text{ in } H_2S \text{ solution} = \sqrt{K_a \times C} = 10^{-4} \text{ M}$$

so $[H^+]$ in HCl solution < $[H^+]$ in H_2S solution

& $[OH^-]$ in HCl solution > $[OH^-]$ in H_2S solution

& degree of dissociation of water in HCl solution > degree of dissociation of water in H_2S solution
pH of HCl solution > pH of H_2S solution.

(B) At pH = 4.74 in CH_3COOH solution, $[CH_3COO^-] = [CH_3COOH]$

$$\therefore \text{degree of dissociation of } CH_3COOH = \frac{1}{2}$$

At pH = 9.26 i.e. pOH = 4.74 in $NH_3.H_2O$ solution, $[NH_4^+] = [NH_3.H_2O]$

$\therefore$ degree of dissociation of CH_3COOH = degree of dissociation of $NH_3.H_2O$

degree of dissociation of water in CH_3COOH solution > degree of dissociation of water in $NH_3.H_2O$ solution.

$$(C) \quad [H^+] \text{ in } CH_3COOH \text{ solution} = \sqrt{1.8 \times 10^{-5} \times 0.1} = \sqrt{1.8} \times 10^{-3} \text{ M}$$

$$[H^+] \text{ in } HCOOH \text{ solution} = \sqrt{1.8 \times 10^{-4} \times 1} = \sqrt{1.8} \times 10^{-2} \text{ M}$$

so, degree of dissociation of CH_3COOH = degree dissociation of $HCOOH$.

$$(D) \quad [H^+] \text{ in } HA_1 \text{ solution} = \sqrt{10^{-5} \times 0.1} = 10^{-3} \text{ M}$$

$$[H^+] \text{ in } HA_2 \text{ solution} = \sqrt{10^{-6} \times 0.01} = 10^{-4} \text{ M}$$

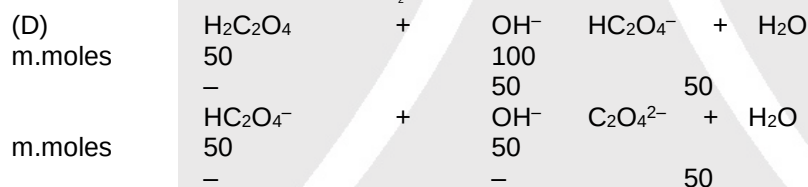
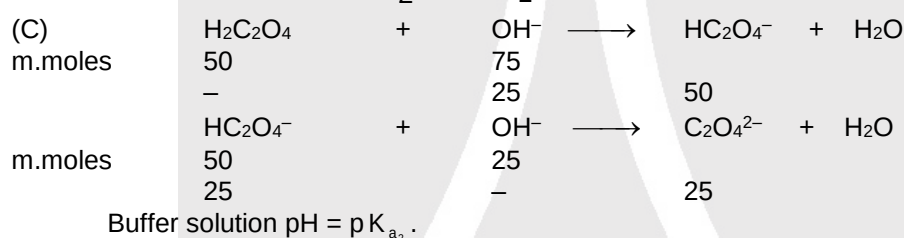
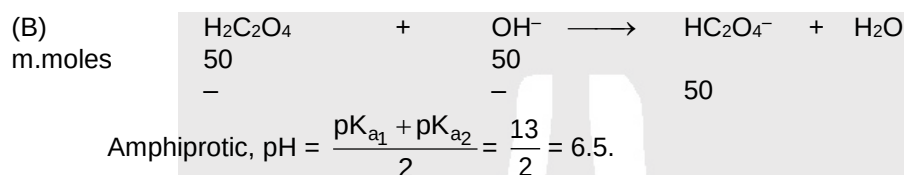
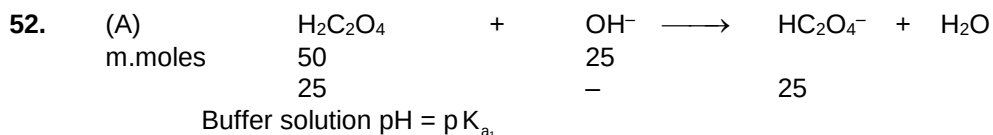
so $[OH^-]$ in solution of HA_1 < $[OH^-]$ in solution of HA_2

pH in solution of HA_1 < pH in solution of HA_2

& degree of dissociation of water HA_1 < degree of dissociation of water HA_2 .



51. (A) $[H^+] = \frac{5 \times 0.08 - 10 \times 0.015 \times 2}{500} = 2 \times 10^{-4} \text{ M} \therefore \text{pH} = 3.7$
- (B) For salt CH_3COONa : $\text{pH} = \frac{1}{2} \left(14 + 4.74 + \log \left(\frac{15}{150} \right) \right) = 8.87$
- (C) For RNH_2 : $\text{pOH} = \frac{1}{2} (\text{p}K_b - \log 0.25)$ & $K_h = \frac{K_w}{K_b} \therefore K_b = 10^{-5}$
 $\text{pOH} = 2.8 \therefore \text{pH} = 11.2$
- (D) For NaH_2A : $\text{pH} = \frac{\text{p}K_{a_2} + \text{p}K_{a_1}}{2} = \frac{4+7}{2} = 5.5$



Salt hydrolysis, $\text{pH} > 7$.

54. (p) $1.794 \times 10^{-11} = [\text{Mg}^{2+}] (0.36)^2 \Rightarrow [\text{Mg}^{2+}] = 1.38 \times 10^{-10} \text{ M}$.
- (q) $[\text{Hg}_2^{2+}] [\text{Cl}^-]^2 = K_{sp} \Rightarrow s(2 \times 0.18)^2 = 1.296 \times 10^{-18} \Rightarrow s = 10^{-17} \text{ M}$.
- (r) $[\text{Ca}^{2+}] = \sqrt{K_{sp_1} + K_{sp_2}} = \sqrt{(2.6 \times 10^{-9}) + (3.8 \times 10^{-9})} = 8 \times 10^{-5} \text{ M}$. (simultaneous solubility)
- (s) $s = \frac{10^{-38}}{(10^{-7})^3} = 10^{-17} \text{ M}$
 (neglecting OH^- coming from $\text{Fe}(\text{OH})_3$ in terms of those coming from water).
- (t)
- | | | | | | | | |
|---------|---------------|---|----------------|----------------------|------------------------------|--|---|
| | Ag^+ | + | 2CN^- | $\rightleftharpoons$ | $[\text{Ag}(\text{CN})_2]^-$ | | $K_f = 2.5 \times 10^{18}$ |
| t = 0 | 0.03 | | 0.1 | | 0 | | |
| t = eq. | x | | 0.04 | | 0.03 | | (Assuming nearly all Ag^+ ions to form complex K_f very large) |
- (≈ 0)
- (Upon mixing equal volumes, volume gets doubled. So initial concentrations get halved).
- $2.5 \times 10^{18} = \frac{0.03}{x(0.04)^2} \Rightarrow x = 7.5 \times 10^{-18} \text{ M}$.

55. $\text{H}_3\text{PO}_4 = 3$, $\text{H}_3\text{BO}_3 = 1$, $\text{H}_3\text{PO}_2 = 1$, rest diprotic



56. $x = 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14$
There are 13 possible values.

57. $K_b(\text{NH}_3) = 1.8 \times 10^{-5}$
 $\text{CH}_3\text{COOCH}_3(\text{aq}) + 2\text{H}_2\text{O}(\text{aq}) \longrightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq}) + \text{CH}_3\text{OH}(\text{aq})$
 $[\text{NH}_4^+]_0 = 0.1 \text{ M}, [\text{NH}_3] = 0.06 \text{ M}, [\text{CH}_3\text{COOCH}_3]_0 = 0.02 \text{ M}$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_3]} = 4.74 + \log \left(\frac{0.1}{0.06} \right)$$

$$(\text{pOH})_{\text{initial}} = 4.74 + 0.22 = 4.96 \quad \therefore (\text{pH})_{\text{initial}} = 9.04$$

$$\text{NH}_3(\text{aq}) + \text{H}^+(\text{aq}) \longrightarrow \text{NH}_4^+(\text{aq})$$

0.06	0.02	0.1	mole
0.04	–	0.12	mole

$$(\text{pOH})_{\text{final}} = 4.74 + \log \frac{0.12}{0.04} = 4.74 + \log \left(\frac{0.12}{0.04} \right) = 4.74 + \log 3 = 4.74 + 0.48 = 5.22$$

$$\therefore (\text{pH})_{\text{final}} = 8.78$$

$$\Delta \text{pH} = 9.04 - 8.78 = 0.26$$

Yes this is satisfactory buffer.

58. $\text{H}_3\text{PO}_4 = \frac{0.98}{98} = 10^{-2} \text{ Mole} \Rightarrow [\text{H}_3\text{PO}_4] = \frac{10^{-2}}{100} \times 10^3 = 0.1 \text{ M}$
 $\text{pH} = 5 \quad [\text{H}^+] = 10^{-5}$
 $\text{H}_3\text{PO}_4 \text{ \& } \text{H}_2\text{PO}_4^-$

$$\text{pH} = \text{p}K_{a1} + \log \frac{[\text{H}_2\text{PO}_4^-]}{[\text{H}_3\text{PO}_4]} \Rightarrow 5 = 3 + \log \frac{[\text{H}_2\text{PO}_4^-]}{[\text{H}_3\text{PO}_4]} \Rightarrow 10^2 = \frac{[\text{H}_2\text{PO}_4^-]}{[\text{H}_3\text{PO}_4]}$$

$$[\text{H}_2\text{PO}_4^-] = 10^2 [\text{H}_3\text{PO}_4] \Rightarrow \text{But } [\text{H}_3\text{PO}_4] + [\text{H}_2\text{PO}_4^-] = 0.1$$

$$\text{H}_3\text{PO}_4 + 10^2 [\text{H}_3\text{PO}_4] = 0.1 \Rightarrow [1 + 10^2] [\text{H}_3\text{PO}_4] = 0.1$$

$$[\text{H}_3\text{PO}_4] = 10^{-3} \text{ M} \Rightarrow [\text{H}_2\text{PO}_4^-] = 0.1 \text{ M}$$

$$\text{For } \text{H}_2\text{PO}_4^- \text{ \& } \text{HPO}_4^{2-} \Rightarrow \text{pH} = \text{p}K_{a2} + \log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}$$

$$5 = 8 + \log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]} \Rightarrow -3 = \log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}$$

$$\frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]} = 10^{-3} \Rightarrow [\text{HPO}_4^{2-}] = 10^{-3} [\text{H}_2\text{PO}_4^-]$$

$$[\text{H}_2\text{PO}_4^-] + [\text{HPO}_4^{2-}] = 0.1 \Rightarrow [\text{H}_2\text{PO}_4^-] + 10^{-3} [\text{H}_2\text{PO}_4^-] = 0.1$$

$$[1 + 10^{-3}] [\text{H}_2\text{PO}_4^-] = 0.1 \Rightarrow [\text{H}_2\text{PO}_4^-] = 0.1 \text{ M} \Rightarrow [\text{HPO}_4^{2-}] = 10^{-4} \text{ M}$$

Again

$$\text{pH} = \text{p}K_{a3} + \log \frac{[\text{PO}_4^{3-}]}{[\text{HPO}_4^{2-}]} \Rightarrow 5 = 12 + \log \frac{[\text{PO}_4^{3-}]}{[\text{HPO}_4^{2-}]}$$

$$10^{-7} = \frac{[\text{PO}_4^{3-}]}{[\text{HPO}_4^{2-}]} \Rightarrow [\text{PO}_4^{3-}] = 10^{-7} [\text{HPO}_4^{2-}]$$

$$[\text{HPO}_4^{2-}] + [\text{PO}_4^{3-}] = 10^{-4} \Rightarrow [1 + 10^{-7}] [\text{HPO}_4^{2-}] = 10^{-4}$$

$$[\text{PO}_4^{3-}] = 10^{-7} \Rightarrow [\text{HPO}_4^{2-}] = 10^{-11} \text{ M}$$

$$\text{pC}_1 + \text{pC}_2 + \text{pC}_3 + \text{pC}_4 = 11 + 4 + 1 + 3 = 19$$

59. (i), (ii) & (vii) are not matched with suitable indicators as the end point of these titrations doesn't lie within the pH range of the indicator





60. Na_2HPO_4 , Na_2S , $\text{K}_2\text{C}_2\text{O}_4$, K_2CO_3 , Ethylene diamine

61. $\text{H}_2\text{CO}_3 + \text{KOH} \longrightarrow \text{KHCO}_3 + \text{H}_2\text{O}$ $V_{\text{KOH}} = 50 \text{ mL}$ needed

0.2 M, 50 mL

$\text{KOHCO}_3 + \text{KOH} \longrightarrow \text{K}_2\text{CO}_3 + \text{H}_2\text{O}$ $V_{\text{KOH}} = 25 \text{ mL}$ needed for mid point

pH = 10.7 is mid titration point of IInd reaction as $\text{pH} = \text{pK}_{a_2}$ so it is possible at $V_{\text{KOH}} = 75 \text{ mL}$.

62.

	$\text{Ag}_2\text{CO}_3 + \text{K}_2\text{C}_2\text{O}_4 \rightleftharpoons \text{Ag}_2\text{C}_2\text{O}_4 + \text{K}_2\text{CO}_3$
Mole at start	Excess 0.1520 0 0
Mole after reaction	0.1520 - 0.0358 0.0358 0.0358
	= 0.1162

Molar concentration of $\text{K}_2\text{C}_2\text{O}_4$ or $\text{C}_2\text{O}_4^{2-}$ left unreacted = $\frac{0.1162}{0.5} = 0.2324 \text{ moles l}^{-1}$

$[\text{K}_2\text{CO}_3] = [\text{CO}_3^{2-}]$ at equilibrium = $\frac{0.0358}{0.5} = 0.07156 \text{ moles l}^{-1}$

Given that K_{sp} for $\text{Ag}_2\text{C}_2\text{O}_4 = 1.29 \times 10^{-11} \text{ mol}^3 \text{ l}^{-3}$ at 25°C

So $[\text{Ag}^+]^2 [\text{C}_2\text{O}_4^{2-}] = 1.29 \times 10^{-11}$

or $[\text{Ag}^+]^2 \times 0.2324 = 1.29 \times 10^{-11}$

Hence $[\text{Ag}^+]^2 = \frac{1.29}{0.2324} \times 10^{-11}$

Then K_{sp} for $\text{Ag}_2\text{CO}_3 = [\text{Ag}^+]^2 [\text{CO}_3^{2-}] = \frac{1.29 \times 10^{-11}}{0.2324} \times 0.0716 = 3.794 \times 10^{-12} \text{ mol}^3 \text{ l}^{-3}$

63.

$$K_{sp} = \frac{S^2}{1 + \frac{[\text{H}^+]}{K_{a_2}} + \frac{[\text{H}^+]^2}{K_{a_1} \cdot K_{a_2}}}$$

$$7.5 \times 10^{-14} = \frac{S^2}{1 + \frac{10^{-3}}{5.5 \times 10^{-11}} + \frac{10^{-6}}{25 \times 10^{-18}}}$$

$$7.5 \times 10^{-14} = \frac{S^2}{1 + \frac{10^8}{5.5} + \frac{10^{12}}{25}}$$

$$7.5 \times 10^{-14} = \frac{25 \times S^2}{10^{12}}$$

$$S^2 = 0.3 \times 10^{-14} \times 10^{12}$$

$$S^2 = 0.003$$

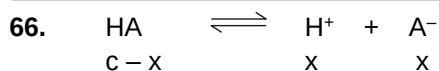
$$S = 0.055 \text{ M}$$

64.

	Cd^{2+}	+	H_2S	$\longrightarrow$	$\text{CdS} \downarrow$	+	2H^+
m.moles	0.1						0.2

Total m.moles of H^+ in solution after the reaction = $0.2 + 0.8 = 1$

$\therefore [\text{H}^+] = \frac{1}{100} = 0.01 \text{ M} \quad \Rightarrow \quad \text{pH} = 2.$



$$[\text{H}^+] = x = 1.5 \times 10^{-3} \text{ M}$$

$$K_a = \frac{x^2}{c - x}$$

$$2.5 \times 10^{-3} = \frac{(1.5 \times 10^{-3})^2}{c - 1.5 \times 10^{-3}}$$

$$C = 2.4 \times 10^{-3} \text{ M.}$$

67. Initial: $8 \times 10^{-5} = \frac{1.6 \times 10^{-3} \alpha^2}{1 - \alpha} \Rightarrow \alpha = \frac{1}{5}$

$$\therefore [\text{OH}^-] = 3.2 \times 10^{-4} \Rightarrow \text{pH} = 10.5$$

$$8 \times 10^{-5} \times 2.7 \times 10^{-8} = \frac{(3.2 \times 10^{-4})^2 [\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_i}{1.28 \times 10^{-3}}$$

$$\Rightarrow 2.7 \times 10^{-8} = [\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_i$$

Final pH = 11; $8 \times 10^{-5} = 10^{-3} \times \alpha \Rightarrow \alpha = 0.08$

$$[\text{C}_2\text{N}_2\text{H}_9^+] = 1.28 \times 10^{-4}$$

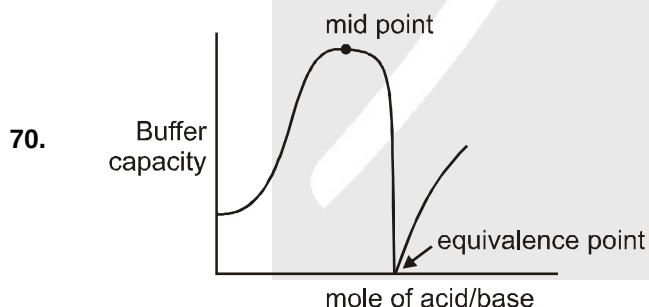
$$2.7 \times 8 \times 10^{-13} = \frac{(10^{-6}) [\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_f}{1.6 \times 10^{-3}} \Rightarrow [\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_f = 1.28 \times 2.7 \times 10^{-9} \text{ M}$$

So, (A), (B) and (D) are correct but $\frac{[\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_i}{[\text{C}_2\text{N}_2\text{H}_{10}^{2+}]_f} = \frac{1}{0.128}$

68. Order of basic strength $\text{O}^{2-} > \text{S}^{2-} > \text{Se}^{2-} > \text{Te}^{2-}$

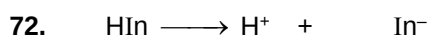
69. $1.2 \times 10^{-7} = \frac{7.2 \times 10^{-2} \alpha^2}{1 - \alpha} \Rightarrow 6\alpha^2 + \alpha - 1 = 0 \Rightarrow \alpha = \frac{1}{3}$

So, $\text{pH} = -\log \frac{0.072}{3} = 1.62$



When conc. of base or acid increases buffer capacity increases.

71. Facutal



$$K_{\text{In}} = \frac{[\text{H}^+][\text{In}^-]}{[\text{HIn}]} \Rightarrow \frac{[\text{In}^-]}{[\text{HIn}]} = \frac{K_{\text{In}}}{[\text{H}^+]} = \frac{10^{-9}}{10^{-9.6}} = 10^{0.6} = 4$$

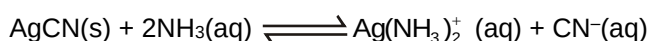
$$\therefore \% \text{ of } [\text{In}^-] \text{ in solution} = \frac{4}{5} \times 100 = 80\%$$

so pink colour will be visible.



$$K_1 = 1.6 \times 10^{-10} \times 10^7 = 1.6 \times 10^{-3}$$

$$= \frac{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}{[\text{NH}_3]^2}$$



$$K_2 = 2.5 \times 10^{-16} \times 10^7 = 2.5 \times 10^{-9}$$

$$\frac{[\text{Cl}^-]}{[\text{CN}^-]} = \frac{1.6 \times 10^{-3}}{2.5 \times 10^{-9}} = 6.4 \times 10^5$$

$$K_1 = \frac{x^2}{(1-2x)^2} \Rightarrow \frac{x}{1-2x} = 0.04$$

$$x = 0.037$$

$$\therefore [\text{CN}^-] = \frac{0.037}{6.4 \times 10^5} = 5.78 \times 10^{-8} \text{ M}$$

PART - IV

1. Upon adding two of the above solutions to an original solution, pH increase in both cases can take place only if original solution is of HCl (since NaCl solution addition will cause its dilution, thus increasing its pH while NaOH solution addition will cause its neutralization, thus increasing its pH upto 7). Further, pH of NaOH solution will decrease upon addition of NaCl solution (because of dilution). So, solutions I, II and III contain HCl, NaCl and NaOH respectively.
 $\text{HCl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$
 Only HCl and NaOH solutions can be used as standard solutions in an acid-base titration.

2. $\text{RNH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{RNH}_3^+ + \text{OH}^-$

$$\begin{array}{ccc} t = \text{eq} & 0.01(1-\alpha) & 0.01\alpha \\ & 0.01\alpha & 0.01\alpha + 10^{-4} \end{array}$$

$$\frac{0.01\alpha(0.01\alpha + 10^{-4})}{0.01(1-\alpha)} = 2 \times 10^{-6}$$

$$\therefore \alpha \approx 0.01$$

$$\therefore [\text{OH}^-] = 0.01(0.01) + 10^{-4}$$

$$= 2 \times 10^{-4} \text{ M.}$$

3. $\text{PuO}_2^{2+} + \text{H}_2\text{O} \rightleftharpoons \text{PuO}_2(\text{OH})^+ + \text{H}^+$

$$\begin{array}{ccc} t = 0 & 0.01 & 0 \\ t = \text{eq} & 0.01(1-h) & 0.01h \end{array}$$

$$\begin{array}{ccc} & 0 & 0 \\ & 0.01h & 0.01h \end{array}$$

$$\begin{array}{ccc} & & (=10^{-4}) \end{array}$$

Therefore, $h = 0.01 = 1\%$. So, $1-h \approx 1$.

$$K_b \text{ of } \text{PuO}_2(\text{OH})^+ = \frac{[\text{PuO}_2^{2+}][\text{OH}^-]}{[\text{PuO}_2(\text{OH})^+]} = \frac{0.01 \times 10^{-10}}{0.01h} = \frac{10^{-10}}{0.01} = 10^{-8}.$$

Hydrolysis is expected to be endothermic because it is reverse of neutralization (exothermic process).

4. $\text{pOH}_i = \text{pK}_b + \log_{10} \frac{[\text{NH}_4^+]_i}{[\text{NH}_4\text{OH}]_i} = 4.7$. Therefore, $\text{pH}_i = 9.3$.
 and $\text{pOH}_f = \text{pK}_b + \log_{10} \left(\frac{90 \times 0.1 + 10 \times 0.1}{90 \times 0.1 - 10 \times 0.1} \right) = 4.7 + \log \frac{5}{4} = 4.8$. Therefore, $\text{pH}_f = 9.2$

$$\% \text{ decrease} = \frac{9.3 - 9.2}{9.3} \times 100 = \frac{10}{9.3} \%$$



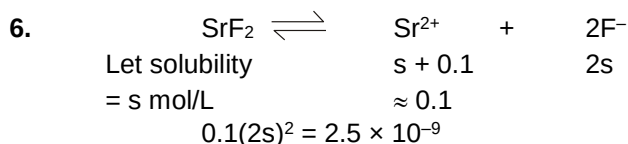
5. (A) Equivalence point pH will be less than 7, but HPh indicator has pH range = 8.3 – 10. So, HPh cannot be used as indicator in given titration.

$$(B) pK_{In} + 1 = pK_{In} + \log_{10} \frac{[In^-]}{[HIn]}$$

$$\frac{[In^-]}{[HIn]} = 10. \text{ So, \% ionisation } \frac{10}{10+1} \times 100 = \frac{1000}{11} \%$$

(C) Above formula is valid only when degree of hydrolysis is negligible.

(D) At second equivalent point, NaH_2PO_4 (amphiprotic species) is produced. So, $pH = \frac{1}{2} [pK_{a_1} + pK_{a_2}]$.



$$\therefore s = 5 \sqrt{\frac{5}{2}} \times 10^{-5} \text{ mol/L.}$$



$$\begin{array}{lclcl} t=0 & c & & 0 & 0 \\ t=eq & c(1-h) & & ch & ch \\ \frac{ch^2}{1-h} = 10^{-4}. & \text{But } [OH^-] = ch = 10^{-5} \text{ M. (} \because pH = 9 \text{)} \end{array}$$

$$\text{So, } \frac{10^{-5} h}{1-h} = 10^{-4}. \text{ Therefore, } h = \frac{10}{11}.$$

$$\text{So, } c = 1.1 \times 10^{-5} \text{ M.}$$

$$\begin{aligned} K_{sp} &= [Ag^+][A^-] = c \times c(1-h) \\ &= 1.1 \times 10^{-5} \times 10^{-6} \\ &= 1.1 \times 10^{-11}. \end{aligned}$$

8. In an alkaline aqueous solution, $[H^+] < [OH^-]$.

$$\therefore pH > pOH.$$

$$\text{Also, } pH + pOH = pK_w \text{ and for pure water, } pH = pOH = \frac{pK_w}{2}.$$

$$\text{In an alkaline solution, } [H^+] < [H^+]_{H_2O} \text{ and } [OH^-] > [OH^-]_{H_2O}.$$

$$\therefore pH > \frac{pK_w}{2} \text{ and } pOH < \frac{pK_w}{2}.$$

9. pH of the salt solutions of SA-SB and WA-WB is independent of concentration of solution.

10. As indicator (InOH) is base type indicator, so $pK_{InOH} = pOH$ at half ionisation point = 6.8.

$$pOH_1 = pK_{InOH} + \log_{10} \frac{5}{1} = 7.5. \quad \therefore pH_1 = 6.5$$

$$\text{and } pOH_1 = pK_{In^*OH} + \log_{10} \frac{4}{1} \quad \therefore pK_{In^*OH} = 6.9 = pOH_2$$

$$\therefore pH_2 = 7.1.$$



11. At 20% neutralisation, $\text{pOH} = \text{pK}_b + \log_{10} \frac{20}{80} = 5$.

$$\therefore \text{pK}_b = 5 - \log \frac{1}{4} \quad \therefore \text{K}_b = 2.5 \times 10^{-6}$$

At equivalence point, $\text{pH} = \frac{1}{2} (\text{pK}_w - \text{pK}_b - \log C)$

$$\therefore 4.5 = \frac{1}{2} (14 - 5.6 - \log C) \quad \therefore C = 0.25 \text{ M}$$

$$\text{Now, } C = \frac{n_{\text{HCl}}}{V_f(\text{L})} \quad \therefore 0.25 = \frac{0.5 \times V_L}{V_L + 0.1} \quad \therefore V_L = 0.1 \text{ L} = 100 \text{ mL}$$

At equivalence point, $n_{\text{WB}} = n_{\text{HCl}}$

$$\frac{m}{45} = 0.5 \times 0.1 \quad \therefore m = 2.25 \text{ g}$$

$$\therefore \text{Weight \% of base in sample} = \frac{2.25}{2.5} \times 100 = 90\%$$

12. For getting selectively precipitated, $\text{IP} > K_{\text{sp}}$.

$$[\text{M}^{2+}] [\text{S}^{2-}] > K_{\text{sp}}$$

$$[\text{M}^{2+}] \frac{K_{a_1} K_{a_2} [\text{H}_2\text{S}]}{[\text{H}^+]^2} > K_{\text{sp}}$$

$$\frac{[\text{M}^{2+}][\text{H}_2\text{S}]}{[\text{H}^+]^2} > K_{\text{eq}} \therefore \frac{0.01 \times 0.1}{1^2} > K_{\text{eq}}$$

$$\therefore K_{\text{eq}} < 10^{-3}$$

For PbS , above condition is satisfied.

13. $K_a(\text{NH}_4^+) = \frac{1}{K_{\text{eq}}} = 5 \times 10^{-10}$

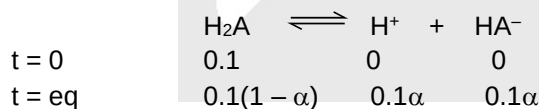
$$\text{Now, } K_b(\text{NH}_3) \times K_a(\text{NH}_4^+) = K_w = 10^{-14}$$

$$\text{So, } \text{pK}_b(\text{NH}_4\text{OH}) = -\log_{10} (K_b(\text{NH}_3)) = -\log_{10} (2 \times 10^{-5}) = 4.7$$

14. $\text{p}_{\text{H}_2} = 10 \text{ atm}$ (as 1 mole H_2 in 2.24 L at 0°C) & $[\text{I}^-] = 0.5 \text{ M}$

$$K = \frac{[\text{OH}^-]^2 \text{p}_{\text{H}_2}}{[\text{I}^-]^2} \Rightarrow [\text{OH}^-]^2 = \frac{4 \times (0.5)^2 \times 10^{-9}}{10} \Rightarrow [\text{OH}^-] = 10^{-5} \text{ M} \quad \therefore \text{pOH} = 5$$

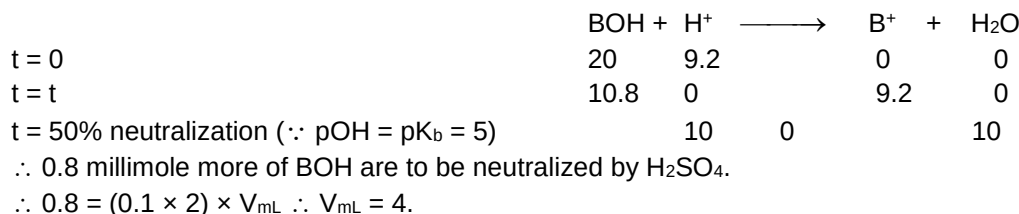
15.



$$K_{a1} = 0.05 = \frac{0.1 \alpha^2}{1-\alpha} \quad \therefore \alpha = 0.5$$

$$\text{Now, } [\text{A}^{2-}]/[\text{H}^+] = K_{a_2}/[\text{H}^+] = \frac{2 \times 10^{-6}}{0.05} = 4 \times 10^{-5}$$

16.





$$17. \quad s = \sqrt{K_{sp}} = 7 \times 10^{-7} \text{ mol/L} = \frac{7 \times 10^{-7} \times 188 \times 10^3}{4} \text{ mg / 250 mL.}$$

$$= 3.29 \times 10^{-2} \text{ mg / 250 mL.}$$

$$\therefore \text{Percentage saturation} = \frac{3.29 \times 10^{-3}}{3.29 \times 10^{-2}} \times 100 = 10\%.$$

$$18. \quad pK_a (\text{CH}_3\text{COOH}) = 4.74$$

$$[\text{CH}_3\text{COOH}] = 0.25 \text{ M}, \quad [\text{CH}_3\text{COONa}] = 0.15 \text{ M}$$

$$[\text{H}^+] = \frac{K_a [\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-]} = \frac{1.8 \times 10^{-5} \times 0.25}{0.15} = 3 \times 10^{-5} \text{ M}$$



$$[\text{S}^{2-}] = \frac{9 \times 10^{-21} \times 0.1}{9 \times 10^{-10}} = 10^{-12} \text{ M}$$

$$\text{IP} (\text{MnS}) = [\text{Mn}^{2+}] [\text{S}^{2-}] = 1.5 \times 10^{-2} \times 10^{-12} = 1.5 \times 10^{-14}$$

$$\text{IP} < K_{sp} \quad \Rightarrow \quad \text{No ppt is formed.}$$

For precipitation of MnS, the minimum concentration of $[\text{S}^{2-}]$ can be obtained as follows :

$$[\text{Mn}^{2+}] [\text{S}^{2-}] = K_{sp}$$

$$1.5 \times 10^{-2} \times [\text{S}^{2-}] = 2.4 \times 10^{-13} \quad \Rightarrow \quad [\text{S}^{2-}] = 1.6 \times 10^{-11} \text{ M}$$

For this $[\text{S}^{2-}]$,

$$[\text{H}^+]^2 = \frac{K_a [\text{H}_2\text{S}]}{[\text{S}^{2-}]} = \frac{9 \times 10^{-21} \times 0.10}{1.6 \times 10^{-11}} = 7.5 \times 10^{-6} \text{ M}$$

$$[\text{H}^+] = \frac{K_a [\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-]} \Rightarrow 7.5 \times 10^{-6} = \frac{1.8 \times 10^{-5} \times 0.25}{[\text{CH}_3\text{COO}^-]}$$

$$[\text{CH}_3\text{COONa}] = 0.6 \text{ M}$$

$$19. \quad \text{In all the other cases, pH} = 1.$$

When all taps are opened,

Millimoles of H^+ = 100

Volume of solution = 2000 mL

$$\therefore [\text{H}^+] = \frac{100}{2000} = 0.05 \text{ M.}$$

$$\text{pH} = -\log[\text{H}^+] = -\log(5 \times 10^{-2}) = 2 - \log 5 \neq 1.$$

$$20. \quad \text{pH}_i = 7.$$

After opening S_1 , solution in bucket would contain $[\text{CH}_3\text{COOH}] = 0.1 \text{ M}$. So, pH after opening

$$S_1 = \frac{1}{2} (\text{p}K_a - \log C) = 2.8775.$$

After opening S_2 , solution in bucket would contain $[\text{CH}_3\text{COONa}] = \frac{1}{15} \text{ M}$. So, pH after opening

$$S_2 = \frac{1}{2} (\text{p}K_w + \text{p}K_a + \log C) = 8.7875.$$

After opening S_3 , solution in bucket would contain $[\text{CH}_3\text{COONa}] = \frac{1}{40} \text{ M}$ & $[\text{CH}_3\text{COOH}] = \frac{1}{40} \text{ M}$. So, pH

$$\text{after opening } S_3 = \text{p}K_a + \log\left(\frac{1/40}{1/40}\right) = \text{p}K_a = 4.755. \text{ (Buffer properties)}$$



After opening S_4 , solution in bucket would contain $[\text{CH}_3\text{COOH}] = \frac{1}{25} = \text{M}$ & $[\text{HNO}_3] = \frac{1}{50} \text{ M}$. So, pH after

opening $S_4 = -\log \frac{1}{50} = 1.7$. (Minimum pH)

Clearly, opening S_2 stop cock would cause maximum pH change compared to the previous pH in bucket.

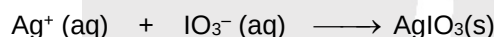
21. Refer solution of previous question.

22. Total added moles of $\text{Ag}^+ = 0.01$; Initial moles of $\text{IO}_3^- = 0.005$
Initial moles of $\text{CrO}_4^{2-} = 0.1$

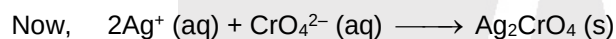
$$[\text{Ag}^+] \text{ required for } \text{AgIO}_3 \text{ precipitation} = \frac{K_{\text{sp}}(\text{AgIO}_3)}{[\text{IO}_3^-]} = \frac{10^{-13}}{0.005} = 2 \times 10^{-11} \text{ M}$$

$$\text{and } [\text{Ag}^+] \text{ required for } \text{Ag}_2\text{CrO}_4 \text{ precipitation} = \sqrt{\frac{K_{\text{sp}}(\text{Ag}_2\text{CrO}_4)}{[\text{CrO}_4^{2-}]}} = \sqrt{\frac{10^{-8}}{0.1}} = 3.162 \times 10^{-4} \text{ M}$$

So, Ag^+ first combines with IO_3^- , almost whole of Ag^+ gets precipitated out, then precipitation of Ag_2CrO_4 starts.



Moles of Ag^+ left after the above reaction = $0.01 - 0.005 = 0.005$.



Moles 0.005 0.1

Moles of CrO_4^{2-} left after the above reaction = $0.1 - 0.0025 = 0.0975$.

At equilibrium,

$[\text{CrO}_4^{2-}] = 0.0975 \text{ M}$

$$[\text{Ag}^+] = \sqrt{\frac{K_{\text{sp}}(\text{Ag}_2\text{CrO}_4)}{[\text{CrO}_4^{2-}]}} = \sqrt{\frac{10^{-8}}{0.0975}} = 3.2 \times 10^{-4} \text{ M.}$$

$$[\text{IO}_3^-] = \frac{K_{\text{sp}}(\text{AgIO}_3)}{[\text{Ag}^+]} = \frac{10^{-13}}{3.2 \times 10^{-4}} = 3.125 \times 10^{-10} \text{ M.}$$

Total moles of precipitate formed = $0.005 + 0.0025 = 0.0075$.